# Introduction to RISC BASED COMPUTER ARCHITECTURE

## AArch32 and AArch64


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# Contents

# 1

# Introduction

**Learning Objectives**

- Compare CISC and RISC architectures in terms of instruction complexity and design goals.
- Explain why ARM RISC processors became important in mobile and energy-efficient computing.
- Describe the advantages of fixed-length instructions in ARM-based RISC architectures.
- Distinguish between CPU simulation and full-system emulation using CPUlator and `qemu-system`.
- Identify why this book uses ARM32 and ARM64 to study assembly and computer architecture.

## 1.1 Historical Context

The evolution of CPU architectures over the years since the 1970s has taken two distinct pathways. The first pathway, and the one taken by Intel with their x86, [1] was to squeeze more and more [circuitry] onto the silicon fabric of the CPU. Starting from around 29000 transistors in 1978, the Arrow Lake variant (launched in 2024) has over 4 billion transistors today [Com25].

The astonishing growth of transistor density is summarized in Fig. 1.1. As you can see from this log-linear graph, growth in transistor density followed the Moore' Law doubling model quite closely, only beginning to slow then plateau in from 2008 onwards.

In addition to the growth in transistor density, the instruction set of the x86 in 1978 had about 100 instructions, whereas today, this number is closer to 1600.

In addition, clock speeds have increased by a factor of 1200 - from 5 MHz to 6 GHz today. Finally pipeline depth: has gone from 4 to about 30 stages in 2024. Pipeline microarchitectures will be discussed in more detail in Chapter 11.

As you can see, the Intel x86 trend has been to continuously pack more and more complex circuitry onto the CPU silicon.

## 1.2 Complex Instruction Set Computers

The Intel x86 as mentioned in the previous section is an example of a *CISC* (Complex Instruction Set Computing) architecture This architecture uses a large set of *complex* instructions, each of which is capable of performing multiple operations in a single instruction. In other words, a CISC instruction is often the *encapsulation* of a set of lower-level operations.

[1] first introduced in 1978

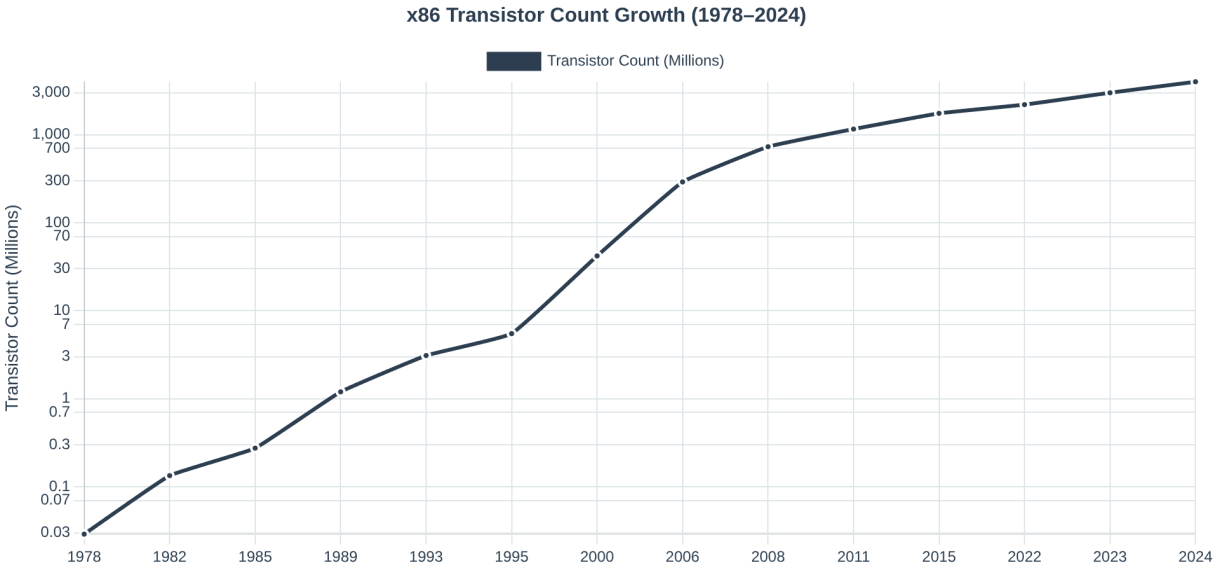



Figure 1.1: The growth of the number of transistors in the x86 architecture from 1978 to the present time. This is a log scale with the y-axis showing a measurement in millions.

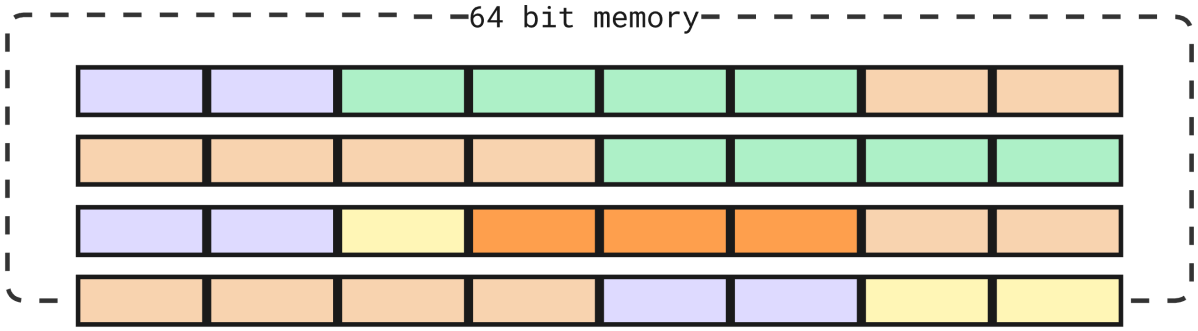



Figure 1.2: Variable length instructions in 64-bit x86 memory. Illustrating 2-byte (purple), 4-byte instructions (green), 6-byte in orange and 1-byte (yellow).

CISC instructions typically offer the programmer a higher level of *abstraction* than, as we shall see, the RISC (Reduced Instruction Set Architecture) discussed next. The Intel x86 Instruction Set Architecture (ISA) supports *variable length* instructions. This means that instructions can occupy less than the full word size of the CPU. Variable length instructions are a space saving solution - reducing the overall footprint of the executable and thereby reducing the amount of memory required to store the process cide. X86 instruction lengths can vary in the range 1 to 15 bytes. This means that the number of bytes used to encode a single instruction can differ, unlike some other architectures that use fixed-length instructions (eg, RISC ISAs)

So instead of padding all instructions out to the same length, variable length instructions allow for more efficient use of main memory.

We present an illustrative model of variable length memory in Fig .1.2. in 64-bit x86 memory. Here, 2-byte instructions in purple, 4-byte instructions in green, 6-byte in orange and one byte in yellow.

## 1.3 Reduced Instruction Set Computers

CISC Instruction Set Architectures (ISAs) were, until relatively recently, the dominant consumer CPU platform. Excluding desktop Apple Mac systems (which used the PowerPC), Windows desktop computers used Intel x86 exclusively. For most end-users, the CISC capabilities of the ISA were an ever-increasing advantage. It made the standard desktop PC more and more powerful and capable, and allowed software engineers to design and build ever more CPU demanding applications (CAD/CAM, animation, modelling and simulation, numerical analysis and so on).

However, as the x86 became the dominant desktop solution, there were plenty of alternative CPU architectures available. These architectures (for example, Apple PowerPC, IBM RS/6000) were notable for one major difference to CISC: they were designed with a *reduced* instruction set (RISC) architecture. For example, the Apple Mac Powerbook G5 used a RISC CPU (from IBM) had only around 500 instructions in the totality of the ISA.

Why might a computer architecture be designed with a deliberately smaller instruction set size than a comparable Intel CPU? There are several reasons for this approach, but there are 3 keys ones:

1. **Energy:** RISC CPUs consume less energy because the architecture has much less circuitry
2. **Speed:** RISC instructions *generally* take fewer CPU cycles to execute than CISC instructions because each instruction has less overhead as is considerably simpler in its circuitry implementation.
3. **Complexity:** RISC instructions are always the same size (the word size of the CPU - 32/bit or 64/bit). Because there are no variable length instructions in CISC, there is no need for the complex circuitry to deal with this.

The fixed length instruction size model is shown in Fig. 1.3. In this example all instructions (purple) are exactly 64-bits in length, with any unused space padded (gray) out. The same logic applies to RISC 32-bit ISAs (although to a lesser extent).

Not only is the ISA reduced in terms of the number of instructions, but we may say that each of these instructions is simpler (this being a relative term of course) to design and implement. Indeed, early versions of the ARM RISC did not contain native CPU support for floating point numbers. It was not until 2001 that on-CPU support for floating point numbers were added to the ARMv5TE architecture.

### 1.3.1 The Mobile Revolution

Of course, we are all familiar with the phenomenal growth in adoptation of mobile devices. From the early 1990s models (typically running custom OS code), to the complex and capable Android and Apple devices of today. There are now only *two* mobile OS standards - Android and iOS. In parallel to the rise of mobile devices, a new RISC mobile architecture emerged from a company called Advanced RISC Machines (ARM).

Although ARM was around even in the 1980s, the release of ARM7TDMI 32-bit RISC architecture was an enabling factor for OEMs to launch complex mobile architectures. ARM7TDMI had only 200 unique instructions. This feature alone gave ARM architectures an unrivalled competitive advantage over Intel in the mobile market.

ARM CPUs consumed far less energy than CISC cpus, requiring very little cooling

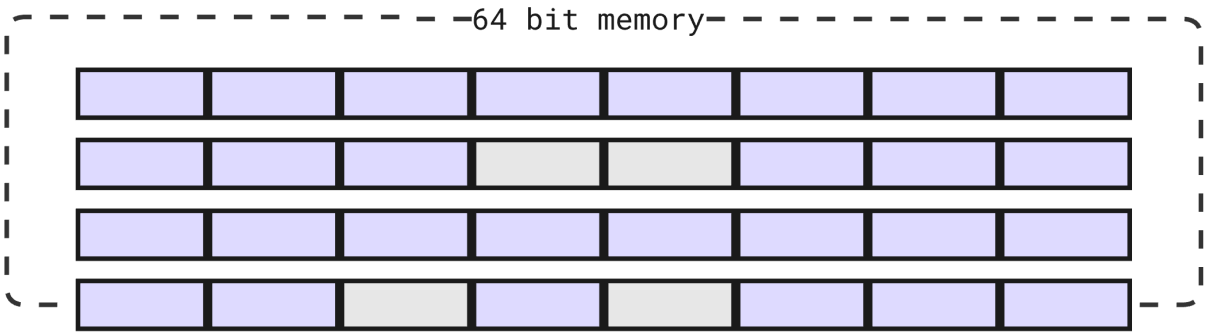



Figure 1.3: RISC fixed length instructions in 64-bit x86 memory. All instructions (purple) are exactly 64-bits in length, with any unused space padded (gray). The same logic applies to RISC 32-bit ISAs (although to a lesser extent)

hardware within the device, and enabling battery powered devices to run complex operating systems such as Android (with the linux kernel). Today, ARMv9-A still has only 500–600 instructions, a substantial difference to Intel, and one that ensures that ARM continues to dominate both the mobile market, and the productivity laptop market seqment (ARM powers Mac hardware).

## 1.4 This book

### 1.4.1 Approach

This book introduces the reader to Computer Architecture. Any book that covers this subject matter needs to decide if it will focus on CISC or RISC ISAs. We have decided to use RISC in the chapters dedicated to assembly programming (Chapters 5 through to Chapter 8), and in the chapters dedicated to microarchitectures (Chapters 10-10).

There are several reasons for this decision, viz:-

1. ARM32 assembly, in particular, is a straightforward language and one that is easy to understand.
2. Both ARM32 and ARM64 are *load and store* architectures (more on that later). So the instructions are WYSIWYG [2], unlike intel x86, where a single instruction can mask a complex implementation.
3. ARM is making ever greater inroads into the data centre. For example, AWS reported in December 2024 that more than 50% of its new CPU capacity added in the previous two years was ARM based. Percentages form the other cloud providers is likely to be similar.

As energy consumption continues to grow in significance for data centre operators, the role of RISC based systems, and ARM64 in particular, is becoming more and more significant.

### 1.4.2 Book Resources

All the code samples used in this book, along with scripts, READMEs and documentation, can be freely cloned from the Book Github Repo.

[2] WYSIWYG - *what you see is what you get*

**▶ Try this: Clone the Book Github Repo**

*Before we begin, make sure you've cloned the Book Github Repo*
Browse the repo, and take a look at the various samples contained there. For example,

1. In the `C` directory we have some code for you to compile and analyze.
2. In the `arm-assembly` directory there is some examples of different ARM32 assembly programs for you to compile and run.

In the assembly language chapters that follow, we will assume the reader is using either `https://cpulator.01xz.net/?sys=arm` (for an in-browser, ARM32 emulator with limited functionality), or they have installed `qemu-sysem` emulator for a complete and accurate emulation of ARM32/ARM64 CPUs. Of course, we do not assume the reader has an Apple Mac (Air or Powerbook). But if you do use Apple, then you do not need an emulator, all you need is a text editor and compiler (`gcc`) will do fine.

### 1.4.3 CPUlator

We recommend two strategies for those starting to learn assembly language programming for the first time. The first resource is called *CPUlator*. CPUlator is a standard web brower *simulator* for the M7/ARM32 ISA. It is an excellent introduction to assembly programming because it has a decent UI that allows you to step through code, add breakpoints, step into and out of label code etc. CPUlator also allows the programmer to view and modify registers, the flag register, stack pointer, and program counter registers, as well as viewing memory during and after program execution.

There are many useful resources to help the new programmer get started with CPUlator. One youtube channel we recommend is called *LaurieWired*. Laurie has an excellent ARM32 video playlist we recommend you work through these carefully in order to better understand CPUlator.

You will find the link to *LaurieWired* along with some other useful web resources in the `README.md` file in the `csci-comp-arch` repository mentioned already.

Please note, although CPUlator provides an excellent emulation learning environment, by its nature, it cannot fully emulate system calls (and does not support any references to `libc`). Therefore, once you develop your competence in ARM assembly programming, you will need to compile and run actual binaries of your own. This is not possible in CPUlator. For this, we need either a true ARM platform (such as Raspberry Pi 3/4/5, Apple Macbook or Air), or a complete system emulator that allows you to boot into full hardware emulation. For this we recommend using `qemu-system`, which we discuss next.

### 1.4.4 `qemu-system`

Finally, let us conside a situation where we want to develop software for an architecture different to the one we are working at. For example, consider a situation where a developer is building software for a ARM64/ARM7TDMI Raspberry Pi, but works on a Linux x86 laptop. How can this be achieved? In this situation, virtual machines are not useful as they cannot execute software that is compiled for a different CPU architecture.

In this situation, there are only two solutions:

- Provide every developer with their own Raspberry Pi hardware. But what do we do when we are building OS-level software for expensive mobile platforms? We cannot give every developer their own expensive mobile device for testing.
- Instead, we need to use a software layer that acts as an *emulator* which fully reproduces the platform we wish to develop on. This includes kernel, file system, binary tool chains and so on. This is the purpose of `qemu`.

As can be seen in Fig. 1.4, we use `qemu-system-aarch64` to run a Debian Linux (kernel 5.12) Emulating ARM7DTI/2 core on top of the host running Arch Linux 6.14 (kernel) on a 4-core CPU x86 ISA. This combination of different kernels, file systems and ISAs is not possible in containerization or virtualization.

### 1.4.5 `qemu-system-aarch64` example

We conclude this section with a concrete example of how to use `qemu-system-aarch64`. Firstly, note that `qemu-system-*` is an full system emulator for a specific ISA. In this case, `aarch64` which is the ARM64 architecture. There are many useful `qemu` resources for you to study - but start at `https://www.qemu.org/` for an excellent overview of the tools and tool chains available.

Consider the following `qemu-sysem-aarch64` command:

```
qemu-system-aarch64
  -machine raspi3b
  -cpu cortex-a72
  -smp 4 -m 1G
  -kernel kernel.img
  -dtb treeblob.dtb
  -drive "file=bullseye.img,format=raw,index=0,if=sd"
  -append "rw earlyprintk loglevel=8 console=ttyAMA0,115200
   dwc_otg.lpm_enable=0 root=/dev/mmcblk0p2 rootdelay=1"
  -device usb-net,netdev=net0 -netdev user,id=net0,hostfwd=
   tcp::5555-:22
  -nographic
```

When we execute this `qemu-system-aarch64` instruction on our host (and assuming `kernel.img`, `treeblob.dtb` and [Debian] `bullseye.img` are available locally), then we will shortly start a Raspberry Pi 3B ARM32 cortex-a72 with the kernel and filesystem of our choice. You can find this script in the book github repository (located in `qemu/start.sh`).

We can then use `ssh` to connect to `localhost` on port 5555 and our instance is ready to begin using for our project developement!

## 1.5 Chapter Summary

In this chapter we have introduced the reader to CISC versus RISC ISAs. We discussed the design principles that motivate the developement of CISC ISAs: that ISA complexity and circuitry complexity outweigh energy consumption performance. Conversely, we noted that RISC ISAs are relatively less complex and significantly less energy demanding.

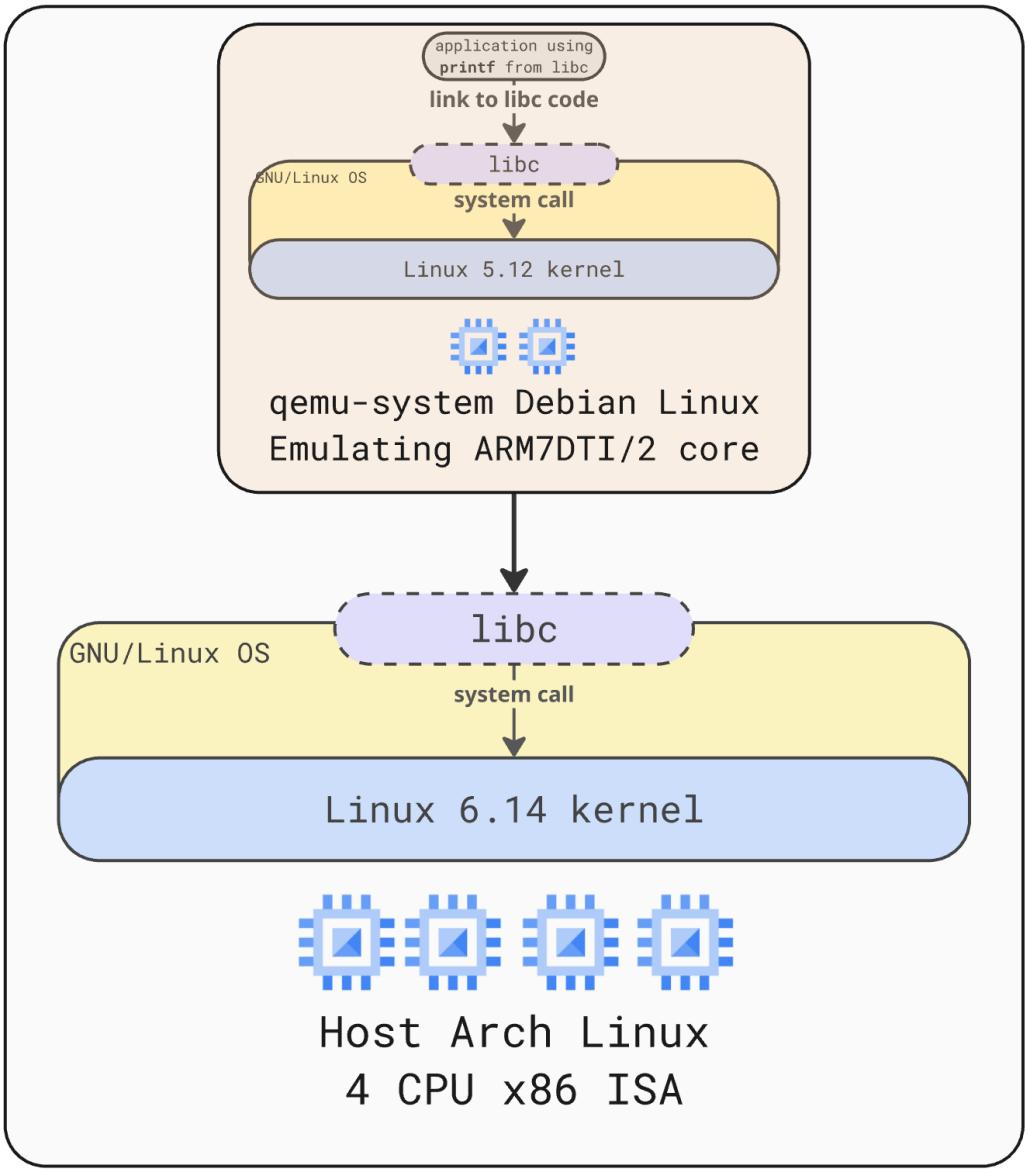



Figure 1.4: Using `qemu-system-aarch64` to run a Debian Linux (kernel 5.12) Emulating ARM7DTI/2 core on top of the host running Arch Linux 6.14 (kernel) on a 4-core CPU x86 ISA

We do not need to consider CISC ISAs any further as the remainder of this book will look exclusively at either ARM32 or ARM64 systems in general.

You are advised to develop a strong understanding of these two architectures and the advantages and disadvantages of each. In this chapter we also introduced you to the source code resources for you to use, along with some suitable emulation platforms. In the early chapters of this book, `CPUlator` is ideal because it gives us a nice develoopement and debugging experience inside a standard web browser. But as we develop our program complexity, we will start using `qemu-sysem` in order to use this full set of `libc` tools present in the Linux OS of our choice.

In the following chapter we will discuss the internal structure of an executable program. It is important to understand this structure well, because when we begin assembly language programming the structures of our executable will be explicitly named and referenced.

## 1.6 Chapter Exercises

**Exercise 1.6.1** *Both ARM7DTI and ARM64 are **load and store** ISAs. Write a brief note comparing load and store with the "traditional" CISC memory access ISA instructions. Give and example to illustrate your answer.*

**Exercise 1.6.2** *The ARM specification discusses emulation. It refers to a limited emulation mode known as **ARM Semihosting**. Write a brief explanation on this ane give an example to support your answer. How does CPUlator support **ARM Semihosting**?*

**Exercise 1.6.3** *CPUlator is an ARM **simulator** whereas `qemu-system-aarch64` is an ARM **emulator**. Write a brief note explaining the difference between these two concepts and give some examples.*

**Exercise 1.6.4** *Here is a full list of the `qemu-system` emulators on our system:*

```
qemu-system-aarch64          qemu-system-ppc
qemu-system-alpha            qemu-system-ppc64
qemu-system-arm              qemu-system-riscv32
qemu-system-avr              qemu-system-riscv64
qemu-system-hppa             qemu-system-rx
qemu-system-i386             qemu-system-s390x
qemu-system-loongarch64      qemu-system-sh4
qemu-system-m68k             qemu-system-sh4eb
qemu-system-microblaze       qemu-system-sparc
qemu-system-microblazeel     qemu-system-sparc64
qemu-system-mips             qemu-system-tricore
qemu-system-mips64           qemu-system-x86_64
qemu-system-mips64el         qemu-system-xtensa
qemu-system-mipsel           qemu-system-xtensaeb
qemu-system-or1k
```

*Taking any two of these platforms (except `qemu-system-i386` and `qemu-system-x86_64`) compare and contrast them under the following headings:*

1. *Clock speeds*

2. *ISA size (number of instructions)*
3. *Energy consumption*
4. *Operating System / kernel*
5. *Number of units still running*
6. *typical workload*

# 2

# Program Representation and Execution

**Learning Objectives**

- Explain the compilation stages from C source to assembly, object files, and ELF executables.
- Identify major ELF sections, including `.text`, `.rodata`, `.data`, `.bss`, and `.dynstr`.
- Map C program components to their corresponding ELF sections and runtime memory regions.
- Distinguish between ELF sections and segments used by the operating system loader.
- Describe the roles of the stack, heap, writable segments, read-only segments, and memory-mapped regions.

**UNDERGRADUATE**
*2.1–2.7*

## 2.1 Introduction to the Executable and Linkable Format

The process by which a C program is eventually transformed into a binary image (that can be loaded and run on a modern computer), is rather complex.

In simple terms, and for a C program called `main.c`, we can define 3 key phases that turn your C programs into executable code:

1. Compiler to Assembly (output: `main.s`) The compiler translates the program into low-level assembly code specific to the target CPU.
2. Assembler to Object File (output: `main.o`) The assembler converts assembly into machine code, producing a file that may have unresolved external references (symbols).
3. Linker to Executable (output: `a.out` ELF file) The linker combines one or more object files and libraries, resolves symbols, lays out memory, and produces a final executable in the ELF format. The basic flow of this process is shown in figure 2.1.

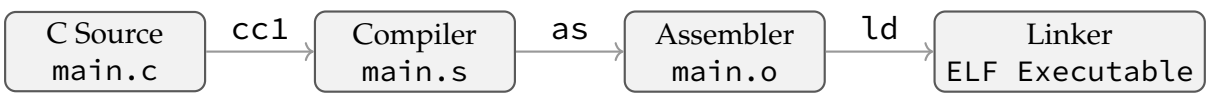



Figure 2.1: C program to ELF executable compilation stages (simplified)

Irrespective of the programming language, almost all unix-like systems (with the exception of OSX) store executable code in the **Executable and Linkable Format** (ELF).

```c
/*
 * simple.C
 * a file to show the layout of objects
 * in the a.out executable
 * using objdump -s -C
 */
#include <stdio.h>

// initialized global variable (r/w)
int global_var1 = 100;

// uninitialized global variable (r/w)
int global_var2;

// initialized global variable (r/w) string
char global_var3[] = "This is a global string";

int main(int argc, char* argv[]) {
  int local_var1 = 99;
  int local_var2;
  // reference to prinf is in .dynstr
  // the string ("This is ..") is r/o in .data
  printf("Locals are %d %d\n", local_var1, local_var2);
  printf("Globals are %d %d %s\n",
    global_var1,
    global_var2,
    global_var3);
  return 0;
}
```

Figure 2.2: Simple C to show the layout of objects in an ELF file

[1] The ELF format is relatively complex, containing a number of sections that describe a program's contents and a smaller number of segments that determine how those contents are loaded into memory. The discussion that follows is necessarily a simplification. Readers wishing to understand more should start with the `objdump` man pages on their local Linux OS.

To demonstrate the layout of an ELF file, we have developed a simple simple C program (shown in Fig 2.2), that has a number variable types, including, global initialized int, global uninitialized int, global initialized character array, initialized function local variable, and uninitialized function local variable.

## 2.2 ELF Layout

When the C program shown in Fig. 2.2 is compiled and linked, ready for execution, the file (by default called `a.out`), will have the structure shown in Fig. 2.3

As you can see from the ELF layout shown in Fig 2.3, the structure of the ELF file contains the *static* properties of the C code - in other words - the properties of the

[1] ELF gABI Specification

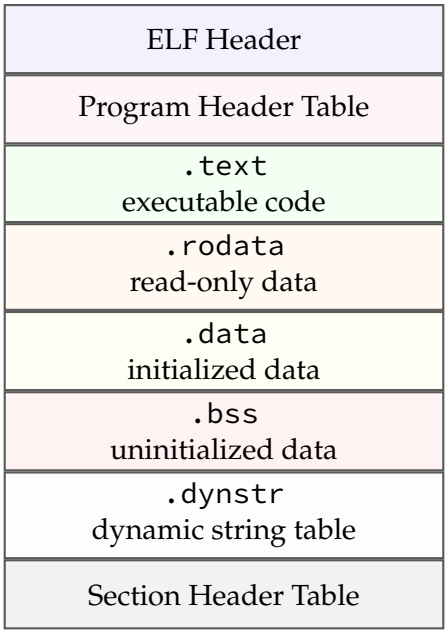



Figure 2.3: Simplified ELF file layout

program at rest on the disk. What's missing from this picture is the *dyanmic* properties of the program when it gets loaded into memory at *process* start time. We discuss this in the next section.

Next we will map the various parts of our program into the ELF sections as defined in Fig. 2.3

### 2.2.1 Mapping C Program Components to ELF Sections

The C program defined in Fig. 2.2 is not especially interesting or useful. The purpose of the program is to show you where the various elements of a C program are placed in the ELF section model introduced in Fig. 2.3. We should also remember, according to the ELF standard, some sections are *read only (ro)* whereas some are *read write (wr)*

1. `.text`: This *read-only* section stores all the instructions from the program in the order in which they are defined in the C program. For example:

```
int local_var1 = 99;
int local_var2;
printf(...)
printf(...)
```

2. `.rodata`: This section also stores *read-only* data such as strings that are passed into a function (for example, `printf`). So this section contains `"Locals are %d %d \n"` and `"Globals are %d %d %s \n"`. These strings are read-only because they can not be modified once compiled into the program code.
3. `.data`: This section is used to store initialized global data, so the variables `global_var1` and `global_var3` are placed into this section. Note that both these variables are modifiable. This is true of any variable data placed into `.data`. We will see this section in much more detail in the coming chapters.
4. `.bss`: This section contains *uninitialized* global data. So in this section we expect to find `global_var2`

5. `.dynstr`: This section is used to record what dynamic runtime interpreter is needed during program execution. For a program that contains dynamically linked shared libraries, for functions such as `printf`, the section `.dynstr` will typically contain the string `printf.libc.so.6`, along with several other entries

### 2.2.2 Unmapped C Program Data

If you've been following the preceeding section closely, you will notice that some of our C program data was not mapped to an ELF section. For example:

```
int local_var1 = 99;
int local_var2;
```

The variables `local_var1` and `local_var2` are not stored in any data section in the ELF file. So where are they stored? They are both stored on the *stack*.

Each thread executing inside a process gets its own stack in memory. This means that should a process have more than one thread, each thread will be allocated its own stack segment in virtual memory. The stack is a very importand and specialized region of memory. It is highly *dynamic*, growing and shrinking over the lifetime of the process. In section 2.3.2, we briefly review the key ideas of the stack, but we dedicate a complete chapter to it - Chapter 7, where important matters such as local variables, saving and restoring registers, and the implementation of procedures, is discussed in detail.

**▶ Try this: Using `objdump` to view ELF sections**

If you have access to a Linux OS you can use `objdump` to view the ELF contents of an executable file. If you don't use a Linux OS, you can still do this at the GCP command line!
*Before you begin, make sure you've cloned the Book Github Repo*

- Change into the directory `C` directoy
- Compile `simple.C` using `gcc` or `clang` - for example: `gcc simple.C`
- Use `objdump -s -C a.out` to view the ELF sections of the file.
- Look out for the `.dynstr` section ... what do you see?
- Look out for the `.data` section ... what do you see?

*Fun question - do `gcc` and `clang` produce significantly different object files?*

## 2.3 Process address space of an ELF program

When the ELF executable becomes a *process*, it gains a number of additional memory resources, such as stack, heap and a shared library section. These are the sections needed for correct process initialization and execution.

### 2.3.1 Sections to Segments

Sections describe the logical contents of a program within the file. They are defined in the Section Header Table and are used primarily by the compiler and linker.

Segments, on the other hand, define how the program is loaded into memory. They are specified in the Program Header Table and are the structures actually interpreted by the operating system loader at runtime.

The connection between the two is established by the linker: The linker groups sections into segments based on their required memory properties (e.g., read, write, execute). Each segment may contain parts of one or more sections. The operating system loads segments, not sections, into memory. This is why the terminology in Fig. 2.4 switches from *sections* to *segments*

The heap and stack sections are *dynamic* regions: they grow and shring as the process executes. When the process exits, the stack, heap and shared library sections are *unmapped*, in other words, they are returned to the kernel. This arrangement is shown in Fig. 2.4.

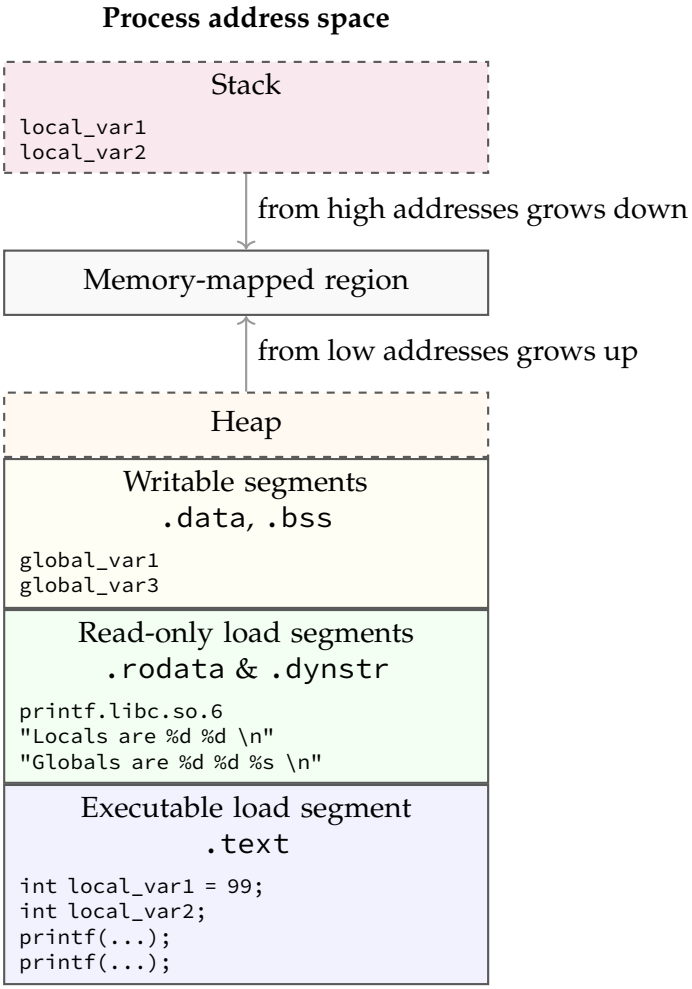



Figure 2.4: Simplified process address space for an ELF program Fixed size segments are shown with solid borders, dynamic sized segments (stack and heap) are shown with dashed borders. A simplified placement of data and code is also shown.

### 2.3.2 The Stack

Function local variables are always placed on the stack. As each thread gets its own stack space, such local variables are always private the the thread and are not shared between threads. Contrast this with data stored in writable segments (`.data`, `.bss`). These data are seen by all threads and, unless access is properly controlled, a race condition may arise. A race condition is a situation where two or more threads are altering a value at the same time, possibly resulting corruption or incorrect values.

In most ISAs, the stack grows downward, from high addresses to low addresses. Technically this called the Full Descending stack model, and in Chapter 7 we discuss the correct way to add and remove data from the stack. For now, let us just make a few observations about thread stacks

1. Stacks are a finite resource. By default, on a typical Linux based system, the default size is 8192kb, or 8MB, per thread
2. Stacks are used for *temporary* or *short-lived data*, for example data needed only for the duration of a function. This short-lived data must be returned to the operating system via proper management of the stack pointer (`sp`) before the funtion returns.
3. Failure to manage the stack pointer correctly can lead to a stack leak or a stack overflow. We will discuss these further in Chapter 7, along with the ARM instructions we use to manage the stack.

### 2.3.3 Memory Mapping

The memory-mapped region of a process is the portion of the virtual address space used for mappings created with system facilities such as mmap. In an Executable and Linkable Format program, this region sits between the heap and the stack (conceptually), though its exact position and growth pattern are determined by the kernel's virtual memory layout.

One of the primary uses of the memory-mapped region is to map shared libraries into the process address space. When an ELF executable depends on dynamic libraries, the dynamic linker loads these libraries by mapping their segments directly into the process's address space.

This allows multiple processes to share the same physical memory for read-only portions (such as code), improving efficiency. These mappings typically include read-only segments (e.g., `.text`, `.rodata`) and sometimes read-write regions for relocation and global state.

### 2.3.4 The Heap

The heap is a region of dynamic memory that typically grows upward, from lower to higher virtual addresses. Memory in the heap is allocated at runtime using functions such as `malloc`, which are implemented in the C standard library and ultimately rely on system mechanisms like `brk` or `mmap`.

Heap allocations are often long-lived, potentially persisting for the lifetime of the process unless explicitly freed. The heap is a per-process resource, shared by all threads within the same process. As a result, concurrent access to heap-allocated data must be properly synchronized (e.g., using mutexes).

In principle, there is no fixed upper bound on heap size. In practice, it is constrained by factors such as the process's virtual address space limits (e.g., `ulimit`), operating system policies (including `cgroups`), and available physical memory.

We will not focus extensively on heap allocation in this book, as it requires interaction with either system calls or the C standard library. However, Chapter 8 introduces how to integrate the C library into an ARM assembly program (e.g., for using `printf`), and the same approach can be extended to support dynamic allocation functions such as `malloc`.

## 2.4 Chapter Summary

## 2.5 Exercises

**Exercise 2.5.1** *Why is it beneficial for multiple processes to share the same physical memory pages for shared libraries? In your answer, include implications for memory usage and overall system performance.*

**Exercise 2.5.2** *How does virtual memory allow each process to have its own address space while still sharing physical memory? In your answer, explain the role of address translation.*

**Exercise 2.5.3** *Why is it important that the stack grows downward and the heap grows upward in a process address space? What problem does this design help prevent?*

**Exercise 2.5.4** *What would happen if each process had to load its own private copy of every shared library it used? Discuss both memory usage and system scalability.*

**Exercise 2.5.5** *How does the operating system ensure that shared memory (such as shared libraries) is protected from unintended modification by processes?*

**Exercise 2.5.6** *Why are memory-mapped files considered more efficient than traditional file I/O in some cases? Provide at least two reasons.*

**Exercise 2.5.7 (Using `ulimit`)**

**Exercise 2.5.8 (Using `objdump`)**

# 3

# Assembly Language Programming

**Learning Objectives**

- Explain how ARM assembly uses registers to store and manipulate word-size data.
- Apply basic arithmetic instructions, including `mov`, `add`, `sub`, `mul`, and `mla`.
- Interpret signed and unsigned integer values using two's complement representation.
- Use logical instructions such as `and`, `orr`, `eor`, and `bic`.
- Describe how the program counter `pc` controls sequential execution and program exit.

We begin our study of assembly by focusing on ARM7DTI which is a ARM32-bit platform. The ARM64 platform contains many additions and updates, and in most real-world scenarios you will be building and deploying software on this ISA. However, the 32-bit variant is interesting from the pedagogical point of view because it is a smaller and significantly simpler ISA.

As we proceed through the following chapters we will compare C programming and assembly to understand how tasks in high level programming languages are implemented in low level assembly.

## 3.1 Basic Instructions: Adding and Subtracting

Let us begin by doing a very simple task: adding two numbers together and storing the result in a variable. Let's choose $5$ and $10$ as our two numbers, and $c$ as the variable. Mathematically this is simply $c = 5 + 10$, and in C:

```
int c = 5 + 10;
```

How do we implement this in assembly? We need to use the assembly assignment instruction (`mov`) and the addition mathematical function (`add`) instruction:

```
mov r1, #5       // move 5 into r1
add r3, r1, #10  // add r1 and 10
```

You notice immediately that in the assembly code we added in *two* steps: fist, assignment using `mov` then add using `add`.

Why did we not do this:

```
add r3, #5, #10  // why not this?
```

This instruction will not assemble and will generate an error. What is the error?

```
Error: bad expression -- 'add r3,#5,#10'
```

It tells us that the *first* operand to `add` cannot be an immediate (in other words, it cannot by a number, it must be a register)

Why this limitation? As we will see in Chapter 9 this constraint is due to the fact that there is *no space* in a 32-bit instruction to encode two arbitrary immediate values. In fact, even encoding one arbitrary immediate has limitations which will be discussed in detail in Chapter 9.

### 3.1.1 Discussion

Notice some interesting features of the assembly code. Let's deal with each of these in turn:

1. We use *registers* to store data. Registers are word-size storage areas on-board the CPU itself. Because they are on the CPU, register updates happen in one CPU clock cycle (in other words, zero latency).
   In ARM7DTI there are 15 *general purpose* registers. These registers are named `r0-r14`. Register `r15` is a special purpose register and should not be used by the programmer. As there are only 15 general purpose registers, we must be careul not to use them unneccessarily, otherwise we may run out of registers.
2. Two new instructions have been introduced: `mov` and `add`. Notice also that when we want to use an integer value, we use the `#` sign in front of it. For example, `#10`. This is a language requirement. Numbers like this are called *immediates*. Later we will explain the derivation of the word, but for now, we will accept this term as-is.
3. In ARM7DTI all instructions are either right-to-left or left-to-right ordered [1]. The `mov` instruction is right-to-left ordered. It means take the *value* from the right hand side and place it into the *location* on the left hand side. Is it possible to have a `mov` that uses *two* locations (eg, `mov r1, r2`)? Yes. Is it possible to have a move with two values `mov #3, #4`)? Of course, no.
   ***Notice that `mov` is a two operand instruction***
4. The `add` instruction is also a right-to-left instruction. Notice that `add` is a *three* operand instruction. So the code `add r3, r1, r2` is understood as *"add the value in r2 to the value in r1 and place the result in r3"*. Does this instruction change the value of either `r1` or `r2`? No. Does it change the value of `r3`? Yes.
   Some variants of `add` that achieve the same result:

   ```
   mov r1, #5
   add r3, r1, #10
   ```

   and

   ```
   mov r1, #5
   add r3, r1
   ```

   Notice here that `r3` does not contain 15 but rather it contains 10.
5. Finally, notice the following illegal instructions:

[1] most are right-to-left ordered

```
// illegal
add r3, #10, r1 // cannot add a reg to a constant
// causes this compiler error
Error: constant expression expected -- ‘add r3,#10,r1’
```

and

```
// illegal
add r3, #10, #5
// causes this compiler error
Error: bad expression -- ‘add r3,#10,#5’
```

We cannot add a register to a constant, nor can we add a constant to a constant. The reason for these contraints is not exactly as you might suppose. We will discuss these types of constraints in Chapter 9 (Machine Code), but suffice it to say, the constraints are due to the nature of ARM7DTI RISC ISA, and in particular, the constraint of fixed word-size instruction length.

### 3.1.2 Two's Complement

In C, we use the type `unsigned` to specify that the variable cannot take on a value of less than zero. [2]

If we deal only with unsigned integers, then all the bits of the word size are allocated in representing the number. However, as you know, there are many negative-type numbers, and we must have a way of representing them within the 32-bit word.

Over the years, there were many attempts to represent negative numbers inside computing hardware. Today, the standard is *two's complement* method which we outline here:

A two's complement negative number is computed as follows:

1. Take the bit pattern of the positive number and invert all the bits ($0 \rightarrow 1$ and $1 \rightarrow 0$)
2. Add one (bit) to the resulting number.

Let us look at an example: we want to represent $-5$. To make the study of this a litle easier, let us deal with a 6-bit computer (the principles generalize to $32-$ and $64-$bit systems in exactly the same way):

1. $5 = 000101 \rightarrow 111010$
2. Now, add $1$ to $111010 = 111011$

The largest non-negative number expressible in a computer of word size $N$ bits, is $2^N - 1$. This means in a six bit CPU, the largest non-negative number is $2^6 - 1 = 63$.

Notice that $111011$ can be thought of in two different ways. As a non-negative number it is

$1 * (2^5) + 1 * (2^4) + 1 * (2^3) + (0 * 2^2) + (1 * 2^1) + (1 * 2^0) = 59$

However, because the most-significant bit (MSB) is $1$, this number is *also* a negative number. The negative number is computed as follows:

$-1 * (2^5) + 1 * (2^4) + 1 * (2^3) + (0 * 2^2) + (1 * 2^1) + (1 * 2^0) = -5$

[2] However, due to the implicit type conversion rules in C, an `unsigned` can silently change to `signed` if it is later assigned a negative value

Let us test to see if *addition* under two's complement works correctly. We know that $27 + -5 = 22$. Does it work? Let's see:

$$\begin{array}{r} 011011 \\ +111011 \\ \hline\hline \\ 1 \overset{\text{carry}}{\longleftarrow} 010110 \end{array}$$

As $010110 = 22$ we can see that addition and subtraction of posive and negative two's complement works correctly.

Note, in the example above, we had a *carry-out* of the MSB. This arises because *there was no more space* left in the 6-bit computer for the full answer. (in other words, the result needed a 7-bit computer, which we were not using)

### 3.1.3 Mathematical Instructions

Registers can hold positve or negative numbers without the need to use words such as `signed`, or `unsgined`:

```
mov r1, #10 // move 10 into r1
mov r2, #-5 // move negative 5 into r2
mov r3, r2  // move (r2) negative 5 into r3
```

Let us look briefly at `sub` instructions.

```
mov r1, #10 // move 10 into r1
mov r2, #-5 // move negative 5 into r2
mov r3, #5 // move 5 into r3
add r4, r1, r2 // result is 5, stored in r4
sub r4, r1, r3 // result is 5, stored in r4
sub r5, r2, r1 // result is -15, stored in r4
// etc
```

Table 3.1 shows some of the more commonly used ARM7DTI mathematical operations. It is by no means complete. Please refer to [Ins25] for a comprehensive treatment of ARM7DTI instructions.

### 3.1.4 A note on `{cond}{S}`

You may notice in Table 3.1 and Table 3.2 the text `{cond}{S}` appended to the end of the mathematical instruction (for example `sub{cond}{S}`). We should first note that there are two different thiings going on here, both of which are optional:

1. `instr{cond}` indicates the instruction should execute only if `{cond}` is *true*. This is called a conditional instruction. For example `mullt` consists of `mul` and `lt`. Here `lt` means *less than*, and signifies that the `mul` operation should only execute *if the condition flags permit*.
2. `instr{s}` indicates the instruction outcome of the instruction (less-than, zero, greater-than etc) should be sent to the *conditional registers* so that other instructions may access the information.

| Instruction | Type | Description |
|---|---|---|
| `add{cond}{S}` | Arithmetic | Adds two registers or a register and an immediate value, storing the result in a destination register. |
| `sub{cond}{S}` | Arithmetic | Subtracts one register or immediate value from another, storing the result in a destination register. |
| `mul{cond}{S}` | Multiplication | Multiplies two registers and stores the 32-bit result in a destination register. |
| `mla{cond}{S}` | Multiplication | Multiplies two registers, adds a third register to the product, and stores the result in a destination register. |
| `umull{cond}{S}` | Multiplication | Unsigned multiply long, multiplies two 32-bit registers to produce a 64-bit result, stored in two registers. |
| `smull{cond}{S}` | Multiplication | Signed multiply long, similar to `UMULL` but for signed integers. |
| `adc{cond}{S}` | Arithmetic | Adds two registers or a register and an immediate value with carry, storing the result in a destination register. |
| `sbc{cond}{S}` | Arithmetic | Subtracts one register or immediate value from another with borrow, storing the result in a destination register. |

Table 3.1: Top 10 ARM32 Mathematical Instructions

We will discuss both `instr{cond}` and `instr{s}` in the next chapter. For now, to keep things simple, we omit these optional appendages and focus on the mathematical operation itself.

### 3.1.5 Multiplication using the `mul` family

We now turn to multiplication operations. There are two main multiplication instructions in ARM assembly ISA: `mul` and `mla`. We will briefly discuss each. We use `mul` when we want to store the result of the operation into some register as follows:

```
mov r1, #10
mov r2, #5
mul r4, r1, r2
```

We can understand this line 3. as *"multiply the value in `r1` by the value in `r2` and store the result in `r3`"*

Here, `r4` now contains 50. This is equivalent to the C code:

```
int c = 10 * 5;
```

In C we often want a *multiplication-accumulation* function as:

```
int c = 10 * 5;
d += c;
```

Because multiplication-accumulation is such a common mathematical operation, ARM assembly has a special *four* operand instruction, (`mla`) to do this:

```
mov r1, #10
mov r2, #5
mla r4, r1, r2, r5
```

Note here that `r5` is the accumulation register, and is an accumulation of the product of `a` and `b` plus whatever was in `r5` previously.

### 3.1.6 Logical Operators

Finally, let us briefly mention the logical Operators in ARM7DTI ISA:

Table 3.2: ARM32 Logical Operator Instructions

| Instruction | Description |
|---|---|
| `and{cond}{S} Rd, Rn, <Operand2>` | Bitwise AND between `Rn` and `Operand2`, result in `Rd` |
| `orr{cond}{S} Rd, Rn, <Operand2>` | Bitwise OR between `Rn` and `Operand2`, result in `Rd` |
| `eor{cond}{S} Rd, Rn, <Operand2>` | Bitwise XOR between `Rn` and `Operand2`, result in `Rd` |
| `bic{cond}{S} Rd, Rn, <Operand2>` | Bitwise AND of `Rn` with NOT of `Operand2`, result in `Rd` |
| `orn{cond}{S} Rd, Rn, <Operand2>` | Bitwise OR of `Rn` with NOT of `Operand2`, result in `Rd` |

## 3.2 Program Flow and the Program Counter

### 3.2.1 Sequential Execution

Assembly programs have a wide number of structural characteristics in common with high level languages (such as C). These structural similarities are not total. There are plenty of points of difference. For the programmer writing assembly code for the first time, one of the most obvious differences is in program flow. In assembly, there is no *scope-control* equivalent.

This means that your assembly code will start at the entry point and continue sequentially until your program either exits or branches. We will discuss branching in Chapter 4.

In the example below, the program begins executing at line 1. and stops at line 6.

```
[numbers=left, stepnumber=1]
mov r1, #10
mov r2, #-5
mov r3, #5
add r4, r1, r2
sub r4, r1, r3
sub r5, r2, r1
```

### 3.2.2 The Program Counter Register (pc)

In ARM7DTI and ARM64 there is a dedicated register known as the *Program Counter* register (pc register), the job of which is to point to the address in memory of the currently executing instruction [3]. Suppose in the above example, the instructions are placed into memory in the following sequential hexadecimal addresses:

```
//Address  Instruction
00FFFF00   mov r1, #10
00FFFF04   mov r2, #-5
00FFFF08   mov r3, #5
00FFFF0C   add r4, r1, r2
00FFFF10   sub r4, r1, r3
00FFFF14   sub r5, r2, r1
```

Then the program counter will change over time as shown in Fig. 3.1:

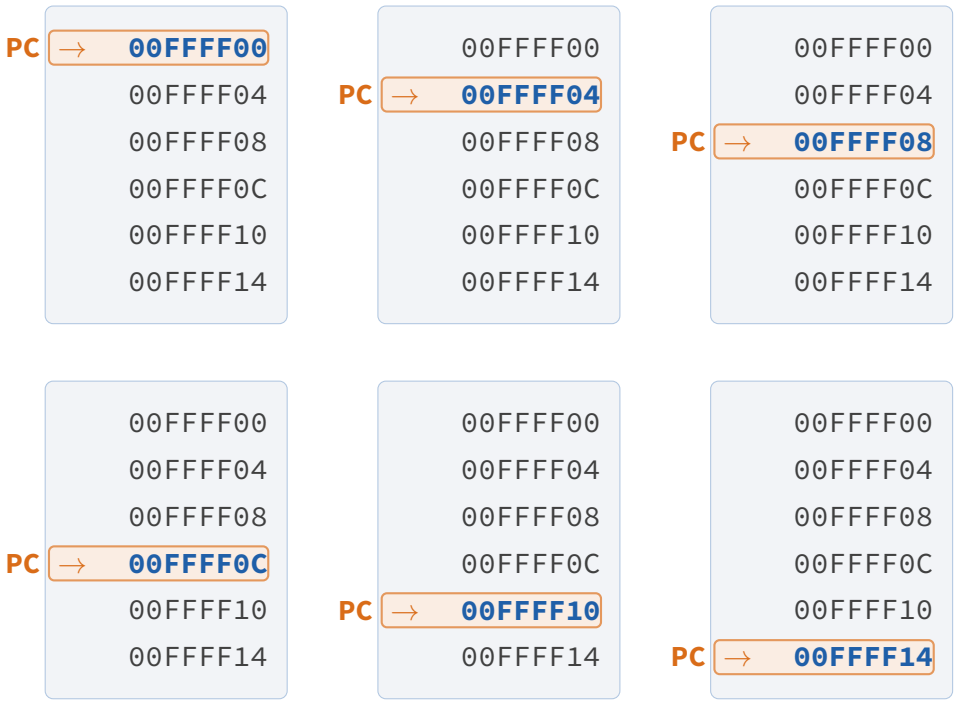



Figure 3.1: Program counter movement through memory addresses.

The pc is modifiable by the programmer. But we need to be extremely carefull in doing so, because changing the programmer counter to address $A$ will cause the

[3] this is a slight simplification which we refine in Chapter 10

```c
int main(int argc, char* argv[] {
  // method 1 - use "return"
  return 0;

  // method 2 - use "exit"
  exit(0);
}
```

Figure 3.2: We can use `return 0` or `exit(0)` to return from `main`. The best practice is to use `return`

```c
int main(int argc, char* argv[] {
    A();
}
void A() {
    B();
}
void B() {
    C();
}
void C(){
    if (fatal_error) {
        // just exit here
        exit(2);
    }
}
```

Figure 3.3: Due to the deep nesting of function calls, it is easier here to `exit(2)` rather than unwinding the function call stack

instruction at address $A$ to be read and processes by the CPU. $A$ should always contain a valid instruction - if it does not - the program will crash.

### 3.2.3 Program Exit

A C program can exit and return control to the operating system in one of two ways, both shown in Fig 3.2.

Note in Fig. 3.2 - by convention - a program that returns $0$ to the operating system (eg, `return 0`) indicates it exited normally. A non-zero value (eg, `exit(3)`) indicates the program encountered an error or some abnormal ending.

If the function call stack is deep and unwinding it via return values is inconvenient, a function can use `exit()` to terminate the program in the function. For example see Fig 3.3:

In assembly there is a direct equivalent to the C `return` and `exit()`. We will briefly show the options in assembly, but the instructions we will explain in detail in future chapters.

```
main:
  // the "return" pattern
  push {lr} // save lr on the stack
  // implement your application
  mov r0, #0 // set the return value in r0
  pop {pc} // set the pc to the lr

  // the "exit" pattern
  // implement your application
  mov r0, #0 // set the return value
  bl exit // call exit from libc
```

Figure 3.4: In assembly we can `return 0` by using using the first pattern and call `exit(0)` using the second pattern.

## 3.3 Chapter Summary

In this chaper we introduced the basic concepts of ARM7DTI assembly programming. We reviewed the main mathematical and logical operations, the role of the Program Counter (`pc`). We also discussed how to terminate an assembly language program (`return` versus `exit()`) and the tradeoffs of each approach. We introduced some important topics that will be covered later in this book, in particular:

1. Stack manipulation using `push` and `pop`.
2. Branch with link using `bl`
3. Setting the `pc`.
4. Conditional execution and/or conditions register updates using `instr{cond}{S}`

We will return to these interesting and importand topics in future chapters.

## 3.4 Chapter Exercises

**Exercise 3.4.1** *Implement the following C program in assembly:*

```
int main(int argc, char* argv[] {
  // we encountered an error
  return 16;
}
```

**Exercise 3.4.2** *Convert this assembly program to C and compile it:*

```
mov r0, #5
mov r1, #50
mov r2, #0
mla r3, r2, r1, r4
```

*Use `printf` to test the result is correct.*

**Exercise 3.4.3** *Modify the previous program to "return" to the operating system with a* 0*.*

**Exercise 3.4.4** *Write the assembly code to use the logical `and` operator to test if a number is odd or even. If the number is odd place* 0 *into `r0` and if it's even, place* 1 *in `r0`.*

**Exercise 3.4.5** *This code does not properly "return" to the opeating system. Why not? What does this code actually do?*

```
push {lr}
mov r0, #0
pop {lr}
```

# 4

# Conditional Execution

**Learning Objectives**

- Explain how ARM implements conditional execution using comparisons and condition codes.
- Interpret the `N`, `Z`, `C`, and `V` flags in the CPSR.
- Use the `cmp` instruction to compare values without storing the subtraction result.
- Distinguish between signed and unsigned condition codes, including `gt`, `lt`, `hi`, and `lo`.
- Construct conditional ARM instructions that execute only when the required CPSR flags are set.

## 4.1 Branching

All programming languages allow us to control the flow of execution based on some logical test. Of course, we are all familiar with this typical program form:

```
if (condition_is_true) {
    // execute instructions a,b,c,d
}
else {
    // execute instructions e,f,g,h
}

// execute instructions i,j,k,l
```

In C we use `if` and `else`, along with scope delimiters `{ }`. Scope delimiters are used to specify how many instructions are to be included in the `if` part, and how many in the `else` part. In C, when there are no curly branches present following the `if` and/or `else`, then the scope is limited to the end of the next line delimited by `;`

More formally, we can say that code containing `if` and `else` has a *branch* structure. You may have seen *flowcharts*, which are useful in understanding the logical branches in a program. Let's look at the above C program in the form of a flowchart as shown in Fig 4.1

Although in a C program we use `if/else` keywords to create a branch in the instruction flow, in ARM7DTI assembly we use a combination of instructions to effect this. But before we look at the appropriate ARM instructions, let us think a little more deeply about `if` tests.

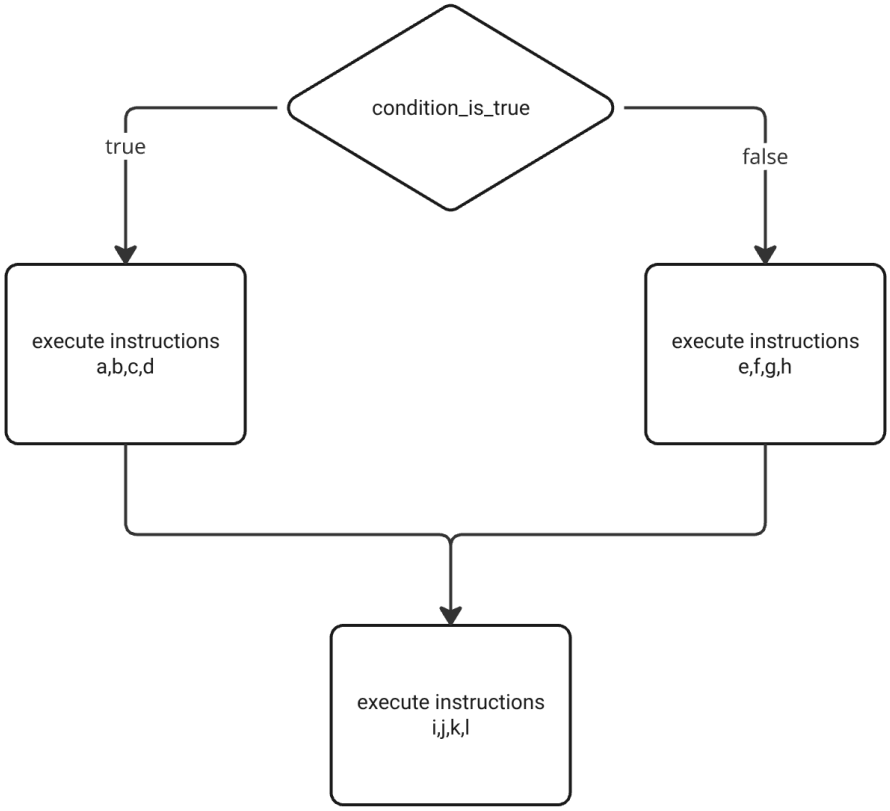



Figure 4.1: Flow chart for the C program showing two *branches*

### 4.1.1 `if` test as subtraction

You may not have realised it, but all `if` tests can be thought of as subtraction operations. For example, suppose we have $a = 5$ and $b = 10$, then

1. `if a > b` is true if $a - b > 0$ which means the result is not zero and not negative
2. `if a >= b` is true if $a - b > 0$ or $a - b = 0$ which means the result is either zero or not negative
3. `if a == b` is true if $a - b = 0$ which means the result is zero
4. `if a < b` is true if $a - b < 0$ which means the result is negative
5. `if a <= b` is true if $a - b < 0$ or $a - b = 0$ which means the result is negative or zero

Notice in the above that we use only two *indicators:* the zero indicator and the negative indicator. In C we don't have to test for these indicators, the compiler generates the code behind the scenes to do the subtraction. But in ARM7DTI assembly we must understand this process, because there is no direct `if/else` equivalent, there is only *subsection* of the operands and the *test* of the outcome. So next we look at these *indicators*. Where are they stored and how do we access them.

## 4.2 The Current Program Status Register

### 4.2.1 NZCV flags

The ARM7DTI processor has a dedicated register known as the Current Program Status Register CPSR, which is updated with the outcome status of mathematical operations. As mentioned above, this outcome status is (a) result was a negative, (b) result was a zero. We also add two new outcome status indicators (c) the result includes a *carry-out*, and (d) the result includes a *overflow*. Although the CPSR is a 32-bit register, only the last 4 least significant bits (LSBs) are of interest to us. Collectively, these 4-bits are often referred to as the NZCV bits of the CPSR register.

The CPSR register cannot be directly read or written to by the programmer. (only the ALU can update the CPSR).

### 4.2.2 Carry-out

Two's complement representation is covered in Chapter 3; here we focus specifically on how carry-out and overflow set the NZCV flags. Consider a 6-bit example: $25 + -5$

$$\begin{array}{r} 011011 \\ +111011 \\ -\,-\,--\\ 1 \overset{\text{carry out}}{\longleftarrow} 010110 \end{array}$$

Note, the can say a carry-out occurs whenever we require one bit *beyond* the MSB to store the result. So in this case, the NZCV register is `0010`

### 4.2.3 Overflow

Although an overflow may seem to be similar to a carry-out, it is completely different. We get an overflow whenever we add two numbers of the same sign, and the result is a number of a different sign. For example, adding $-32 + -1$ in a 5-bit computer:

$$\begin{array}{r} 100000 \\ +111111 \\ -\,-\,--\\ 1 \overset{\text{carry out and overflow}}{\longleftarrow} 011111 \end{array}$$

Notice here that the sign changed - we added two negative numbers and the result was a positive number ($+31$). This is both an overflow and a carry-out. So in this case, the NZCV register is `0011`.

Overflow does not *always* result in a carry out. For example

$$\begin{array}{r} 010000 \\ +010000 \\ -\,-\,--\\ 100000 \end{array}$$

In the above example, we have $16 + 16 = 32$ so there is no carry-out (because 32 can be represented in a 6-bit word), yet the MSB bit has changed from positive to negative. It is important to note that, in this case, an overflow *does not necessarily* mean an error. It is only an error if your calculations are *signed*. If your calculations are *unsigned* then the result is correct and there is no problem.

In conclusion, overflow arises when two operands of the *same* sign are added, and the result is a number of a *different* sign ($+n \ + \ +m = -s$ or $-n \ + \ -m = +s$)

If you are not interested in the operand signs, then the overflow event does not matter but carry-out certainly do matter.

Now that we understand how the `NZCV` gets updated, and what it can be used for (ref. Table 4.1), we now look at the ARM7DTI approach for implementing the humble `if`, by using the `cmp` instruction.

### 4.2.4 Compare and test

To implement `if` we use the `cmp` instruction followed by a test appended to the instruction that follows the `cmp` (see Table 4.1). `cmp` has two operands, the left and the right. It subtracts the right operand from the left operand and updates `NZCV` with the outcome of the operation. The actual numerical result is not stored anywhere.

In `cmp`, both the left and right operands can be registers (this is usually the way), or the left operand can be a register and the right operand an immediate. For example:

```
mov r1, #10
mov r2, #5
mov r3, #20
cmp r3, #20 // NZCV = 0110
cmp r2, r1  // NZCV = 1000
cmp r1, r2  // NZCV = 0010
```

Note that each of the above 3 `cmp` Instructions will update the `NZCV` flags. This means that the result will be overwritten and lost. Therefore, if we use `cmp`, we never follow it immediately with another `cmp`. We first must *examine* the `NZCV` and make a decision. Lets look at an example in C:

```
int i = 10;
int r = 5;

if (i > r) {
  r = i;
}
```

We can say that the `r = i` is a conditional operation - it should only happen if `i > r`. Let's do this in assembly:

```
mov r1, #10
mov r2, #5
cmp r1, r2
movgt r2, r1
```

Note the new version of `mov` here: `movgt` which means *move if greater than*. If what is greater than what? if `r1` is greater than `r2`. How is this determined? by subtracting `r2` from `r1` and checking the `NZCV = 0000` (not negative and not zero).

The full list of conditionals is shown in Table 4.1

| Condition | Meaning | Flags Tested |
|---|---|---|
| eq | Equal (zero) | Z = 1 |
| ne | Not equal (non-zero) | Z = 0 |
| cs/hs | Carry set / Unsigned higher or same | C = 1 |
| cc/lo | Carry clear / Unsigned lower | C = 0 |
| mi | Negative (minus) | N = 1 |
| pl | Positive or zero (plus) | N = 0 |
| vs | Overflow set | V = 1 |
| vc | Overflow clear | V = 0 |
| hi | Unsigned higher | C = 1 and Z = 0 |
| ls | Unsigned lower or same | C = 0 or Z = 1 |
| ge | Signed greater than or equal | N = V |
| lt | Signed less than | N $\neq$ V |
| gt | Signed greater than | Z = 0 and N = V |
| le | Signed less than or equal | Z = 1 or N $\neq$ V |

Table 4.1: Full set of ARM7DTI conditional codes

Almost all ARM7DTI instructions, with he exception of `cmp`, can optionally have *one* conditional code appended. The the conditional code is omitted then the instruction *always* executes.

For example:

```
addlt r2, r4 // add r4 to r2 NVCV = 1000
subeq r2, r4 // add r4 to r2 NVCV = 0110
mulhi r2, r3, r4 // multiply r3 by r4 if NVCV = 0010
```

### 4.2.5 Signed and Unsigned Conditions

To make the following examples easier to understand we will use a simple 3-bit computer. The ideas here apply equally to a computer of word size of any integer.

Let us think through the example of a signed *greater than* test. When we say *signed*, what this means is that *if* the number has the `MSB=1` it must be treated as a negative number. We will later compare this to an unsigned test using the same bit pattern.

Let's look at the first example, using `gt` (for example, `movgt`). As you can see, `gt` is a signed greater than test, and for it to be true the `NZCV` flags must be: `Z=0` and `N=V`. The `Z=0` requirement is obvious. But less obvious is `N=V`. Let's look at a 3-bit example: `r0=-1` and `r1=2`.

When the instruction `cmp r0, r1` runs, we have $-1 + -2 = -3$.

$$\begin{array}{r} 111 \\ +110 \\ \hline 101 \end{array}$$

The result is correct, $101 = -3$. But do the flags agree with requirement for `gt`?

- `Z=0` true
- `N=V` false (negative but no overflow)

No - there is a disagreement, because the result is negative without an overflow. Therefore, $r0 < r1$ and the condition is false.

Therefore, we would expect the instruction `movgt r4, #5` to *not* execute.

How about reversing the test? `cmp r1, r0`. When it runs we have $2 - -1 = 3$.

$$\begin{array}{r} 001 \\ +010 \\ \hline 011 \end{array}$$

The result is correct, $011 = 3$. But do the flags agree with requirement for `gt`?

- `Z=0` true
- `N=V` true (no negative, no overflow)

Yes - there is agreement, as the result is not negative and not overflow. Therefore, $r1 > r0$ and the condition is true. Therefore, we would expect the instruction `movgt r4, #5` to execute.

Just for clarity, let us think again about the previous example from an *unsigned* point of view. Again, `r0=7` and `r1=2`, lets see what happens when we compare them and test if `r0` is `gt r1` (it is). This becomes $7 - 2 = 5$, or

$$\begin{array}{r} 111 \\ +110 \\ \hline 1 \overset{\text{carry out}}{\longleftarrow} 101 \end{array}$$

Does this agree with the definition for unsgined greater than (`hi`)?

- `Z=0` true
- `Z=0` true (no overflow)

Yes, $7 > 2$. Rememeber that although for us, we think of these two numbers as unsigned, when `cmp r0, r1` runs, it multiplies the second operand by $-1$ and then adds it to the first operand. In unsigned arithmetic this always produces a negative second operand.

You can see the contradiction here: when `cmp r0, r1` is followed by `gt`, the result waa false ($-1$ is not greater than $2$), but when we use `hi` we find the result is true! ($7 > 2$).

If our algorithm uses only unsigned arithmetic, then we should use `hi` and `ls` for our greater than and less then or equal to. Conversely, if we are using signed arithmetic, then use `le` and `gt` respectively.

### 4.2.6 Updating NZCV

`cmp` *always* updates the `cpsr` (`NZCV` bits) with the outcome of the instruction. But there are other instructions that can produce Zero, Negative, Carry and Overflow outcomes. For example, a calculation such $m - n > 0$ always results in a carry (`C=1`). How do we update `NZCV` ourselves? It can only be done *indirectly* - by appending `s` to the instruction in question. For example:

```
mov r0, #5
mov r1, #3
subs r0, r1  // results in C=1
sub r0, r1   // does not change NZCV
```

Notice in the example above, `subs r0, r1` should be understood as a substraction instruction that updates `NZCV` with the outcome of `r0` minus `r1` (from the ALU). Whereas `sub r0, r1` performs no such update. Therefore, when you want to make a decision about the outcome of an operation, append `s` to the instruction in question.

### 4.2.7 Combining condition codes and NZCV updates

Is it possible to combine both condition codes and `NZCV` updates in one instruction? Yes. For example: `addlts r0, r1` will update the `NZCV` with the outcome of the `addlt` so long as the *previous* instruction resulted in a signed `lt` outcome (`Z = 0` and `N = V`).

### 4.2.8 Branching

Now that we have seen how `if` is implemented in assembly as the pair `cmp` + *instr*`cond`, we should return to scope control. Let's look at an example: where 5 instructions should execute if the `lt` condition is true:

```
mov r0, #-1
mov r1, #2
mov r3, #0
cmp r0, r1
// the next 5 instructions
// only execute if lt is true
movlt r4, #5
movlt r5, #10
mullt r6, r5, r4
sublt r3, r6
movlt r5, r6
```

The above structure does not resemble how we write `if` statements in C. What we would like to do is to "jump" to the a block of code if the condition is true, and if it's false, then it should continue executing from the condition. This is the important role that *branching* plays. We will discuss branching in Chapter 6.

## 4.3 Chapter Summary

In this chapter we have introduced the ARM7DTI equivalent of the C program `if` keyword. We noted that `if` in ARM7DTI is always at least*two* instructions: the `cmp` instruction followed by one or more instructions with the condition code appended (see Table 4.1).

In this chapter we discussed several examples of how signed and unsigned conditions are handled. We noted that in some cases, signed and unsigned tests of the same condition type (eg, `gt` and `hi`) produce different results. Rememeber that as although the bit patterns in memory are the same, the results in the both `gt` and `hi` are tested against different parts of the `NZCV`. We also looked at how the programmer can indirectly update `NZCV` (by appending `s` to the instruction), and we also briefly noted that condition codes and `NZCV` updates can be combined into one instruction such as `addlts r0, r1`.

## 4.4 Exercises Summary

**Exercise 4.4.1** *Using three few different examples test to see if a carry-out always arises whenever we have a positive and a negative addition where the result is positive. Does it hold true?*

**Exercise 4.4.2** *Derive an ARM7DTI assembly example where* `NZCV` $= 0011$ *(Positive result, with carry-out and overflow). Test it in CPUlator.*

**Exercise 4.4.3** *Derive an ARM7DTI assembly example where* `NZCV` $= 1010$ *(Negative result, with carry-out, no overflow). Test it in CPUlator.*

**Exercise 4.4.4** *Give informal examplea from the real world, where (a) an overflow would be a concern, and (b) where an overflow would* not *be a concern.*

**Exercise 4.4.5** *Write and ARM7DTI program to demonstrate how* `lt` *and* `lo` *conditional codes produce different results for the same bit pattern.*

# 5

# Branching and Iteration

**Learning Objectives**

- Explain how branch instructions modify the program counter to control ARM assembly execution flow.
- Distinguish between unconditional, conditional, direct, and indirect branch instructions.
- Implement `while`, `for`, and `do-while` loops using comparisons and branches.
- Analyze how branch instructions affect pipeline performance, including prediction, flushing, and misprediction penalties.
- Construct nested loop structures in ARM assembly using labels, counters, comparisons, and branch targets.

## 5.1 Introduction to Program Control Flow

In previous chapters, we have examined how to load and store values to and from memory. However, real programs do not simply execute instructions in a straight line; they must make decisions and repeat operations. In higher-level languages, this is accomplished with `if` statements, `while` loops, `for` loops, and function calls. In ARM Assembly, we must implement these control flow mechanisms using *branch* instructions.

### 5.1.1 Control Flow in Assembly

At the heart of control flow in ARM Assembly are instructions that affect the Program Counter (`pc`). The `pc` determines which instruction the CPU executes next. Most instructions implicitly increment `pc` to the next instruction, but branch instructions explicitly update it to a new target.

**Control Flow Instructions in ARM Assembly:**

- `b label` – Unconditional branch to `label`.
- `bl label` – Branch with link; jumps to a function and stores return address in the Link Register (`lr`).
- `bx lr` – Return from function by branching to the address in `lr`.
- `bne`, `beq`, `bgt`, etc. – Conditional branches that execute based on status flags.

### 5.1.2 The Role of the Program Counter (pc)

In ARM32, due to the fetch-decode-execute pipeline, reading from the pc actually gives the address of the instruction *two steps ahead*, i.e., `pc + 8`. In ARM64, this is simplified to `pc + 4`. This offset is important when performing pc-relative calculations such as loading data from memory based on the current program position.

### 5.1.3 Branch Instruction Encoding and Range

The `b` and `bl` instructions use relative addressing with a signed immediate value. In ARM32:

- The offset is 24 bits wide and shifted left by 2, allowing for a range of $\pm 2^{25}$ bytes, or $\pm$32MB.

  In ARM64:

- The offset is 26 bits wide, also shifted left by 2, resulting in a branch range of $\pm$128MB.

If a branch target is outside this range, the address must be loaded into a register, and an indirect branch performed:

```
ldr x0, =target_address
br x0  // ARM64 indirect branch
```

### 5.1.4 Branching and the Pipeline

To understand why branches are special, we first need to look at how a modern processor actually runs instructions.

**Pipeline Architecture: The Core Idea**

Rather than finishing one instruction completely before starting the next, a processor uses a **pipeline**: it splits execution into stages and works on several instructions at once, each at a different stage. Think of a kitchen where one person chops, another fries, and a third plates. No one waits for a dish to be fully served before starting the next one.
A classic ARM pipeline has five stages:

- **Fetch (IF):** read the next instruction from memory.
- **Decode (ID):** figure out what the instruction does and read its source registers.
- **Execute (EX):** run the ALU operation or compute a memory address.
- **Memory (MEM):** read or write data memory (for `ldr` and `str`).
- **Write-Back (WB):** store the result back into the register file.

When all five stages are busy, the processor finishes one instruction every clock cycle. The total time for any single instruction does not change, but the rate at which instructions complete goes up dramatically. Chapter 11 walks through the full pipelined datapath, hazards, and forwarding.

Modern ARM cores use deep pipelines to maximize throughput. However, branches can disrupt the pipeline:

- If a branch is correctly predicted, the pipeline continues smoothly.
- If mispredicted, the processor must flush instructions and refetch from the correct path, causing a delay.

| Core | Branch Misprediction Penalty |
|---|---|
| Cortex-A53 | ~10 cycles |
| Cortex-A76 | ~15 cycles |
| Neoverse V1 | 20+ cycles |

Table 5.1: Misprediction penalties in common ARM cores

**Why the penalty grows with pipeline depth**

These cores have 8- to 15-stage pipelines, so more instructions are fetched on the wrong path before the mistake is detected. The pedagogical 5-stage pipeline in Chapter 11 has a 2-cycle penalty because the branch resolves in the Execute stage with only two instructions fetched speculatively behind it. The hardware mechanism, flushing the Decode and Execute pipeline registers, is identical; only the depth (and therefore the cost) differs.

**▶ What is PlayARM?**

**PlayARM** is a browser-based ARM microarchitecture simulator. Write ARM assembly in the built-in editor, then step through it cycle by cycle, watching the 5-stage pipeline canvas, register file, CPU flags, control signals, memory accesses, and TLB translations update in real time. Throughout this book, purple **Try it in PlayARM** boxes point you to programs you can run directly in the simulator to reinforce each concept.

**▶ Watch a Branch Flush the Pipeline**

Run the program below in PlayARM. Use **Slow** speed and watch the **Pipeline Activity** panel when the `BNE` instruction is in the **EX** stage. If the branch is *taken*, notice that the instructions fetched after it are flushed, the pipeline must restart from the branch target.

```
mov r0, #3
loop:
    sub r0, r0, #1
    cmp r0, #0
    bne loop
mov r1, #99
```

Count how many extra cycles the branch adds each time it is taken compared to when it falls through on the final iteration.

### 5.1.5 Why Control Flow Matters

Control flow constructs (branches, loops, and function calls) enable the creation of algorithms, decision-making structures, and code reuse. Efficient use of branches is critical in performance-sensitive applications, such as signal processing or real-time embedded systems.

In the sections that follow, we will explore how these ideas translate into practical assembly code, including:

- Creating branches based on comparisons.
- Writing loops with different structures (while, do-while, for).
- Minimizing the performance cost of branching using techniques such as conditional execution and loop unrolling.

## 5.2 ARM Branch Instruction Set

In ARM Assembly, branching is the fundamental method of controlling program flow. Branch instructions modify the Program Counter (`pc`), allowing a jump to another instruction in the program. These jumps may be conditional or unconditional, direct or indirect, and are essential for implementing loops, conditionals, and function calls.

### 5.2.1 Core Branch Instructions

The most basic branching operations in ARM Assembly are:

- `b label` – Unconditional branch to `label`.
- `bl label` – Branch to `label` and save return address in the Link Register (`lr`).
- `bx lr` – Return from subroutine (branch to address in `lr`).
- `blx reg` – Branch with link to address in `reg`, also supports interworking between ARM and Thumb states.

```
bl my_function   // Call function
...
my_function:
  ; function body
  bx lr          // Return
```

The 'bl' instruction automatically stores the address of the next instruction in `lr`, making it possible to return using `bx lr`.

### 5.2.2 ARM64 Enhancements

Modern ARM64 introduces new branching capabilities that are both compact and powerful:

- `b.cond label` – Compact conditional branch (e.g., `b.eq`, `b.ne`, `b.ge`).
- `br xN` – Branch to address in register `xN`.
- `ret xN` – Return from subroutine (equivalent to `bx lr`).
- `braa xN, xM` – Branch with return address authentication (used for security).

**Example – ARM64 return mechanism:**

```
bl some_function
...
some_function:
  ; do something
  ret             // return to caller
```

These enhancements are particularly relevant in performance- and security-critical applications, such as operating system kernels and cryptographic routines.

ARMv8.3 also introduces **Pointer Authentication Codes (PAC)**, introducing the use of `braa` for secure, authenticated control transfers.

*we need to develop this braa point - why do we need it? what problem does it address .... does it need its own section?*

## 5.3 Conditional Execution

Modern ARM processors support **conditional execution**, a powerful feature that allows certain instructions to execute only if specific conditions (based on the status flags) are met. This eliminates the need for short branches in many cases and helps to reduce pipeline disruptions.

### 5.3.1 Condition Flags Overview

ARM architecture maintains a special register called the **Application Program Status Register (APSR)**. This register stores condition flags that reflect the result of the most recent arithmetic or logical operation.

- **N (Negative)** – Set if the result is negative (bit 31 of result is 1).
- **Z (Zero)** – Set if the result is zero.
- **C (Carry)** – Set if there was a carry out (for unsigned arithmetic).
- **V (Overflow)** – Set if there was a signed overflow.

**Example – Flag update by comparison:**

```
cmp r0, r1  // Updates N, Z, C, and V
bge next    // Branch if r0 >= r1 (N == V)
```

These flags are implicitly updated by arithmetic instructions such as `add`, `sub`, `cmp`, and `movs`. Conditional branches and conditional instruction execution depend on the values of these flags.

**▶ Watch the N, Z, C, V Flags Update**

Enter the program below and step through it in PlayARM. After each `cmp` or `sub`, check the **CPU Flags** panel and note which of the **N**, **Z**, **C**, **V** flags are set.

```
mov R0, #5
mov r1, #5
cmp R0, r1
mov r2, #10
mov R3, #3
sub R4, R3, r2
cmp R4, #0
```

After `cmp R0, r1`: **Z** should be 1 (equal).
After `sub R4, R3, r2`: **N** should be 1 (negative result).
Verify each flag matches the condition table above.

### 5.3.2 Using Flags in Practice

ARM supports a full set of conditional suffixes that can be applied to many instructions, allowing them to execute only when a condition is true.

**Common condition suffixes:**

| Suffix | Meaning | Condition |
|---|---|---|
| `eq` | Equal | Z = 1 |
| `ne` | Not equal | Z = 0 |
| `lt` | Less than (signed) | N ≠ V |
| `ge` | Greater than or equal (signed) | N = V |
| `cs` | Carry set (unsigned ≥) | C = 1 |
| `cc` | Carry clear (unsigned <) | C = 0 |

Table 5.2: ARM Condition Suffixes

**Example – Using conditional arithmetic:**

```
cmp r0, r1
addgt r2, r0, r1  // if r0 > r1
sublt r2, r1, r0  // if r0 < r1
```

This technique is particularly useful when minimizing branch penalties. Rather than using `bgt` or `blt` to jump around the code, the instructions themselves are conditionally suppressed or executed.

## 5.4 Loop Constructs and Patterns

Looping structures are essential for repeating instructions in programs. In C, loops are expressed as `while`, `for`, and `do-while` constructs. ARM Assembly does not have these keywords, so all loops must be built explicitly using branches and comparison instructions.

This section explains how to implement these high-level loop constructs using ARM's conditional branches and registers.

### 5.4.1 While Loop in Assembly

A `while` loop in C repeatedly checks a condition before executing the loop body.

**C code:**

```
[language=C]
int i = 10;
while (i > 0) {
  // loop body
  i--;
}
```

**ARM Assembly:**

```
mov r1, #10          // i = 10
b test_condition     // jump to condition check

loop_body:
  ; your loop logic here
  subs r1, r1, #1     // i--

test_condition:
  cmp r1, #0
  bgt loop_body       // loop if i > 0
```

Note: ‘subs‘ subtracts and updates flags; ‘cmp‘ checks the loop condition.

### 5.4.2 For Loop Using subS

A `for` loop has explicit initialization, condition, and iteration components.

**C code:**

```
[language=C]
for (int i = 5; i > 0; i--) {
  // loop body
}
```

**ARM Assembly:**

```
mov r1, #5           // i = 5

loop:
  ; loop body here

  subs r1, r1, #1     // i--
  bgt loop            // loop if i > 0
```

This form is compact and efficient. It uses ‘subs‘ to both decrement the counter and update the flags for ‘bgt‘.

**► For Loop: Count Down from 5**

Translate the `for` loop above directly into PlayARM. Use **Step** mode and watch `r1` count down from 5 to 0 in the **Register File** panel. Observe that the **Z** flag goes high on the final iteration (when `r1` hits 0), causing `BNE` to fall through instead of branching.

```
mov r1, #5
loop:
    sub r1, r1, #1
    cmp r1, #0
    bne loop
mov r2, #1
```

After `mov r2, #1` executes, `r1` should be 0 and `r2` should be 1, confirm both in the **Register File**.

### 5.4.3 `Do-While` Loop

The `do-while` loop guarantees that the body executes at least once.

**C code:**

```
[language=C]
int i = 5;
do {
  // loop body
  i--;
} while (i > 0);
```

**ARM Assembly:**

```
mov r1, #5

loop:
  ; loop body here
  subs r1, r1, #1
  bgt loop            // repeat if i > 0
```

Unlike the previous loops, this structure places the condition check *after* the body, ensuring at least one execution.

### 5.4.4 Nested Loops and Flowcharts

Nested loops involve placing one loop inside another. These are common in array or matrix processing.

**C code:**

```
[language=C]
for (int i = 0; i < 3; i++) {
  for (int j = 0; j < 2; j++) {
    // inner loop body
  }
}
```

**ARM Assembly:**

```
mov r0, #0          // i = 0
outer_loop:
  cmp r0, #3
  bge end_outer

  mov r1, #0        // j = 0
inner_loop:
  cmp r1, #2
  bge end_inner

  ; inner loop body here

  add r1, r1, #1
  b inner_loop

end_inner:
  add r0, r0, #1
  b outer_loop

end_outer:
```

Each loop must use separate registers for indexing (‘r0‘, ‘r1‘) and must have clearly marked start and end labels.

## Visualizing Control Flow

To understand nested loop structure better, we can sketch a flowchart:

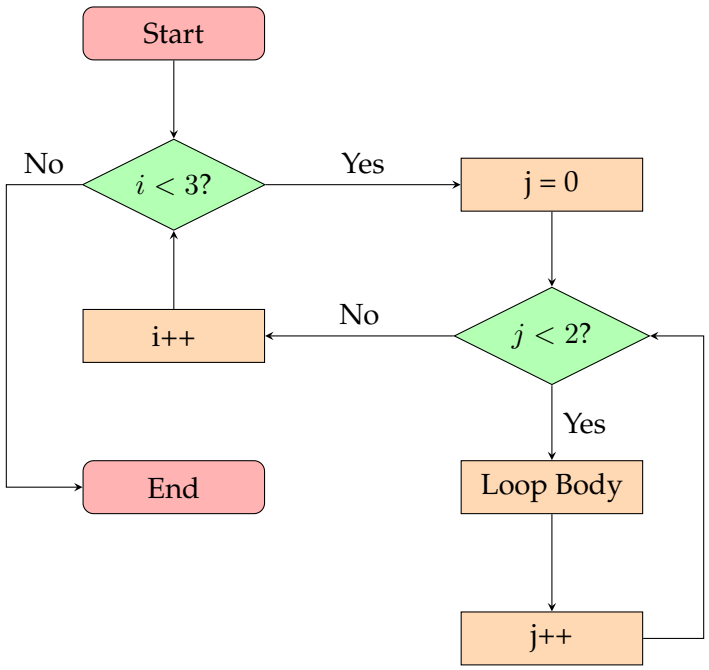



Figure 5.1: Flowchart of Nested Loops

## 5.5 Exercises

### 5.5.1 Basic Load and Store Operations

**Exercise 5.5.1** *Write an ARM assembly program that does the following:*

1. *Declares an integer variable* `x` *initialized to* `25`.
2. *Loads* `x` *into a register.*
3. *Adds* `10` *to* `x`.
4. *Stores the result back in memory.*

*Use* `ldr` *and* `str` *instructions to perform memory operations.*

### 5.5.2 Byte vs Word Access

**Exercise 5.5.2** *Given the following memory declaration:*

```
.data
  array: .word 0x12345678
```

*Perform the following operations in ARM assembly:*

1. *Load the full word into a register and print its value.*
2. *Load only the least significant byte using* `ldrb` *and print its value.*
3. *Store* `0xAB` *into the least significant byte of* `array` *without modifying other bytes.*

### 5.5.3 Offset Addressing

**Exercise 5.5.3** *Given an integer array in memory:*

```
.data
  numbers: .word 5, 10, 15, 20
```

*Write an ARM assembly program that:*

1. *Loads the second element of the array into a register.*
2. *Adds* `5` *to the second element.*
3. *Stores the modified value back into the array.*

*Use offset addressing mode in the* `ldr` *and* `str` *instructions.*

### 5.5.4 Pointer Arithmetic

**Exercise 5.5.4** *Convert the following C code into ARM assembly:*

```
unsigned values[5] = { 2, 4, 6, 8, 10 };
unsigned *p = values;
*(p + 2) = *(p + 2) + 5;  // Modify the third element
```

1. *Identify how memory addressing is done in both C and ARM assembly.*
2. *Use a base register and appropriate addressing mode to modify the third element.*

### 5.5.5 Pointer Traversal in Assembly

**Exercise 5.5.5** *Write an ARM assembly program that:*

1. *Defines an array of 6 integers.*
2. *Uses a base register (pointer) to iterate through each element.*
3. *Doubles each value in the array.*

### 5.5.6 Finding Maximum in an Array

**Exercise 5.5.6** *Write a C function that finds the maximum value in an array using pointers:*

```
unsigned find_max(unsigned *arr, int size) {
    unsigned max = *arr;
    for (int i = 1; i < size; i++) {
        if (*(arr + i) > max) {
            max = *(arr + i);
        }
    }
    return max;
}
```

1. *Translate this function into ARM assembly.*
2. *Use a loop with indexed addressing mode to traverse the array.*

# 6

# Addressing Memory

**Learning Objectives**

- Explain the ARM load-store memory model, including base registers and memory addresses.
- Distinguish between `ldr`, `str`, `ldrb`, and `strb` instructions.
- Apply offset, pre-index, and post-index addressing to access integer arrays correctly.
- Translate C pointer arithmetic into ARM assembly using base registers and byte offsets.
- Use character addressing to iterate through strings, detect null terminators, and load bytes safely.

## 6.1 Load and Store

The ARM32/64 Microarchitecture uses a *load and store* memory access model. This means that before an item from memory can be used in the CPU, the *address* of the item must first be *load*ed into a *base register*. To write (*store*) a value back to memory, we must use the address contained in the base register.

This means obtaining the address of the first element of the variable you wish to access, placing this address in a base register, and computing the access locations of subsequent elements, as an offset of the base register.

Later we will discuss the ARM instructions required for this, but first we will consider a C program as the basis for understanding how memory operations take place.

### 6.1.1 What to know when addressing memory

We have to plan carefully in order to address memory correctly. Our approach needs to be aware of the following factors:

1. *Data Type*: There are two possible choices here. Addressing 32-bit words or addressing 8-bit bytes

   (a) For accessing signed or unsiged integer data we read from memory using `ldr` (LoaD Register) and write to memory using `str` (StoRe Register)
   (b) For accessing ASCII character data we read from memory using `ldrb` (LoaD Register Byte) and write to memory using `strb` (StoRe Register Byte)

2. *Addressing modes*:

   (a) Offset addressing

| Instruction | Meaning |
|---|---|
| `ldr ToReg, [FromAddr]` | Load the integer (word) found at address `FromAddr` into the register `ToReg` |
| `str FromReg, [ToAddr]` | Store the integer found in the register `FromReg` into the value at address `ToAddr` |
| `ldrb ToReg, [ FromAddr]` | Load the byte from address `FromAddr` into the register `ToReg` |
| `strb FromReg, [ ToAddr]` | Store the byte found in the register `FromReg` into the value at address `ToAddr` |

Figure 6.1: Memory Instructions

(b) Post-index Addressing
(c) Pre-index Addressing

Offset, Post- and Pre- index Addressing will be discussed in Section 6.4, but first we will discuss the general syntax of the memory instructions.

### 6.1.2 Instruction Syntax

Let use briefly examine the syntax of the memory access instructions:
Let's consider an example from Table 6.1, we will use the load register (`ldr`)

```
ldr r1, [r0]
```

This should be understood as *read the value from the memory location pointed to by* `r0`, *and place it into* `r1`.

## 6.2 C Program

We will use the C program shown in Fig. 6.2 to motivate our understanding of how this load and store model works.

1. Place the code into a file called `memory.C`
2. Compile it using: `gcc memory.C -o memory`
3. Run it: `./memory`

### 6.2.1 C Pointers

In the C code, we declared and initialized a *pointer* as `unsigned* p = array`. This means that `p` points to the address of the first element in `array`. We can now access any element of `array` using the notation `*p`. For example, the code:

```
p += 3; // move the pointer to the 3rd index element
```

```c
#include <stdio.h>

const int sz = 6;
unsigned array[ sz] = { 9, 2, 3, 4, 8, 10 };
unsigned someValue = 55;

int main (int argc, char** argv) {
  unsigned* p = array;

  // update the 3rd element of the array
  p += 3;
  *p += 10;
  printf("the 3rd element is: %u \n", *p);

  // use p to access / modify the someValue;
  printf("someValue is %u \n", someValue);
  p = &someValue;
  *p = 123;
  printf("now someValue is %u\n", someValue);

  return 0;
}
```

Figure 6.2: Sample C program using pointers

will move p to index 3 in the array of integers, thus it is now pointing to the element that contains the value 4. Notice that `p += 3` is shorthand for `p += (3*w)` where $w$ is the word size of the hardware.

This value can be modified by using the pointer-access notation as follows:

```c
*p += 10; // add 10 to the value at the 3rd index
```

This will change the value to which p points, from 4 to 11.

## 6.3 ARM Assembly "Pointers"

In ARM Assembly, we see that C pointers have a direct counterpart. This is achieved by selecting some register as the *base-register*. The base-register is initialized to the name of the `.data` object, as we show later. This base register is the direct equivalent of our C program pointer.

To begin, assume that the ARM Assembly `.data` section contains the following code:

```
.data
  array: .word 9, 2, 3, 4, 8, 10
  someValue: .word 55
```

When this array is placed into memory at program load time, the elements will be stored in *contiguous* memory, with each element occupying 4 bytes (a `.word`) as shown in Fig. 6.3.

| Address | Value |
|---|---|
| ... | ... |
| ... | ... |
| 00FFAB00 | 00 00 00 09 |
| 00FFAB04 | 00 00 00 02 |
| 00FFAB08 | 00 00 00 03 |
| 00FFAB0C | 00 00 00 04 |
| 00FFAB10 | 00 00 00 08 |
| 00FFAB14 | 00 00 00 0A |
| ... | ... |
| ... | ... |

Figure 6.3: `array` in memory starting at address `00FFAB00`

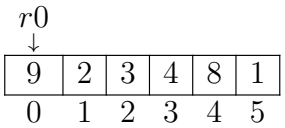
### 6.3.1 Initializing the base-register

We first load the address of the array using the following syntax (we will give `r0` the role of base register), using the instruction `ldr`. For clarity, we also show the C code equivalent to the assembly instructions:

```
// In C: unsigned* p = array;

// In ARM assembly:
ldr r0, =array
```

Note the use of the special character `=`. This character is used for *initializing* the base-register. This initialization does not need to be repeated (in most cases). As `array` starts at the memory location `00FFAB00`, `r0` now contains the value `00FFAB00`. We say that `r0` is *pointing to* the first element in the array:




(6.1)

### 6.3.2 Pointing to the $i$-th element

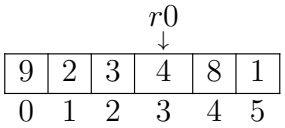
Now we consider how to change the base-register so that it points to some arbitrary location (the $i$-th element).

```
// In C: p += 3;

// In ARM assembly 3 x 4 = 12:
add r0, #12
```

| Mode | Syntax | Address | Base-register `r0` |
|---|---|---|---|
| Offset | `ldr r1, [r0, V]` | `r0+V` | Unchanged |
| Pre-Index | `ldr r1, [r0, V]!` | `r0+V` | Changes *before* the location is read |
| Post-index | `ldr r1, [r0], V` | `r0` | Changes *after* the location is read |

Figure 6.4: Addressing modes with using `ldr` instruction and the base register `r0`. Note `V` is either (a) a register, (for example, `ldr r1,[r0,r2]`), or (b) an immediate (for example, `ldr r1,[r0,#4]`)




(6.2)

Notice that to change the base register so that it points to the $i$-th element (in this example $i = 3$), we must add 12 to it. This is because we need to calculate how many *bytes* to move `r0` on by. Because the word size of our platform is 32-bit, and there are 4 bytes in 32 bits, we multiply our new index (in the example it is 3) by 4 to get the next index position address when accessing integer arrays. This is in contrast to C, where we need to add 3. There are other ways to achieve base register pointer changes. For example:

```
// In ARM assembly 3 x 4 = 12:
mov r1, #4
mov r2, #3
mul r3, r1, r2
add r0, r3
```

## 6.4 Integer Addressing Modes

ARM Assembly has three different modes for accessing memory. You choose the correct mode depending on the situation. It is possible to mix and match them in a program, but this needs to be done with care so that you don't lose track of where the base-register is pointing to. These modes are now summarized in Table 6.4.

### 6.4.1 Offset Addressing

Use this mode when we do not want to change our base-register. For example, in situations where we want to flexibly access any location in the memory object by the addition of some constant.

Suppose our algorithm requires us to set certain elements in our array in a particular order. Let's say the third, first and fourth elements in `array` should be set with some number in `r2`

```
// using offset addressing
ldr r0, =array
mov r2, #0
str r2, [r0, #8]
str r2, [r0, #0]
str r2, [r0, #12]
```

In this sense we can "jump around" our array in any kind of random sequence, only knowing which element we need to access. Because the base-register does not change, there is no need to reset it after each "jump".

**▶ Offset Addressing: Random Access**

The program below mimics the random-access pattern above using only registers and the stack as a stand-in for an array. Write three values to non-sequential offsets from SP, then read them back.

```
mov r2, #0
sub sp, sp, #16
str r2, [sp, #8]
str r2, [sp, #0]
str r2, [sp, #12]
ldr r3, [sp, #8]
ldr r4, [sp, #0]
ldr r5, [sp, #12]
add sp, sp, #16
```

Step through in PlayARM and watch the **Memory** panel after each `str`. Notice that SP never changes during the stores: that is the defining property of offset addressing. Check the **TLB** panel to see the virtual-to-physical address translation for each memory access.

On the other hand, a plain iteration over `array` is troublesome using offset addressing. Consider:

```
// using offset addressing to iterate
// to zero out the array is a bit troublesome
ldr r0, =array
mov r1, #0 // incrementer
mov r2, #0 // value to write
mov r3, #6 // array size

loop:
  cmp r1, r3
  strlt r2, [r0, r1, lsl #2]
  addlt r1, r1, #1
  blt loop
.data
  array: .word 1,2,3,4,5,6,7
```

Here, we use `str r2, [r0, r1, lsl #2]`, which has the effect of shifting left (multiplying by 4), the incrementer in `r1` during each iteration (which generates the

sequence, $0, 4, 8, ...$). Obviously, this is extra work for the programmer to manage, so for plain $0, 1, 2, ..., n-1$ iteration tasks, we can use either pre-indexing or post-indexing. Let's consider post-index addressing first.

### 6.4.2 Post-index Addressing

As mentioned above, offset addressing is not a good choice for a simple iteration over each element of an array (for example, in linear search, or selection sort).

A much simpler approach, and the default approach used in C, is to increment the address stored in the base-register by some constant amount. In C, the notation `*ptr++` is used to dereference the memory address pointed to be `p` *and then* move `p` to the next element.

```
// in C, dereference the pointer
// and then move it on
unsigned r = *ptr++;
```

Assuming `ptr` is contains `00FFAB00`, then the unsigned integer `r` will be assigned $9$, and *immediately after*, `ptr` will be incremented to `00FFAB04`.

Of course, `*ptr++` is just shorthand for:

```
unsigned r = *ptr;
ptr++;
```

This is called *post-index* addressing (or post-fix addressing in C). This should be our default method when iterating over an array in seqential order. In ARM assembly, we implement post-index addressing as follows:

```
// using post-index addressing to iterate sequentially
// over array and zero out each element
ldr r0, =array
mov r2, #0 // value to write
mov r3, #6 // array size

loop:
  cmp r1, r3
  strlt r2, [r0], #4 // much neater syntax
  blt loop
```

Notice the simplification of the loop strucure: we have a much neater syntax for our `str` instuction:

```
// much neater syntax for iteration
strlt r2, [r0], #4
```

*versus*

```
// more awkward syntax for iteration
strlt r2, [r0, r1, lsl #2]
addlt r1, r1, #1
```

## Post-Index Patterns

We now review some of the most common C programming post-index memory access patterns and their equivalent in ARM assembly.

**Pattern 6.4.1** *Initialize-Increment*

In C we often see the following code pattern:

```
unsigned r = *ptr++;
```

You should understand that this code has **two** steps. Step 1., store the value found at the address pointed to by **ptr**, into the variable `r`. Step 2. **then** increment **ptr** to point to the next element in the array.

In ARM Assembly we have the exact same semantics:

```
ldr r2, [r0], #4
```

Step 1., load the value found at the address in `r0` into the register `r2`, and 2., **then** add texttt4 to the value in `r0`.

**Pattern 6.4.2** *Assign-Increment*

Another commonly encountered pattern is the assign increment pattern, for example:

```
*ptr++ = r;
```

becomes:

```
str r2, [r0], #4
```

**Pattern 6.4.3** *Read-Write-Increment*

The following pattern arises where, in one statement, the value from a pointer is read, modified then written. Following this, the pointer is incremented. As follows:

```
*ptr++ += 3;
```

This pattern must be implemented as **three** instructions:

```
ldr r2, [r0] // load using offset addressing
add r2, r2, #3 // add our 3 as an immediate
str r2, [r0], #4 // write results, increment pointer
```

Of course, if you are writing a C program for the ARM platform, the compiler will convert `*ptr++ += 3;` into the three ARM instructions as shown above.

**▶ Post-Index: Watch the Base Register Advance**

Post-index addressing is the natural way to iterate sequentially. Run this program and watch `r0` increment by 4 after each `str` in the **Register File** panel.

```
sub sp, sp, #16
mov r0, sp
mov r2, #10
str r2, [r0], #4
mov r2, #20
str r2, [r0], #4
mov r2, #30
str r2, [r0], #4
mov r2, #40
str r2, [r0], #4
add sp, sp, #16
```

After each `str`, `r0` moves to the next slot *without* a separate `add` instruction. Check the **Memory** panel to confirm all four values (10, 20, 30, 40) are stored at consecutive addresses. Also watch the **TLB** panel: each access hits a different virtual page offset.

### 6.4.3 Pre-index Addressing

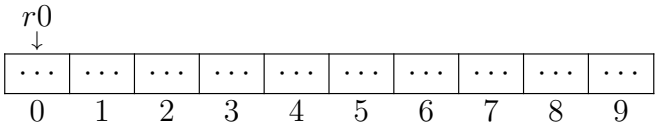
Post-index addressing has the interesting effect of leaving the base-register pointing to the next element in the array. This works well in cases where the next element in the array is *initialized*. However, it does not work well in cases when the next element is not initialized.

Consider a 10-element array of *uninitialized* ($\cdots$) data:

```
ldr r0, =array

// the .bss section for uninitialized data
.section .bss
  array: .skip 40
```

In the above example, 40-bytes have been reserved in the `.bss` segment. As you may remember from Chapter 2, the `.bss` segment is use for the placement of uninitialized data.

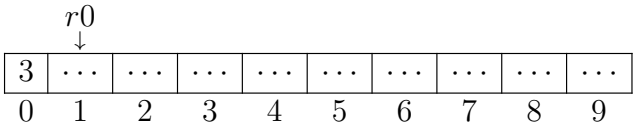



(6.3)

In Fig. 6.3, the base-register is pointing to the first uninitialized element. We can store a value into the first element using post-index addressing:

```
mov r1, #3
str r1, [r0], #4
```

`r0` now points to index 1, but this value is not initialized.




(6.4)

So, what happens here?

```
ldr r1, [r0]
```

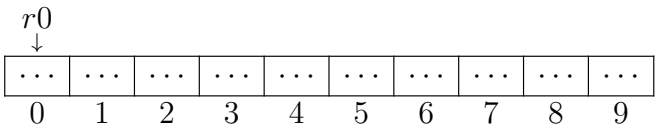
It is obvious that `ldr r1, [r0]` leaves `r1` with garbage. This is not desirable. The base-register is pointing to uninitialized memory. Hence, the post-index addressing into uninitialized produces undesirable results. Of course, we could easily solve this using a work-around:

```
ldr r1, [r0, #-4]
```

But this approach is awkward. We need another approach. Let's reset the base-register `r0`:

```
ldr r0, =array
```

so now things look like this:




(6.5)

Now use pre-index as follows:

```
mov r1, #3
str r1, [r0, #4]!
```

and the array now looks like:


(6.6)

Now a `ldr` from the base-registger will return a valid value:

```
ldr r1, [r0] //places 3 into r1
```

Because the base-register was moved *before* the write, the subsequent read is guaranteed to be initialized. Of course, the effect here is that the first element is skipped!

But the good news is that `ldr r1, [r0]` produces a properly initialized result every time. In fact, every pair of:

```
str r1, [r0, #4]!
ldr r1, [r0]
```

produces a *guaranteed* initialized value in `r1`, whereas, with

```
str r1, [r0], #4
ldr r1, [r0]
```

no such guarantee exists. In what circumstances might we want to use this pre-index addressing pattern? Use pre-index addressing when we want to move the base-register, write to an element and read that element back.

**▶ Pre-Index: Write Then Read Back Safely**

Pre-index addressing moves the base register *before* the memory access, guaranteeing the subsequent read returns an initialized value. Run the program below and compare what happens with pre-index vs post-index for the same slot.

```
sub sp, sp, #16
mov r0, sp
mov r1, #42
str r1, [r0, #4]!
ldr r2, [r0]
mov r1, #99
str r1, [r0, #4]!
ldr r3, [r0]
add sp, sp, #16
```

Step through in PlayARM and watch `r0` advance *before* each store. After each `LDR`, `R2` and `R3` should hold 42 and 99 respectively: never garbage. Check the **Memory** and **TLB** panels to see each translated address.

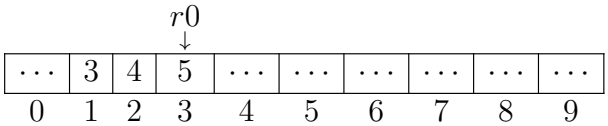
## Stacks

Pre-index addressing is used to insert (or *push*) elements onto *stack*-type data structures. We discuss the details of stack-programming in Chapter 7. But we will briefly look ahead now.

Stacks are sometimes called LIFO (Last-In-First-Out) structures. The stack starts out empty and elements are added using pre-index addressing. Elements are removed (or *popped*) by using post-index addressing.

Consider this example:

```
ldr r0, =array
mov r1, #3
str r1, [r0, #4]! // stack "push"
mov r1, #4
str r1, [r0, #4]! // stack "push"
mov r1, #5
str r1, [r0, #4]! // stack "push"
```

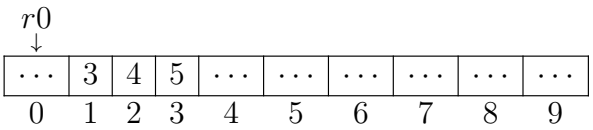



(6.7)

Now the stack has three elements, and the base-register is pointing to the *most recently added* element. We can now remove (*pop*) the elements as follows:

```
ldr r1, [r0], #-4 // stack "pop"
ldr r1, [r0], #-4 // stack "pop"
ldr r1, [r0], #-4 // stack "pop"
```

Notice, this leaves our stack-like structure looking like this:




(6.8)

which is exactly the configuration we had in 6.3.

Notice that when `r0` returns to its initial value (at offset 0), three elements remain in the array at positions 1,2 and 3. Does this matter? No, it does not matter. All that matters is where `r0` is pointing to. Everything following `r0` is considered to be free space.

## 6.5 Character Addressing Modes

In 6.4 we discussed how integer memory can be accessed. Of course, integer data is not the only type of data we may want to use in our assembly language programs. A *string* type is an array of 1-byte alpha-numeric character elements. Consider the following C code fragment:

```
char *string = "Hello World!";
```

The ARM assembly equivalent declaration is:

```
.data
  string: .asciz "Hello World!"
```

Notice the declaration here: `.asciz`. This type is an array of ASCII characters (ie, a string), which may contain one or more elements. The `z` tells us that the string is terminated with a zero. This zero (the *null* character ASCII code 0) is automatically added to the end of our C and assembly strings during the compilation/assembly phase - we do not need to add it ourselves.

### 6.5.1 Iterating over strings

A common task is to iterate over a string looking for a particular value, or maybe looking for an upppercase or lowercase character. One problem we face is *how do we know when to stop iterating?*.

The appeareance of the null character signifies we have reached the end of the string in C:

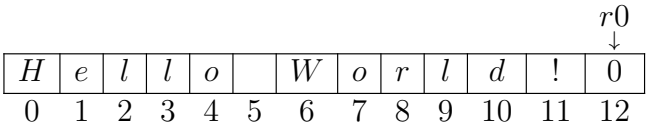
```
char string[] = "Hello World!";
char* cptr = string;
while(*cptr != 0) {
  // process the string
  // increment the pointer
  cptr++;
}
```

The ARM assembly equivalent is very similar - it must also test for the null character (ASCII 0). But because we are loading byte data and not integer data, we use the byte versions load and store instructions shown in Table 6.1:

```
ldrb r0, =string
ldrb r1, [r0], #1 //load the first character
loop:
  cmp r1, #0
  ldrneb r1, [r0], #1 //load the next character
  bne loop

loop_end:
  // the end of the loop
.data
  string: .asciz "Hello World!"
```

The base-register is incremented using post-fix addressing and iteration stops when the null character is encountered:




(6.9)

Let us look carefully at this instruction `ldrneb r1, [r0], #1`. In ARM Assembly, the syntax `ldr<c>b` (where `<c>` is the condition `ne` not equal to) is referred to as a *Pre-UAL* syntax. This syntax has been superceded by UAL syntax, in which case the instructions is `ldrbne` (load register byte if not equal to). However, the pre-UAL syntax is the syntax used by *cpulator*, Hence this is the one we will use in these examples.

We should also note that in load or store byte instructions, the use of the immediate `#1` is correct because we are incrementing the base-register in steps of one byte (`#1`) and not one word (`#4`)

### 6.5.2 Zero-extending bytes

Looking again at this loop:

```
loop:
  cmp r1, #0
  ldrneb r1, [r0], #1
  bne loop
```

of course, we can see that a single byte of data is loaded into `r1` each time `ldrneb r1, [r0], #1` executes. As we know, all the registers in ARM-32 are 32-bits in size. This means that the byte of data stored in `r1` must be *zero-extended* to fit into the

destination register. Consider how the ASCII value of 'A' ($65_{10} = 01000001_2$) would be handled:

| $31-24$ | $23-16$ | $15-8$ | $7-0$ |
|---|---|---|---|
| 00000000 | 00000000 | 00000000 | 01000001 |
| 3 | 2 | 1 | 0 |

(6.10)

As you can see, the $01000001_2$ is right aligned into the LSB (byte 0), and the rest of the register is zero-filled to the MSB (bytes 1-3).

## 6.6 Revisiting the Program Counter

As introduced in Chapter 3 (Section 3.2.2), the Program Counter (`r15`) holds the address used during instruction fetch. However, its exact value at the point of execution is a common source of confusion in ARM assembly, and is worth examining more carefully here.

For many people, the program counter (`r15`) is a confusing register. When we first start programming in ARM assembly, we tend to think of this register as holding the address of the `currently executing instruction`. However, this is not technically accurate.

Consider the following code sample:

```
mov r1, pc
mov r2, #20
mov r3, #30
```

We tend to think that `mov r1, pc` places the address of the instruction `mov r1, pc` into `r1`. This is not quite right. By the time the instruction is actually *executed*, the program counter is no longer holding the address of `mov r1, pc`. In fact, it is holding the address of `mov r3, #30`.

Why? The reason is simple: historically (but not anymore), instructions were processed in three distinct and concurrent stages. The so-called **F**etch, **D**ecode, and **E**execute phases. [1]. Thus, when the instruction was executed, the program counter has already moved to the **F**etch phase.

We can now add these phase letters to our code in order to understand where `pc` is at the point of `mov r1, pc`:

```
[E] mov r1, pc  //execute, pc->Fetch
[D] mov r2, #20 //decode
[F] mov r3, #30 //fetch
```

In summary, when we see instructions like `mov r1, pc`, we should remember that this is really `pc`$+8$

```
@ Pseudo-instruction written by the programmer:
ldr r0, =my_var

@ What the assembler emits (example):
ldr r0, [pc, #offset]   @ load from address (PC + offset)
```

[1] we will learn more about these stages in Chapter 10

Because the offset is computed at *assemble time*, the code is **position-independent**: the same binary works correctly regardless of where in memory it is loaded, as long as the instruction and its data remain the same distance apart.

## 6.7 pc-Relative Addressing

So far we have used `ldr r0, =label` to load the address of a variable into a register. This is a *pseudo-instruction*: the assembler rewrites it behind the scenes using **PC-relative addressing**, a technique where an address is expressed as an offset from the current value of the Program Counter (`pc`).

### 6.7.1 ADR and ADRP

ARM provides two instructions that let you load the *address* of a label directly into a register, without going through a literal pool.

**ADR (ARM32 and ARM64).** `ADR Rd, label` computes the address of `label` relative to the current `pc` and stores it in `Rd`. The offset is encoded inside the instruction itself, so no extra memory access is needed. The catch is range: the offset field is only wide enough to express $\pm 4$ KB, so `label` must sit within 4 KB of the `ADR` instruction.

```
// ARM32/64: load the address of a nearby label
adr r0, my_table   // r0 = address of my_table (must be within +-4
   KB)
```

**ADRP (ARM64 only).** A full 64-bit address space is far too large for any single instruction to encode in one offset field. `ADRP` solves this by splitting the job into two steps.

First, understand what a *page* is in this context: the processor treats memory as a series of 4 KB blocks called pages. Every page starts at an address that is a multiple of 4096, so its bottom 12 bits are always zero. `ADRP` ("Address of Page") loads the *page-aligned base address* that contains `label`, ignoring the low 12 bits. A second `add` instruction then adds the low 12-bit offset to give the exact address.

```
// ARM64: load address of a global variable (step by step)
adrp x0, my_var            // step 1: x0 = address of the 4 KB page
                           //         that contains my_var
                           //         (bottom 12 bits are zero)
add  x0, x0, :lo12:my_var  // step 2: x0 += offset of my_var within
                           //         that page (the low 12 bits)
                           //         now x0 holds the exact
   address
ldr  w1, [x0]              // step 3: load the value stored at that
    address
```

The assembler directive `:lo12:my_var` means "give me only the bottom 12 bits of `my_var`'s address". Together, steps 1 and 2 reconstruct the full address from its upper and lower halves, and this two-instruction pair can reach any address in the 64-bit address space.

**PC-Relative Addressing**

Every address in ARM machine code is ultimately stored as an offset from the Program Counter. The `=label` pseudo-instruction, `ADR`, and `ADRP` are three different ways to do the same thing: compute an address relative to where the current instruction lives. The assembler picks the right form for you when you write `ldr r0, =label`, but knowing the underlying instructions helps you understand range limits, position-independent code, and what the disassembler shows you.

## 6.8 Literal Pools

A **literal pool** is a small region of data that the assembler embeds *inside the code section*, immediately after a function or at the end of a translation unit, to hold constant values that cannot be encoded directly into a 32-bit instruction.

ARM32 instructions are fixed at 32 bits wide. A data-processing instruction uses roughly 12 of those bits for the immediate operand, which means only constants expressible as an 8-bit value rotated by an even number of bits can be embedded directly. A full 32-bit address (e.g. `0x00FFAB00`) cannot fit. The assembler solves this by:

1. Placing the full 32-bit constant in a nearby *literal pool*.
2. Replacing the pseudo-instruction with a PC-relative `ldr` that reads from that pool.

We can let the assembler place the literal pool in our code, or we can decide ourselves where it should be placed. The latter is accomplished by inserting the pseudo-command `.ltorg`

Let's look at an example. Notice, along with our standard ARM code, we see `.ltorg`. When the assembler encounters this it places the literal pool at that exact place.

```
ldr r0, =array
str r2, [r0, #4]
ldr r7, =0X00FFFFAA
add r5, r6, r7
.ltorg          // can be placed anywhere
add r5, r6, r7
ldr r1, =array
labelx:
add r5, r6, r7

.data
array: .word 1,2,3,4,5,6
```

What is in the literal pool? Any objects that we are seeking the address of (using the = prefix). So in the example here, the literal pool should consist of two objects, the constant `0X00FFFFAA` and the *address* of `array`. As we shall see, when looking at the disassembled code, the literal pool objects are computed as an offset from `pc`+8

### 6.8.1 Pool Placement and Range Limits

Now we study the disassembly of the code above. The file here is called `segments.txt` and can be found in the public repo here

```
   0: e59f000c  ldr r0, [pc, #12] @ 14 <main+0x14>
   4: e3a0200a  mov r2, #10
   8: e5802004  str r2, [r0, #4]
   c: e59f7004  ldr r7, [pc, #4]  @ 18 <main+0x18>
  10: e0865007  add r5, r6, r7
  14: 00000000  .word 0x00000000
  18: 00ffffaa  .word 0x00ffffaa
  1c: e0865007  add r5, r6, r7
  20: e59f1000  ldr r1, [pc]  @ 28 <labelx+0x4>

00000024 <labelx>:
  24: e0865007  add r5, r6, r7
  28: 00000000  .word 0x00000000
```

The PC-relative offset in an ARM32 `ldr` is limited to $\pm 4$ KB. If a function grows beyond that distance, the assembler will automatically insert an additional literal pool mid-function to stay within range. The assembler directive `.ltorg` can be used to force a pool at a specific location:

```
my_func:
  ldr r0, =big_constant
  @ ... many instructions ...
  .ltorg            @ force a literal pool here
  @ ... more instructions ...
```

### 6.8.2 Viewing Literal Pools

You can inspect literal pools in a compiled binary with `objdump`:

```
[language=bash]
arm-linux-gnueabi-objdump -d my_program | less
```

Between function bodies you will see lines like:

```
00010234 <.text+0x34>:
   10234:  00ffab00  .word  0x00ffab00
```

These `.word` entries are the literal pool values, sitting in the `.text` section alongside the instructions that use them.

**Literal Pools**

A literal pool is not a separate section: it lives inside `.text`. This is a deliberate design choice: keeping constants close to the instructions that reference them guarantees the PC-relative offset stays small.

## 6.9 Mixed Instructions and Data in ELF

The previous section revealed something surprising: ARM programs routinely store *data* (literal pool entries) inside the `.text` section, which is nominally a *code* section. The ELF format is designed to allow this, and understanding why is important for reading disassemblies and writing correct linker scripts.

### 6.9.1 ELF Sections vs. Segments

Recall from Chapter 2 that an ELF file distinguishes between *sections* (used by the linker) and *segments* (used by the OS loader). The `.text` section is mapped into a read-only, executable `LOAD` segment. Because literal pool entries are also read-only (they are constants baked in at link time), placing them in `.text` is both *safe* and *efficient*: the loader does not need an extra segment.

### 6.9.2 ARM-Specific ELF Annotations

When **Thumb** code is mixed with ARM code in the same object file, the assembler must tell the linker which bytes are instructions and which are data. It does this with `mapping symbols`:

- `$a`: the following bytes are ARM (32-bit) instructions.
- `$t`: the following bytes are Thumb (16/32-bit) instructions.
- `$d`: the following bytes are data (e.g., a literal pool).

These symbols are visible in `objdump -t` output and allow debuggers, disassemblers, and the linker to correctly interpret every byte in the object file.

```
[language=bash]
$ arm-linux-gnueabi-objdump -t my_program | grep '\$'
00010220 l     .text  00000000 $a     @ ARM code starts here
00010234 l     .text  00000000 $d     @ data (literal pool) starts
   here
00010238 l     .text  00000000 $a     @ ARM code resumes
```

### 6.9.3 Implications for the Programmer

- Never branch into a literal pool. The bytes there are data values, not valid instructions; executing them produces undefined behaviour.
- When writing hand-crafted linker scripts, ensure the `.text` output section includes `*(.text*)` *without* splitting it in a way that separates a function from its literal pool, as this can violate the $\pm 4$ KB range constraint.
- On ARM64 the situation is cleaner: `ADRP/add` sequences can reach any page in the 64-bit address space, so the compiler rarely needs to emit literal pools.

**ARM32 vs ARM64**

In ARM32, literal pools inside `.text` are common because the PC-relative range of `ldr` is limited to $\pm 4$ KB. ARM64's `ADRP` + `add` pattern can address the entire 64-bit virtual address space in two instructions, largely eliminating the need for literal pools in normal code.

## 6.10 Chapter Summary

This chapter has introduced the process by which ARM assembly programs access the `.data` (static) segment of a program. We have shown the the programmer must carefully choose the correct data access pattern, as well as understand how memory is accessed depending on the particular data type being used. The two data types are integer (declared as `.word`), and alpha-numeric bytes (declared as `.asciz`).

You should develop an understanding of how the C programming pointer arithmetic model is implemented in ARM assembly. Developing a strong understanding of the similarities between C and assembly will make you a more well rounded software engineer, computer engineer, or general IT expert.

In section 6.4.3 we introduced the topic of *stacks*, and how we can understand these important data structers by studying how to add and remove elements using pre-index and post-index addressing respectively. In the next chapter we will discuss how stacks can be used to help with a variety of programming tasks that would otherwise be impossible without them.

# 7

# Functions and The Stack

**Learning Objectives**

- Explain how the ARM stack supports function calls, including parameters, local variables, saved registers, and return addresses.
- Apply the ARM Procedure Call Standard (APCS) to identify caller-saved and callee-saved registers.
- Trace how the stack pointer `sp` changes during push and pop operations.
- Compare offset addressing and post-index addressing for reading function parameters from the stack.
- Construct an ARM assembly function that saves `lr`, preserves registers, returns a result, and restores the stack correctly.

## 7.1 Introduction

In the previous chapter we introduced the concept of a stack. The idea of a stack should be familiar to all those who have studied discrete data structures.

As introduced in Chapter 6 (Section 6.4.3), the stack follows a **last-in-first-out (LIFO)** protocol: the most recently pushed element is always the first to be removed. In ARM, the stack is accessed via the `sp` (Stack Pointer) register, managed by the operating system.

To this general definition, we can add some ARM-specific stack observations:

1. The stack is used to store *short-lived* word data
2. it is primarily used to:
    - pass *additional* parameters from caller to callee label
    - save and restore general purpose registers
    - store label return address
    - store local variables
3. The programmer does not create the stack, ARM uses the Stack Pointer register `sp` to address the stack. The operating system allocates the stack region for each thread, while program code updates `sp` as temporary values are added to or removed from the stack.
4. The stack grows downward by default (high to low addresses) and is finite in size; for a full discussion of the Full Descending stack model and Linux thread stack limits see Chapter 2
5. As the thread executes, the stack grows and shrinks over time

## 7.2 ARM Procedure Call Standard (APCS)

ARM defines a framework for all compilers and assembly code programmers to follow when passing parameters to a procedure (label/function). This framework is known as the ARM Procedure Call Standard (APCS)

The guide specifies conventions for passing arguments, returning values, using registers, managing the stack, and preserving program state across subroutine calls. By following a common calling convention, separately compiled modules and libraries can interoperate correctly regardless of the compiler or programming language used.

### 7.2.1 Parameter Passing and Return Values

As mentioned in Section 7.2, the APCS defines a standard for passing parameters to procedures and returning values. This standard ensures that both the caller and callee agree on how data is exchanged, which registers are used for arguments, and where the return value will be placed. Before a procedure can execute, the caller must place the parameters (arguments) where the callee expects to find them. The APCS defines this contract: arguments are passed in specific registers, and the procedure returns its result in a specific register. The key concepts are summarized here:

- `r0-r3`: parameter/result passing (extra arguments are stacked). A procedure may overwrite these registers. If the caller needs their values after the call, the caller must save them before calling the procedure.
- If `r0-r3` must be preserved, it is the caller's responsibility to do this
- `r4-r8, r10, r11`: are callee saved. This means that if the `callee` needs to use these registers in the procedure, it must first save them, and later restore them, before the procedure returns. Failure to do this will result in callee corrupted registers.
- If `r9, r12`: must be preserved, it is the caller's responsibility to do this
- Procedure return values are placed in `r0`. For void procedures, the value in `r0` is ignored.

The ARM32 specification can be found here: AAPCS32 Specification Repository

### 7.2.2 Preserved and Unpreserved Register Classes

Formally, we can define the two register classes as follows:

**Caller-Saved vs. Callee-Saved Registers (ARM32)**

- **Unpreserved (caller-saved): `r0–r4`.** The callee may freely overwrite these. If the caller needs their values preserved across the call, the caller must push them onto the stack first.
- **Preserved (callee-saved): `r5–r14`.** If the callee uses any of these, it is responsible for saving and restoring them on the stack, typically immediately after saving `lr`.

Thus, the callee can directly overwrite `r0–r4` without concern for the consequences for the caller. If the caller has some content in `r0–r4` that must not be lost, then the caller must push these registers onto the stack before transferring control to the callee.

The remaining registers (`r5–r14`) must be saved and restored by the callee. That is, if the callee uses any or all of the preserved set, the callee is responsible for saving

and restoring their state by placing them onto the stack right after the link register, and restoring their state before transferring control back to the caller using `bx lr`.

Failure to follow these conventions will result in corrupted registers and unpredictable behavior. For example, if the callee uses `r5` without saving it first, the caller's value in `r5` will be lost when control returns to the caller. There is also a more subtle case where the callee uses `r5` and saves it, but fails to restore it before returning. In this case, the caller's value in `r5` will be lost when control returns to the caller, even though the callee did save it at some point.

As we shaall see, the stack is a crucial data structure, enabling the callee to save and restore registers, pass additional parameters, and manage return addresses. Proper stack management is essential for ensuring that function calls work correctly and that the program behaves predictably.

## 7.3 Pass by Value and Pass by Reference

Before translating function calls into ARM assembly, we need to understand what is actually being passed. There are two important cases: *pass by value* and *pass by reference*. These concepts are fundamental to understanding how data is shared and we discuss some C examples of each case to clarify the difference, before explaining how they are implemented in ARM assembly.

### 7.3.1 Pass by Value

In pass by value, the caller passes a copy of the data to the callee. The callee may change its local copy, but the caller's original variable is unchanged.

```c
void double_value(int x) {
    x *= 2;
}

int main() {
    int a = 5;
    double_value(a);

    // a is still 5
    return 0;
}
```

In this example, `a` is copied into the parameter `x`. The assignment `x *= 2` changes only the local copy inside `double_value`. It does not change `a` in `main`.

In ARM terms, this means the integer value itself is placed in an argument register such as `r0`. The callee receives the data value, not the memory location where the original variable is stored.

### 7.3.2 Pass by Reference

In pass by reference, the caller passes the address of the original data. The callee can use that address to read or modify the caller's variable.

```c
void double_value(int *p) {
    *p *= 2;
}

int main() {
    int a = 5;
    double_value(&a);

    // a is now 10
    return 0;
}
```

In this example, `double_value` receives a pointer to `a`. The expression `*p *= 2` follows the pointer and updates the original variable.

In assembly language terms, pass by reference means that a register such as `r0` contains an *address*, not the data itself. The callee must use `ldr` to read through that address and `str` to write an updated value back to memory.

The key difference is where the modification occurs. Updating `r0` directly changes only the register value. Storing through the address contained in `r0` changes the caller's original memory object.

**Summary: Values, Registers, and References**

- Passing a value means the register contains the data itself.
- Returning a value usually means the callee places the result in `r0`.
- Updating `r0` directly changes the register value seen after return.
- Passing a reference means the register contains an address, not the data itself.
- A referenced memory object is changed only when the callee stores through that address.

We will see examples of both pass by value and pass by reference in Section **??** and Section **??**.

## 7.4 Implementing `call` and `return`

In C, a function call transfers control to another function, then returns to the instruction immediately after the call. ARM32 assembly has no `call` or `return` keywords, so this behavior must be built explicitly using branch instructions and registers.

The `bl` instruction performs the call-like operation: it branches to a procedure label and stores the return address in the link register, `lr`. The procedure then returns by branching back to the address stored in `lr`, usually with `bx lr`. This section explains how these two instructions work together to implement the familiar `call` and `return` pattern.

### 7.4.1 Branch with Link and Returning with `bx lr`

A normal branch instruction, `b`, changes the program counter but does not save a return address. This means the callee has no automatic way to return to the instruction after the branch.

ARM solves this problem with `bl`, meaning branch with link. The instruction `bl procedure` branches to the procedure label and stores the return address in the link

register, `lr`. The callee can then return by executing `bx lr`, which branches back to the address stored in `lr`.

```
main:
  mov r0, #5         // first argument
  mov r8, lr         // save main link register
  bl  double_value   // call procedure
  // when execution returns here,
  // r0 contains the result
  // use r0

  // now return from main
  mov r0, #0  // return value for main - no error
  bx r8

double_value:
  add r0, r0, r0       // r0 = r0 * 2
  bx  lr               // return to caller
```

In this example, `bl double_value` stores the address of the instruction after the call in `lr`. Execution then jumps to `double_value`. When `bx lr` executes, control returns to `main`, and the updated result remains in `r0`.

```
main: //caller
  mov r0, #3
  mov r1, #44
  mov r2, #1
  mov r8, lr
  bl find_max
  // do some print
  bx r8 // end the program

find_max:
```

from a label, and to continue execution from the instruction after the branch instruction. That is, there is no `return` keyword.

## 7.5 Sample C Code

In order to motivate our understanding, we will review a simple C code fragment that shows the key stack features:

```c
// this is the "callee"
int find_max(int a, int b, int c) {
    int max = a;
    if (b > max) max = b;
    if (c > max) max = c;
    return max;
}
// this is the "caller"
int main() {
  int max = find_max(3,44,11);
  printf("The maximum is %d\n", max);
}
```

Although this is easy to implement in C, it is quite difficult to implement in ARM assembly. Let us look at the complexity involved.

1. `caller` prepares 3 parameters to share with the callee
2. `caller` invokes `callee`
3. `callee` retrieves the 3 parameters, one after the other
4. `callee` computes the largest of the three parameters and stores the result in `max`
5. `callee` prepares `max` to make it available to `caller`
6. `callee` returns to `caller`
7. `caller` uses the value returned from `callee`

As you can see, there is a lot going on here. None of the above steps would be possible without the stack. So now we will look at how to "port" this simple C program to ARM assembly by using the Stack Pointer (`sp`) register.

In a high-level language like C, we call a *function*. In ARM assembly, we branch with link to a *label* that marks the entry point of a *procedure*.

Although the assembler target is technically a label, we use the term *procedure* for code that follows the calling convention and returns control to its caller. So in this chapter the C function is synonymous with the procedure entry point.

You can find more details on this approach in the AAPCS32 Specification Repository, which defines the ARM32 calling convention, register usage, argument passing rules, return values, and stack management requirements.

## 7.6 Branch with link `bl` and the link register `lr`

To solve the problem of *where* a label should return to, we replace the `b` instruction with `bl`. The latter means branch *with link*. This causes a special register known as the link register (`lr`) to be updated with the address of the instruction following the `bl`, as shown:

```
main:
00FFFF00 mov r5, #5
00FFFF04 mov r6, #10
00FFFF08 bl label2
00FFFF0C add r7, r5, r6

label2:
00FFFF10 mov r1, #11
00FFFF14 mov r2, #22
00FFFF18 bl label3
00FFFF1C sub r3, r2, r1
00FFFF20 bx lr

label3:
00FFFF24 mov r1, #45
00FFFF28 mov r2, #2
00FFFF2C mul r3, r2, r1
00FFFF30 bx lr
```

Notice `bl` replaces `b` and `bx lr` is used to branch back to the address stored in `lr`. So we place `bx lr` at the end of our label.

Unfortunately, this does not quite work. Each time `bl` is used, it replaces the `lr` with the address of the instruction following. So although the instruction `bl label2` cause `lr` to be updated correctly, *the subsequent* `bl` instructions will overwrite it. The link register is assigned `00FFFF0C`, then it is assigned `00FFFF1C`. If a label calls a label, which calls a label etc. then *only the most recent* `lr` value is preserved, and all the others are lost. How should we solve this? By using the stack to store `lr` as the first thing a label does. We can rewrite the code as follows:

```
main:
00FFFF00 mov r1, #5
00FFFF04 mov r2, #10
00FFFF08 bl label2
00FFFF0C add r3, r2, r1

label2:
00FFFF10 str lr, [ sp, #-4 ]!
00FFFF14 mov r1, #11
00FFFF18 mov r2, #22
00FFFF1C bl label3
00FFFF20 sub r3, r2, r1
00FFFF24 ldr lr, [ sp ], #4
00FFFF28 bx lr

label3:
00FFFF2C str lr, [ sp, #-4 ]!
00FFFF30 mov r1, #45
00FFFF34 mov r2, #2
00FFFF38 mul r3, r2, r1
00FFFF3C ldr lr, [ sp ], #4
00FFFF40 bx lr
```

The above solution now properly handles the case of nested labels, and how to return from them using the stack to store temporary `lr` values. Notice the stack pointer is correctly reset at the end, and the callee label always saves the state of the `lr` as the *very first thing it does*

## 7.7 `callee` saves its working set of registers

When `callee` begins execution, it should, as mentioned in Section **??**, save the `lr` on the stack as its *first* instruction. It should *then* save the set of working registers it will need in order to complete the label code *without clobbering the callers registers*. As mentioned previously, this is the task of saving the *preserved* registers.

This is a very important responsibility of `callee`. A callee that omits to save its working set of preserved registers will almost certainly clobber the registers of caller.

Consider again the code from Example **??**:

```
main:
00FFFF00 mov r1, #5
00FFFF04 mov r2, #10
00FFFF08 b label2
00FFFF0C add r3, r2, r1

label2:
00FFFF0C mov r1, #11
00FFFF10 mov r2, #22
00FFFF14 b label3
00FFFF18 sub r3, r2, r1

label3:
00FFFF1C mov r1, #45
00FFFF20 mov r2, #2
00FFFF20 mul r3, r2, r1
```

Notice how, in each label, `callee` is overwriting the values that the `caller` has placed in the registers (`r1` and `r2`). Of course, the `caller` cannot predict how `callee` will use the registers. The code may be written by different programmers, or the code for `caller` may not be available in source code format to the developer of `caller` The ARM best practice guide recommends that the caller should save any unpreserved registers it uses, and the callee is responsible for saving and restoring the preserved registers it will be using in the label code. For example, the code in `label2` overwrites `r1` and `r2` of `caller`, thus clobbering them. We see that the code in `label3` does exactly the same thing for its `caller`. This kind of problem is easy to solve: `callee` should save its working set of registers on the stack before it begins, and restore the register set when it finishes:

```
main:
00FFFF00 mov r1, #5
00FFFF04 mov r2, #10
00FFFF08 b label2
00FFFF0C add r3, r2, r1

label2:
00FFFF20 str r1, [sp, #-4]
00FFFF24 str r2, [sp, #-4]
00FFFF28 str r3, [sp, #-4]
00FFFF2C mov r1, #11
00FFFF30 mov r2, #22
00FFFF34 b label3
00FFFF38 sub r3, r2, r1
00FFFF3C ldr r3, [sp], #4
00FFFF40 ldr r2, [sp], #4
00FFFF44 ldr r1, [sp], #4

label3:
00FFFF48 str r1, [sp, #-4]
00FFFF4C str r2, [sp, #-4]
00FFFF50 str r3, [sp, #-4]
00FFFF54 mov r1, #45
00FFFF58 mov r2, #2
00FFFF5C mul r3, r2, r1
00FFFF60 ldr r3, [sp], #4
00FFFF64 ldr r2, [sp], #4
00FFFF68 ldr r1, [sp], #4
```

Notice how the save/restore sequence follows the Last-In-First-Out (LIFO) semantics of the stack: save $[r1, r2, r3, ..., rn]$ then restore $[rn, ..., r2, r2, r1]$. Of course, you should only save and restore the set of registers you will be using in your label. There is no need to save and restore all of them.

## 7.8 Stack Overflow and Underflow

The stack is a critical part of the ARM architecture, and managing its space effectively is vital to prevent runtime errors such as **stack overflow** and **stack underflow**. These errors can disrupt the execution of a program and lead to unpredictable behavior.

## 7.9 Stack Overflow

A **stack overflow** occurs when more data is pushed onto the stack than the available space can handle. This situation typically arises in recursive functions or when a large amount of memory is allocated to the stack without considering its limitations. In ARM, since the stack grows downwards, exceeding its bounds can cause it to overwrite other important data, leading to crashes or undefined behavior.

### 7.9.1 Causes of Stack Overflow

The primary causes of stack overflow are:

- **Deep Recursion**: Recursive functions call themselves, and each call pushes its return

address and local variables onto the stack. Without a base case or if recursion depth exceeds the stack capacity, the stack will overflow as more data is pushed onto it.

- **Large Local Variable Allocation**: Functions that allocate large arrays or variables on the stack may exceed the available stack space. If this happens, the stack will overflow.

In ARM assembly, the stack grows from high memory to low memory addresses. When the stack grows beyond its allocated size, the `sp` (stack pointer) may overwrite crucial data, leading to undefined behavior and program crashes.

### 7.9.2 Example of Stack Overflow in Recursive Function

A common scenario that leads to stack overflow is a recursive function that does not have a base case or has excessive recursion depth. Each recursive call pushes more data to the stack, and if the function calls itself indefinitely or too many times, the stack will overflow.

**Example:** A function that recursively calls itself without a base case:

```
void recursive_call() {
    recursive_call();
}
```

In ARM assembly, this can look like:

```
recursive_call:
    push {lr}          // Save lr on to the stack
    bl recursive_call // Call the function recursively
    pop {lr}           // Restore lr from the stack
    bx lr              // Return from the function
```

- Each recursive call adds its return address onto the stack using `push lr`.
- Without a base case, the function continues calling itself, leading to the stack growing larger
- Eventually the stack exceeds the kernel allocated size, causing a stack overflow.

### 7.9.3 Effects of Stack Overflow

When a stack overflow occurs, several issues can arise:

- **Memory Corruption**: The stack is supposed to store critical data such as return addresses, local variables, and saved registers. If the stack overflows, it overwrites this data, leading to corrupted values and unpredictable behavior.
- **Program Crashes or Undefined Behavior**: As the program may attempt to execute corrupted return addresses or access invalid memory locations, it can result in crashes or undefined behavior. This can cause the program to behave in unexpected ways or terminate unexpectedly.
- **Security Vulnerabilities**: Stack overflows are a common vulnerability exploited in buffer overflow attacks. An attacker may intentionally overflow the stack to inject malicious code and gain control over the program.

### 7.9.4 How to Prevent Stack Overflow

There are several strategies to prevent stack overflow in ARM assembly programming:

#### Limit Recursion Depth

Ensure that recursive functions have a proper base case to terminate the recursion. Avoid infinite recursion, and limit the recursion depth to prevent excessive stack growth.

**Example:** Recursive function with a base case:

```
recursive_call:
    cmp r0, #0          // Compare argument with 0
    beq end_recursion   // If base case reached, return
    push {lr}           // Save return address
    sub r0, r0, #1      // Decrement counter by 1
    bl recursive_call   // Recursive call
end_recursion:
    pop {lr}            // Restore return address
    bx lr               // Return from the function
```

In this example, the recursion halts once the argument `r0` reaches zero, preventing infinite recursion and stack overflow.

#### Optimize Stack Usage

Minimize stack usage by avoiding large local variables. Instead of allocating large arrays on the stack, allocate them dynamically in the heap.

**Example:** Using dynamic memory allocation instead of stack allocation:

```
// Avoid allocating large arrays on the stack:
sub sp, sp, #1000  // 1000 bytes on the stack
// Instead, use dynamic memory allocation on
// the heap to avoid stack overflow.
// Example in C: malloc() for large arrays
// along with the code to ingegrate libc calls
```

**▶ Safe Iterative Sum, No Stack Growth**

Instead of a recursive function (which needs `bl`/`bx lr`), use a loop to compute the sum $1 + 2 + \cdots + 5$. This uses only the stack for storing the running total, with no risk of overflow.

```
mov r0, #5        // n = 5
mov r1, #0        // sum = 0
loop:
    add r1, r1, r0    // sum += n
    sub r0, r0, #1    // n--
    cmp r0, #0
    bne loop          // repeat while n != 0
str r1, [sp, #-4]!  // push final sum onto stack
```

Step through in PlayARM and watch `r1` accumulate the sum. After the `str`, check the **Memory** panel to confirm the result (15) is stored at the top of stack.

## Use Tail Recursion

Tail recursion allows a function to reuse its current stack frame for recursive calls, thus avoiding the growth of the stack. A tail-recursive function does not require additional stack space for each recursive call, which helps prevent stack overflow.

**Example of Tail Recursion:**

```
tail_recursive:
    cmp r0, #0          // Check if base case
    beq end_tail_recursion
    sub r0, r0, #1      // Update argument
    bl tail_recursive   // Tail call (no extra stack frame needed)
end_tail_recursion:
    bx lr               // Return from function
```

In tail recursion, the function does not create a new stack frame for each recursive call, thus preventing stack overflow.

## Use the Stack Wisely

Avoid unnecessary usage of the stack. Only push registers to the stack when they are necessary to preserve, and use registers for temporary variables when possible.

**Example:** Pushing only necessary registers to the stack:

```
    // Avoid pushing unnecessary registers to the stack
    push {r4, r5}         // Only push necessary registers
    // Function logic
    pop {r4, r5}          // Restore only the necessary registers
```

By limiting the number of pushed registers, the stack usage is minimized, reducing the risk of overflow.

### Monitor Stack Usage

In embedded systems, monitoring the stack usage through system logs or dedicated tools can help detect potential stack overflows before they happen. A stack guard mechanism can be implemented to check for overflow conditions and raise exceptions if the stack is nearing its limit.

### 7.9.5 Conclusion on Stack Overflow

Stack overflow is a critical issue in ARM assembly and other low-level programming environments. It is commonly caused by excessive recursion or large allocations on the stack. By limiting recursion depth, using the stack wisely, and employing techniques like tail recursion, you can minimize the risk of stack overflow and ensure that your program executes efficiently and securely.

## 7.10 Prologue and Epilogue

Each function typically includes a **prologue** and an **epilogue** that manage the stack, save and restore registers, and adjust the stack pointer for local variable allocation.

The **prologue** and **epilogue** are essential for managing the stack during function calls in ARM assembly. These two sequences of instructions handle saving and restoring registers, allocating and deallocating space for local variables, and ensuring that control is properly returned to the caller.

### 7.10.1 Prologue: Setting Up the Stack Frame

The **prologue** is the first part of a function, responsible for setting up the stack frame. The stack frame stores the function's local variables, return address, and callee-saved registers. The prologue is crucial for establishing the function's execution environment.

- **Saving the `LR`**: The link register (`LR`) holds the return address, which tells the processor where to resume execution after the function finishes. It must be saved on the stack to ensure the function can return to the correct location.
- **Saving Callee-Saved Registers**: The registers `r4`–`r11` are callee-saved, meaning the callee is responsible for saving and restoring these registers if it uses them. This prevents the callee from inadvertently modifying the caller's state.
- **Adjusting the Stack Pointer (`sp`)**: The `sp` is adjusted to allocate space for local variables. The amount of space depends on how many local variables the function needs to store.

### 7.10.2 Epilogue: Returning Control to the Caller

The **epilogue** is executed at the end of a function to clean up the stack and return control to the caller. It undoes the actions performed by the prologue and ensures that the function call does not leave the stack in an inconsistent state.

- **Restoring Callee-Saved Registers**: The registers saved in the prologue are restored to their original values. This ensures that the caller's state is preserved across function calls.
- **Deallocating Space for Local Variables**: The `sp` is adjusted to release the space allocated for local variables. This ensures the stack is returned to its previous state.

- **Restoring the LR and Returning**: The return address stored in `lr` is restored, and the program jumps back to the caller using the `bx lr` instruction.

**Example of Prologue and Epilogue**

Here is a simple example of how the prologue and epilogue are implemented in ARM assembly:

```
    // Function Prologue
function_prologue:
    push {lr}            // Save return address (Link Register)
    push {r4-r7}         // Save callee-saved registers
    sub sp, sp, #16      // Allocate space for 16 bytes of local
   variables

    // Function Body (do some work here)

    // Function Epilogue
function_epilogue:
    add sp, sp, #16      // Deallocate 16 bytes of local variable
   space
    pop {r4-r7}          // Restore callee-saved registers
    pop {lr}             // Restore the return address (Link
   Register)
    bx lr                // Return to the caller
```

## 7.11 Register Allocation and Stack Management

ARM's 16 general-purpose registers (`r0`–`r15`) are introduced in Chapter 3. Their calling-convention roles—which registers are caller-saved, which are callee-saved, and how parameters are passed—are specified by APCS as described in Section 7.2 of this chapter. Keeping register usage within APCS guidelines is essential: accessing registers is far faster than memory, and minimising unnecessary stack spills improves performance significantly.

## 7.12 ARM32 vs. ARM64 Stack Conventions

**ARM32 vs. ARM64: Stack & Calling Conventions**

| Feature | ARM32 (AArch32) | ARM64 (AArch64) |
|---|---|---|
| Stack pointer | `r13` / `sp` | `x28` or `sp` (`x31`) |
| Link register | `r14` / `lr` | `x30` / `lr` |
| Frame pointer | `r11` / `fp` (optional) | `x29` / `fp` |
| Return | `bx lr` | `ret` (branches to `x30`) |
| Save/restore pair | `push`/`pop` (single regs) | `stp`/`ldp` (pairs, e.g. `stp x29,x30,[sp,#-16]!`) |
| Stack alignment | 4-byte (word) | **16-byte mandatory** at all public call boundaries |
| Argument regs | `r0`–`r3` | `x0`–`x7` |
| Callee-saved | `r4`–`r11` | `x19`–`x28` |

**Key implication:** AArch64 always saves `x29` (FP) and `x30` (LR) together as a pair, and `sp` must be 16-byte aligned before any `bl`/`blr`.

## 7.13 Exercises on Programming the Stack

### 7.13.1 Exercise 1: Stack Management in Recursive Functions

Write a recursive function in ARM assembly to calculate the factorial of a number. Implement the function without a base case and observe how the stack grows, leading to a stack overflow. Then, modify the function to include a base case and explain how adding the base case prevents the overflow.

**Instructions:**

- Write the recursive function to compute the factorial without a base case.
- Simulate how the stack overflows when the recursion depth exceeds the available space.
- Modify the function to include a base case and explain how it fixes the overflow.

### 7.13.2 Exercise 2: Understanding Caller-Callee Stack Interaction

Given the C code below:

```
void caller() {
  int sum = callee(4,5,6);
}
int callee(int a, int b, int c) {
  int tmp = 0;
  tmp = a + b + c;
  return tmp;
}
```

**Task:** Translate this C code to ARM assembly by using the stack to pass parameters from the caller to the callee, store local variables, and manage the return address.

**Instructions:**

- Implement the function ‘caller‘ and ‘callee‘ in ARM assembly, passing parameters through the stack.
- Use the ‘sp‘ (Stack Pointer) and ‘LR‘ (Link Register) properly to manage the stack and return address.

### 7.13.3 Exercise 3: Stack Overflow Prevention

Write a function in ARM assembly that uses a loop to sum integers from 1 to n. Compare the iterative solution with a recursive one and analyze how stack overflow could occur in the recursive version.

**Instructions:**

- Implement both iterative and recursive versions of the summing function.
- Simulate how the recursive function could lead to a stack overflow if ‘n‘ is too large.
- Modify the recursive version to use tail recursion and explain how it reduces stack usage.

### 7.13.4 Exercise 4: Prologue and Epilogue Function Design

Write an ARM assembly function that uses a prologue to save registers and allocate space for local variables, and an epilogue to restore registers and deallocate the space.

**Instructions:**

- Implement a function that saves and restores registers (‘r4‘ to ‘r7‘) as part of the prologue and epilogue.
- Allocate space for local variables in the prologue and deallocate them in the epilogue.
- Ensure the function returns control to the caller correctly using the ‘bx lr‘ instruction.

### 7.13.5 Exercise 5: Exploring Stack Underflow

Write a program that demonstrates stack underflow in ARM assembly. Underflow occurs when the stack pointer is adjusted incorrectly, and the program attempts to pop more data than was pushed onto the stack.

**Instructions:**

- Write a function that incorrectly manipulates the stack pointer by popping data without pushing it first.
- Simulate stack underflow by observing incorrect data restoration, or by analyzing how crashes occur when the stack is manipulated incorrectly.
- Explain how correct stack management can prevent stack underflow.

### 7.13.6 Exercise 6: Stack Size Limitation and Stack Guard

Implement a simple ARM assembly program that simulates exceeding the stack size limit. Show how a guard mechanism (e.g., checking if the stack pointer exceeds a set limit) can prevent stack overflow.

**Instructions:**

- Write a function that simulates allocating large data on the stack and exceeding the stack’s size limit.

- Implement a stack guard that checks if the stack pointer exceeds the set limit and prevents overflow.
- Demonstrate how the guard mechanism can safely terminate the program when a potential overflow is detected.

# 8

# Integrating libc

**Learning Objectives**

- Explain how ARM assembly programs can call Linux system calls and libc functions.
- Compare direct system calls using `svc` with higher-level libc function calls.
- Prepare arguments correctly for `printf` using ARM32 register-based calling conventions.
- Access command-line arguments through `r0` and the `argv` pointer stored in `r1`.
- Construct an ARM assembly program that calls libc, prints formatted output, returns through `main`, and reports status through `r0`.

## 8.1 Linux System Calls

In the previous chapters we learned the basics of ARM32 assembly techniques and guidelines. The code we looked at was limited in terms of its *functionality* - it was limited to logical, mathematical and flow control instructions. The code did not require any services from the operating system.

These services are known as *system calls*, and a typical application will use many system calls in order to achieve its objectives. Indeed, an application program that does not use system calls can not achieve anything useful.

There are two ways for an assemble program to access the operating system services (1) by using system calls directly from the assembly, and (2) linking to a library of functions that provide an abstraction layer above the system call level. Such libraries are very useful because it allows us to reuse high-level abstractions rather having to build the scaffolding code around the service call. The difference is summarized in Fig. 8.1.

Directly using system calls from an application process is a difficult task. It requires the developer to execute a software interrept (`svc #0`) instruction, along with populating specific registers correctly. For example, the system call to print `"Hello World\n"` to the screen shown in Fig 8.2, and as you can see, the implementation in `print` is rather complex for such as simple taks.

Using system calls for I/O, process management, memory allocation etc. is perfectly possible, but requires us to reinvent functionality that has already been solved and implemented elsewehere. As showin in Fig. 8.2, we will have difficulty creating a `printf` type function - one that supports parameter substitution within the string. For example, consider this simple C program:

As we can see clearly from Fig. 8.3, `printf` supprts parameter substitution by way of the `%s, %d, ...` modifiers. How could we achieve this functionality using the `svc` instruction? With great difficulty!

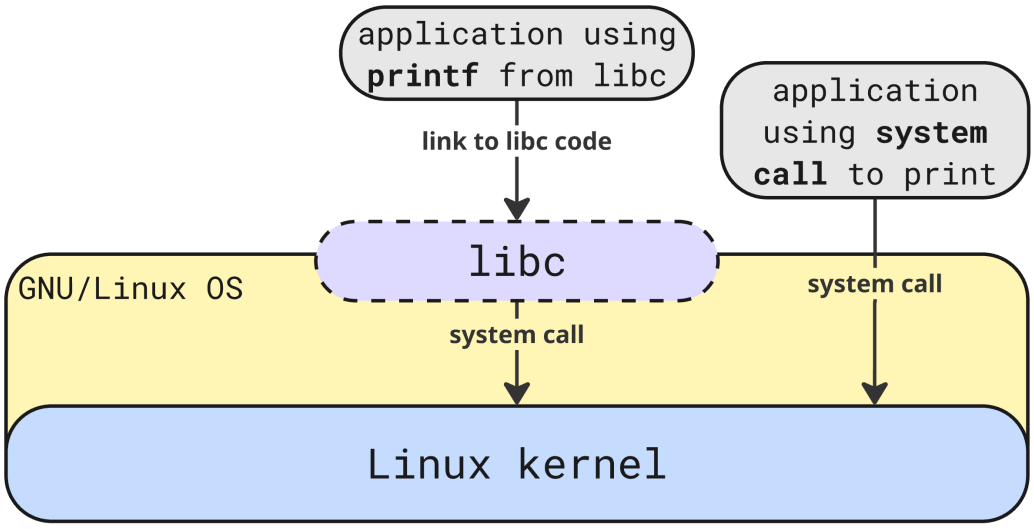



Figure 8.1: Application code can use either `libc` or system calls (`svc` to print to the screen)

```
print:
    mov r0, #1      // r0 gets 1=stdout
    ldr r1, =string // r1 gets the string address
    mov r2, #12     // r2 is the string length
    mov r7, #4      // r7 gets 4=print system call
    svc #0
.data
  string: .asciz "Hello World\n"
```

Figure 8.2: Assembly code for printing to the screen using the system call instruction `svc`

```
int main(int argc, char* argv[]) {
  printf("num params: %d, \
    program name %s, \
    first param: %s\n", \
    argc, argv[0], argv[1]);
}
```

Figure 8.3: C code for printing to the screen using `printf` from `libc`

In summary, although all the kernel functions for I/O, memory management, process management etc are accessible via system calls, we tend to avoid using this method because it requires us to build complex scaffolding code that adds to the overall complexity and management of the assembly application. Therefore, we leave it up to you to build your own `printf` function using service calls only.

### 8.1.1 Linux libc

GNU/Linux `libc` contains over 400 functions [1] functions that can be called from user-layer applications. A complete discussion on `libc` is beyond the scope of this book. Readers interested in looking at `libc` in detail should refer to `https://www.gnu.org/software/`

We will discuss one `libc` example next: the `printf` function shown in Fig. 8.3. `printf` can be used to print a string along with any number of *arguments* to the string. See `https://man7.org/linux/man-pages/man3/printf.3.html`.

### 8.1.2 Printing to the screen using `printf`

To make the example more interesting, we will user `printf` to print the following information to the screen:

- the number of command line parameters to the program
- the name of the program
- the first parameter to the program

Assuming that the program is called `a.out`, the running the following command:

```
$ ./a.out 55
num params: 2, program name ./a.out, first param: 55
```

This example demonstrates some interesting features. First, notice that we need to obtain command line parameters the OS passes into our application. In C, we use the standard main function signature `int argv, char *argc[]` to obtain the command line parameters passed in. But how do we get the equivalent of `int argv, char* argc[]` in ARM assembly? The convention is that:

- `r0` contains the `argv`
- `r1` is a base-register which points to an array of `char*` pointers. This is the definition of `char* argv[]`
- The stack pointer `sp` is *not* used to pass items from the command line into `main`

Based on the code in Fig 8.4 we see how to integrate `printf` from `libc`. We will briefly discuss this solution next.

## Discussion of Fig 8.4

Let's focus on these two instructions with line numbers:

```
1. ldr r2, [ r1 ]
2. ldr r3, [ r1, #4 ]
```

[1] excluding the _ functions. If we include these functions, this number rises to over 3000

```
main:
  // save the return address from main
  push {lr}

  // in ARM32 r1 is the base-pointer
  // to an array of pointers to strings
  // zero-ith parameter - program name
  ldr r2, [ r1 ]
  // first parameter - program parameters
  ldr r3, [ r1, #4 ]

  // if have n parameters they are
  // accessed as r1 + (n * 4)
  // we are finished accessing r1
  // so we can prepare it for printf
  mov r1, r0
  ldr r0, =output_string

  // use puts or printf from libc
  bl printf

  // set up mains return parameters
  mov r0, #0

  // now return from main
  pop {pc}

.data
  output_string:
    .asciz "num params: %d, program name %s, first param: %s\n"
```

Figure 8.4: Assembly code for printing to the screen using `printf` from `libc`

line 1: `r1` is dereferenced (`[r1]`) we get `argv[0]` (which is *always* the name of the program).
line 2: Using offset addressing to point to the next pointer and dereferencing (`[ r1, #4 ]`) brings us to `argv[1]`.

Next, we look at how to prepare parameters for passing information `libc` functions. To better understand how to prepare parameters into `libc` you must carefully review the function prototype from the man pages:

```
int printf(const char *restrict format, ...);
```

Notice that the first parameter to `printf` is the formatting string. In our case, this string is:

```
num params:  %d, program name %s, first param:  %s
n
```

```
3. mov r1, r0
4. ldr r0, =output_string
5. bl printf
```

ARM32 has some specific rules around passing parameters into functions. For the first *four* parameters, registers `r0-r3` should be used. This means the stack is not involved. If there are more than four parameters, then all subsequent parameters should be pushed on to the stack from right-left ordering. However, here we have only 4 parameters, so `r0-r3` will be used, as follows:

1. line 3: place the value of `argv` into `r1`. (
2. line 4: set `r0` as the address of the formatting string
3. line 5: branch with link to `printf`

Graphically this is summarized in Fig 8.5, where the application (`a.out`) is invoked and the OS passes in command line parameters (in `r0` and `r1`). The application then process the parameters and sets up the registers correctly (`r0-r3`), then invokes `printf`

## Finally

Finally, note this code fragment:

```
push {lr}
// .. call to printf
mov r0, #0
pop {pc}
```

as we know, all lables (and this includes `main`), should save the link register (`push {lr}`) as the first instruction. Notice when the `printf` routine returs we set `mov r0, #0`. This is used by the OS to understand if our program completed successfully or not. [2]). Finally, the link register value is popped and placed in the program counter (`pop {pc}`). This is effectively how a return takes place. Of course, when `main` returns the program is finished executing.

How is the value in `r0` used? We can use the linux parameter `$?` to check:

[2]where a non-zero value is used to indicate an error

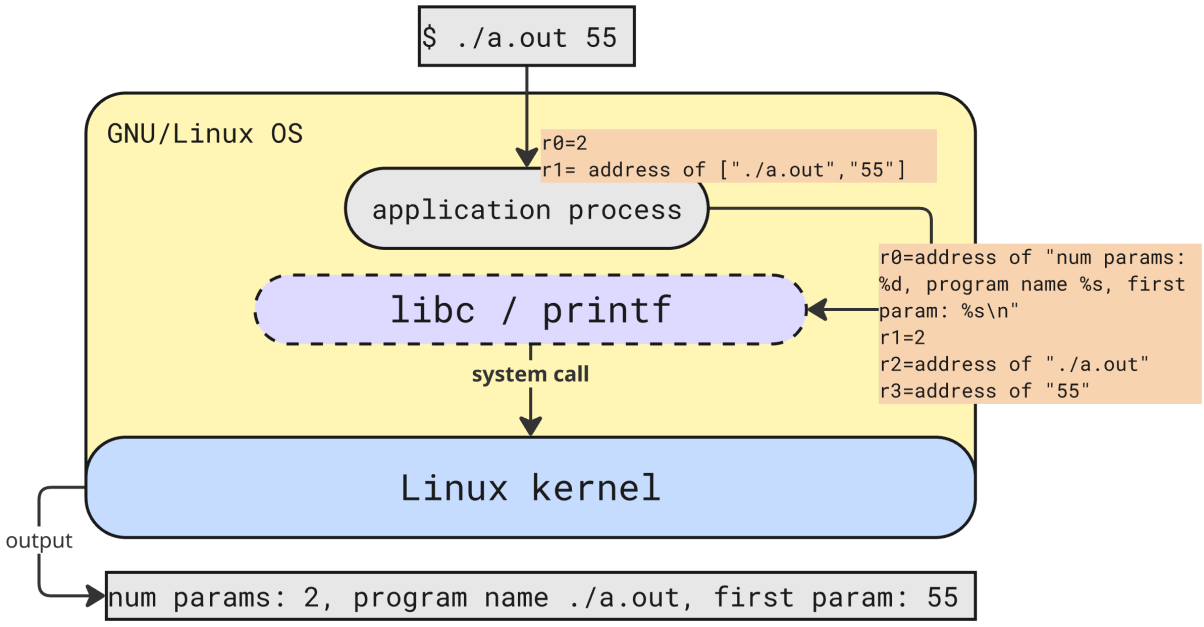



Figure 8.5: The application (`a.out`) is invoked and the OS passes in command line parameters (in `r0` and `r1`). The application then process the parameters and sets up the registers correctly (`r0-r3`), then invokes `printf`

```
$ ./a.out 55
num params: 2, program name ./a.out, first param: 55
$ echo \$?
0
```

When we run `echo $?` and get 0, we are receiving the 0 placed into `r0` prior to the return from `main`

## 8.2 Chapter Exercises using `libc` functions

**Exercise 8.2.1**

*Write an assembly program to iterate over `argv` printing out each parameter (using `printf`) on a new line. For example:*

```
$ ./a.out 55 66 77 88 99
param 1: 55
param 2: 66
param 3: 77
param 4: 88
param 5: 99
```

*In `libc`, upper and lower case string manipulation is handled by `int toupper(int c)` and `int tolower(int c)` functions respectively. Using , write an assemply program to convert its use input from upper or mixed case, to all lower case. For example:*

```
$ ./a.out MY NamE iS TOLOwer
lower_case: my name is tolower
```

*The `libc` function `int atoi (const char *string)` takes a string pointer and converts it into an integer. For example, calling*

```
int r = atoi("55")
```

*would convert "55" to 55 in a C program. Write an assembly program that takes two parameters and adds them together, printing the result output:*

```
$ ./a.out 55 65
Total is: 120
```

*For this to work correctly you will need to use both `atoi` and `printf` from `libc`*

*Write an assembly language program to read from a the users name from the command line, and print a friendly greeting to the user. For example:*

```
$ ./a.out
Hi! - what is your name? [ user enters Bob ]
Hi Bob, nice to meet you!
```

*You will need to use `printf` and `gets` from `libs`*

*Write and assembly program to give the user 3 random numbers $r_1, r_2, r_3$ where $0 \leq r_k \leq 100$. You can store the random values in `r4-r6`. Print the results to the user. For example, here is some sample output:*

```
$ ./a.out
Your 3 numbers are: 55, 22, 10
```

*You will need to use `int rand (void)` and of course, `printf`*

# 9

# Binary Representation of Instructions

**Learning Objectives**

By the end of this chapter you should be able to:

- Explain why ARM instructions are represented as fixed-width 32-bit binary values and describe the constraints this places on instruction encoding.
- Describe the three major classes of ARM instructions—data processing, memory, and branch instructions—and explain the role of each class within an assembly language program.
- Interpret the binary representation of common data processing instructions and identify the instruction operation and registers being used.
- Interpret the binary representation of common memory instructions and explain how they specify memory accesses and addressing modes.
- Decode simple hexadecimal or binary machine code into its corresponding ARM assembly instruction and verify the result using an assembler or simulator.

So far in this book we have been focused on the *higher-level* challenges of selection, iteration, branching, accessing memory, reading and writing the stack, and so on. Of course, even assembly language program consists of textual instructions that a CPU simply does not understand. To the CPU, the only understandable input is a word-size binary (or hexadecimal) number.

This naturally raises two related questions:

1. By what process is an assembly instruction translated into binary machine code?
2. What design constraints shape the representation of textual instructions in binary form?

In this chapter, we will address both of these important questions together as we proceed. We first begin with a motivating example.

## 9.1 Motivating Example

So we begin this chapter by noting that **each and every line of code we write in ARM32/64 *must* fit into one 32-bit field.**

For example, every line of code from this fragment (taken from Chapter 6) must fit into one 32-bit word in ARM7DTI:

```
ldr r0, =array
mov r1, #0
mov r2, #0
mov r3, #6

loop:
cmp r1, r3
strlt r2, [ r0, r1, lsl #2 ]
addlt r1, r1, #1
blt loop

.data
array: .word 1,2,3,4,5,6,7
```

It is the objective of this chapter to first explain the *process* by which instructions (like those shown) above, can be encoded in a 32-bit field. It should be noted that in both ARM32- and 64- bit platforms, instruction length is 32-bits. So on 64-bit platforms one memory word contains two instructions, whereas on the 32-bit platform, one memory word contains one instruction.

As you will see, there is nothing mysterious about encoding instructions into *machine code*,

if we follow some basic structural patterns. These patterns are based on the *class* of instruction: being encoded. There are only 3: (a) **data processing** (b) **memory**, and (c) **branching**. Each of these will be covered in detail in Section 9.2

In the above example, we have eight instructions. At the end of the assembly process there will be 8 32-bit instructions in our ELF file too, although there will also be a lot of set-up and configuration instructions added by the assembler (/compiler) and the linker. However, *conceptually*, our 8 instructions in a text file will result in 8 executable 32-bit fields in our ELF file (again, assuming we are on the ARM7DTI platform).

So our 8 instructions will look like:

```
e59f0018
e3a01000
e3a02000
e3a03006
e1510003
b7802101
b2811001
bafffffb
```

or in binary format:

```
11100101100111110000000000011000
11100011101000000001000000000000
11100011101000000010000000000000
11100011101000000011000000000110
11100001010100010000000000000011
10110111100000000010000100000001
10110010100000010001000000000001
10111010111111111111111111111011
```

Now that we know *what* is happening, we will focus on *how* it happens: how a text instruction (eg `ldr r0, =array`) is converted into a binary representation (in this example, `11100101100111110000000000011000`)

## 9.2 Instruction Classes

ARM, in common with most ISAs, defines three classes of instructions:

1. **Data Processing (DP) Instructions**: the instructions that use the Arithmetic and Logical Unit (ALU) of the CPU to perform mathemtical or register move operations. There are 16 different such instructions (in ARM32), for example: `add,sub,mul,mov` and so on. It also includes logical instructions such as `and,orr` along with all the bitwise instructions, for example, `lsl,lsr,asr,ror`.
2. **Memory Instructions**: this smaller set of instructions, excluding the condionals, signed and flag setting variants, has ony four main classes that we will consider (again, there are others): `ldr,ldrb,str,strb`
3. **Branching Instructions**: this set of instructions contains three main variants (there are some others): `b, bl, bx`

## 9.3 Data Processing Instructions

The class of *data processing* instructions are those whose job it is to move and operate on data. For example, these are the all the `mov{cond}{S}` variants, along with all mathematical and logical instructions (the more common of which are shown in Table 3.1).

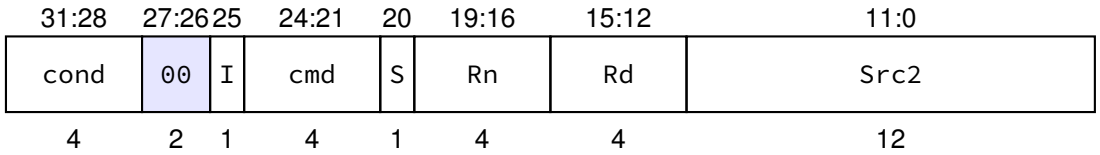
In some ways, data processing instructions are the most complex in terms of their 32-bit binary encoding. The encoding must allow for a great deal of variation in instruction format, for example:

```
mla r0, r1, r2, r3
addlts r4, r5, r6
and r5, r6, #1
sub r7, #4
```

Notice the variation here: `sub r7, #4` and `mla r0, r1, r2, r3` for example. Both of these instructions must be encoded using the data processing layout that we will discuss next. How this happens is very interesting.

### 9.3.1 Non-Multiply Instruction Format

There are eight different parts to the general (non-multiply) data processing instructions shown here. Along the top row, for each field, is the bit from:to pair, and along the bottom row is the number of bits for the field.




We now discuss each of these in turn, from the most significant bit (31) to the least significant bit (0).

`[31:28]` `cond`: the *condition* code for the instruction. Used to determine if the instruction is to be executed. The decision is made by the micro-architecture in reference to the state of the `cpsr/nzcv` state. The full list of condition codes is shown in Table 9.1, left hand side
`[27:26]` `00`: the *operation* code for the instruction. `00` means *Data Processing*
`[25]` `I` the *immediate* bit. Where $I = 1$ the instruction contains an immediate value (for example `add r0, r1, #5`) and $I = 0$ there is no immediate (for example, `add r0, r1, r2`, `add r0, r1, r2, lsl #2`
`[24:21]` `cmd`: the data processing operation to perform. With 4 bits for `cmd` there are only 16 possible variants as shown in Table 9.1 right hand side
`[20]` `S`: the *update-flags* bit, when $S = 1$ the output form the operation (for example, a negative number) is updated to the `NZCV` condition register (eg, `adds r0, r1, #-10`). Where $S = 0$ the register is not updated (eg, `add r0, r1, #-10`)
`[19:16]` `Rn`: the *first source register*. For example, in the instruction `add r0, r1, r2`, `Rn` is `r1`
`[15:12]` `Rd`: the *destination register*. For example, in the instruction `add r0, r1, r2`, `Rd` is `r0`
`[11:0]` `src2`: the second *source* parameters. There are 4 possible arrangements for `src2`, detailed in Section 9.3.1

Notice an especially important fact: the ARM32 non-multiply data-processing instructions have a 12-bit field called the second source, or, more simply, `Src2` (`[11:0]`) which is used to encode a variety of operands to the instruction. `Src2` is discussed in the next section.

## `Src2`: The Second Source

The non-multiply data processing instructions have a region of the instruction known as second source (`src2`) from bit `[11:0]`. The reason its called second source, is because `Rn` is the first source! The purpose of this 12-bit section is to accommodate a variety of different second source arrangements. Here are 4 examples to illustrate this diversity, where the `[31:12]` is the same, and `[11:0]` is different. Note that sections in blue are hard-coded and not dependent on the operands of the instruction:

1.

| 11:8 | 7:0 |
|---|---|
| ror | Imm |

Example: `add r5, r1, #2`, $I = 1 \therefore$ **"Immediate"** variant:

2.

| 11:8 | 7 | 6:5 | 4 | 3:0 |
|---|---|---|---|---|
| 0000 | 0 | 00 | 0 | Rm |

Example: `add r5, r1, r2`, $I = 0 \therefore$ **"Register"** variant:

3.

| 11:7 | 6:5 | 4 | 3:0 |
|---|---|---|---|
| shamt5 | sh | 0 | Rm |

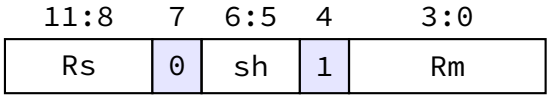
Example: `add r5, r1, r2, lsl #3`, $I = 0$ and Immediate Shift $\therefore$ **"Immediate Shifted Register"** variant

| Code | Meaning |
|---|---|
| 0000 | Equal to (eq) |
| 0001 | Not equal (ne) |
| 0010 | Carry set (cs/hs) |
| 0011 | Carry clear (cc/lo) |
| 0100 | Negative (minus) (mi) |
| 0101 | Positive or zero (plus) (pl) |
| 0110 | Overflow set (vs) |
| 0111 | Overflow clear (vc) |
| 1000 | Unsigned higher (hi) |
| 1001 | Unsigned less than or same (ls) |
| 1010 | Signed greater than or equal to (ge) |
| 1011 | Signed less than (lt) |
| 1100 | Signed greater than (gt) |
| 1101 | Signed less than or equal to (le) |
| 1110 | Unconditional - always execute |

| Code | cmd |
|---|---|
| 0000 | and |
| 0001 | eor |
| 0010 | sub |
| 0011 | rsb |
| 0100 | add |
| 0101 | adc |
| 0110 | sbc |
| 0111 | rsc |
| 1000 | tst |
| 1001 | teq |
| 1010 | cmp |
| 1011 | cmn |
| 1100 | orr |
| 1101 | mov |
| 1110 | bic |
| 1111 | mvn |

Table 9.1: On the left, the condition codes for $31:28$ (there is no $1111$ pattern). On the right, the arrangement for `cmd` $(24:21)$

4.




Example: `add r5, r1, r2, lsl r3`: $I = 0$ and Register Shift $\therefore$ **"Register Shifted Register"** variant

## Worked Examples

We will work through the examples from here to show how the in hexadecimal/binary generated by the assembler can be analyzed. As all four instructions share the same `[31:12]` we will briefly cover them first before looking at the `src2` variants.

1. *Immediate:* `add r5, r1, #2`

   (a) $31:28$ =`1100` - this is unconditional execution (Table 9.1, LHS)
   (b) $27:26$ =`00` - this is a data processing instruction
   (c) $25$ =`1` - there is an immediate present
   (d) $24:21$ =`0100` - the `add` function is `0100` (Table 9.1, RHS)
   (e) $20$ =`0` - there is no update to the flags register (`S`)
   (f) $19:16$ =`0001` - register `Rs` is `r1`

(g) $15:12 =$`0101` - register `Rd` is `r5`
(h) $11:8 =$`0000` - no rotate right required
(i) $7:0 =$`00000010` - immediate value $= 2$

Solution: `1110 00 1 0100 0 0001 0101 0000 00000010` ($e2815002$)

2. *Register:* `add r5, r1, r2`

(a) $31:28 =$`1100` - this is unconditional execution (Table 9.1, LHS)
(b) $27:26 =$`00` - this is a data processing instruction
(c) $25 =$`0` - there is no immediate present
(d) $24:21 =$`0100` - the `add` function is `0100` (Table 9.1, RHS)
(e) $20 =$`0` - there is no update to the flags register (`S`)
(f) $19:16 =$`0001` - register `Rn` is `r1`
(g) $15:12 =$`0101` - register `Rd` is `r5`
(h) $11:8 =$`0000` - hard-coded by the assembler as there is no shift operation
(i) $7 =$`0` - hard-coded by the assembler as there is no shift operation
(j) $6:5 =$`0` - hard-coded by the assembler as there is no shift operation
(k) $4 =$`0` - hard-coded by the assembler as there is no shift operation
(l) $3:0 =$`0010` - encodes `r2` for register `Rm`

Solution: `1110 00 0 0100 0 0001 0101 0000 0 00 0 0010` ($e0815002$)

3. *Immediate shifted Register:* `add r5, r1, r2, lsl #2`

(a) $31:28 =$`1100` - this is unconditional execution (Table 9.1, LHS)
(b) $27:26 =$`00` - this is a data processing instruction
(c) $25 =$`0` - there is no immediate present (the shift immediate does not count)
(d) $24:21 =$`0100` - the `add` function is `0100` (Table 9.1, RHS)
(e) $20 =$`0` - there is no update to the flags register (`S`)
(f) $19:16 =$`0001` - register `Rn` is `r1`
(g) $15:12 =$`0101` - register `Rd` is `r5`
(h) $11:8 =$`0000` - hard-coded by the assembler as there is no shift operation
(i) $7 =$`0` - hard-coded by the assembler as there is no shift operation
(j) $6:5 =$`0` - hard-coded by the assembler as there is no shift operation
(k) $4 =$`0` - hard-coded by the assembler as there is no shift operation
(l) $3:0 =$`0010` - encodes `r2` for register `Rm`

Solution: `1110 00 0 0100 0 0001 0101 00010 00 0 0010` ($e0815102$)

4. *Register Shifted Register:* `add r5, r1, r2, lsl r3`

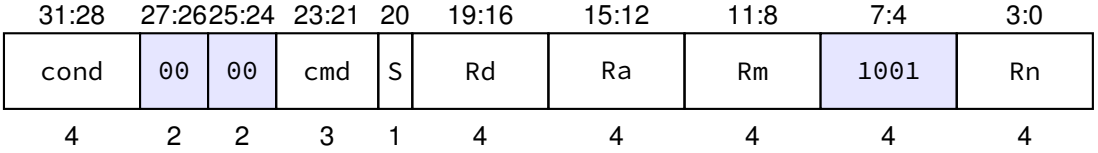
(a) $31:28 =$`1100` - this is unconditional execution (Table 9.1, LHS)
(b) $27:26 =$`00` - this is a data processing instruction
(c) $25 =$`0` - there is no immediate present (the shift immediate does not count)
(d) $24:21 =$`0100` - the `add` function is `0100` (Table 9.1, RHS)
(e) $20 =$`0` - there is no update to the flags register (`S`)
(f) $19:16 =$`0001` - register `Rn` is `r1`
(g) $15:12 =$`0101` - register `Rd` is `r5`

(h) $11:8 =$`0011` - the shift register `Rs` is `r3`
(i) $7 =$`0` - hard-coded by the assembler
(j) $6:5 =$`00` - `lsl` is encoded as 00
(k) $4 =$`1` - hard-coded by the assembler
(l) $3:0 =$`0010` - encodes `r2` for register `Rm`

Solution: `1110 00 0 0100 0 0001 0101 0011 0 00 1 0010` ($e0815312$)

### 9.3.2 Multiply instructions

In contrast to the non-multiply instructions, ARM7DTI multiply instructions have 10 different parts, two of which (`[25:24]` and `[7:4]`) are hard-coded by the assembler and cannot be modified by the programmer. Again, we discus each of them in turn, from the most significant bit (31) to the least significant bit (0). The structure is shown here. Note that these instructions do not have any `src2` sections.




It is now obvious why any kind of multiply instruction with an immediate, for example, `mul r0, r2, #3` will not assemble because, as we can see, there is no immediate anywhere in the instruction. This was in order to make room for `mla` multiply-accumulate, which is extremely useful for dot products, filters, matrix operations, address calculations, and DSP-like code. Multiply by a small constant often has good alternatives. For constants like $2, 3, 4, 5, 8, 10, ..$, ARM can often use shifts and adds:

```
// implementing mul with a constant
add r1, r2, r2, lsl #1 ; r1 = r2 + r2*2 = r2*3
```

## Worked Examples

Let us work through two examples of multiply. The first example is `mullt r1, r2, r3`. This is a conditional multiply that will be executed if `NZCV=1000` and demonstrates the plain multiply without accumulate. The second example, `mlage r1, r2, r3, r4` uses `r4` as an accumulator register such that the result of `r2` multiplied by `r3` is stored in `r1` and *added to* `r4`.

1. `mullt r1, r2, r3`

   (a) `[31:28]`: `1011` - conditional execution (Table 9.1, LHS).
   (b) `[27:26]`: `00` - a data processing instruction
   (c) `[25:24]`: `00` - hard-coded by the assembler
   (d) `[23:21]`: `000` - code is used to represent `mul`. Note - this leads to potential ambiguity discussed below.
   (e) `[20]`: `0` - there is no update to the flags register (`S`).
   (f) `[19:16]`: `0001` - register `Rd` is `r1`

(g) `[15:12]`: `0000` - there is no accumulator register `Ra` is `0000`
(h) `[11:8]`: `Rm` is the second source register (`r3`), `0011`.
(i) `[7:4]`: `1001` - hard-coded by the assembler
(j) `[3:0]`: the first source register (`r2`) is `0010`

Solution: `1011 00 00 000 0 0001 0000 0011 1001 0010` ($b0010392$)

2. `mlage r1, r2, r3, r4`

(a) `[31:28]`: `1110` - conditional execution (Table 9.1, LHS).
(b) `[27:26]`: `00` - a data processing instruction
(c) `[25:24]`: `00` - hard-coded by the assembler
(d) `[23:21]`: `001` - is used to represent `mla`. Note - this leads to potential ambiguity discussed below.
(e) `[20]`: `0` - there is no update to the flags register (`S`).
(f) `[19:16]`: `0001` - register `Rd` is `r1`
(g) `[15:12]`: `0100` - the accumulator register `Ra` is `r4`
(h) `[11:8]`: `Rm` is the second source register (`r3`), `0011`.
(i) `[7:4]`: `1001` - hard-coded by the assembler
(j) `[3:0]`: the first source register (`r2`) is `0010`

Solution: `1110 00 00 001 0 0001 0100 0011 1001 0010` ($e0210392$)

## Avoiding Ambiguity using `[7:4]`

An important feature of all instructions is that they must *avoid ambiguity*. This means, of course, exactly *one* unambiguous instruction is possible for a given 32-bit pattern. Unfortunately, multiply instructions *are* ambiguous in their first 12-bits. This means the first 12-bits may match a totally different instruction. Let's look at two examples, and see how this ambiguity is resolved:

- Both `and r1, r2, r3` and `mul r1, r2, r3` share the same first 12-bits ($e00$) `[31:20]` is `1110 00 0 0000 0` for the `and` and `1110 00 00 000 0` for the `mul`. In this case, the only way to resolve this ambiguity is by looking at `[7:4]`. In the case of `and` this section *cannot* be `1001`. [1]. If `1001` is found at position `[7:4]` then this is a `mul`, and if it is not found, then it is an `and`.
- The same reasoning applies to `eor` and `mla`. For example, `eor r1, r2, r3` and `mla r1, r2, r3` share the same first 12-bits `111000000010`. Again, there are *two* ways to view this pattern: as either an exclusive-or, or, a multiply accummulate. As in the previous discussion, this can only be resolved by reading `[7:4]`. If `1001` is found at position `[7:4]` then this is a multiply accumulate, otherwise, it is an exclusive-or.

[1] you should able to prove this to yourself using the various structures for `src2` discussed in Section 9.3.1

**▶ Try this: Check Your Knowledge**

We have just shown you how to go from assembly → binary → hex. But we can easily go from hex → binary → assembly instruction. Let's see how: Suppose we are given $e0535004$ and we want to know what the assembly instruction is, we can follow these steps.

- Use Rapid Tables to convert $e0535004$ to binary: you should get `11100000010100110101000000000100`
- We can space it out to make it easier to understand. The first 6 bits are: `1110 00`. So this is an unconditional data processing instruction.
- Next we look at `[7:4]` for the multiply pattern. `[7:4]` = `0000` so this is not a multiply instruction.
- Next, using the structure discussed in Section 9.3.1 we look at `[25:20]` - `000101` - which tells us this there is no immediate, and the operation is `0010` which is `sub`. Note the the S bit is 1 so now we have `subs`
- All that remains is to work out the 3 registers: `Rn` = `0011` so this means `r3`
- `Rd` = `0101` so this means `r5`
- Finally, according to Section 9.3.1, we are dealing with #2 the "Register" variant, which means we look at the last four bits for `Rm` = `0100` which is `r4`

Now we have assembled the instruction: `subs r5, r3, r4`.
But, is it correct? Put `subs r5, r3, r4` in to CPUlator, then click compile and load. In the Disassembly view you should see the original hexadecimal number we started with, alongside the instruction: $e0535004$.
Congratulations - you just reverse engineered a binary instruction!

## 9.4 Memory Instructions

We will now look at the *memory instruction* class. This class consists of a much smaller number of instructions. There are only *two*. The load register instruction `ldr`, and the store register instruction `str`. However, as you may remember from Chapter 6, there are quite a few variants possible.

In Section 9.4.1 we discuss the details of each field and the meaning of each configuration combination. But first, let us list a few examples of the diversity of memory instructions:

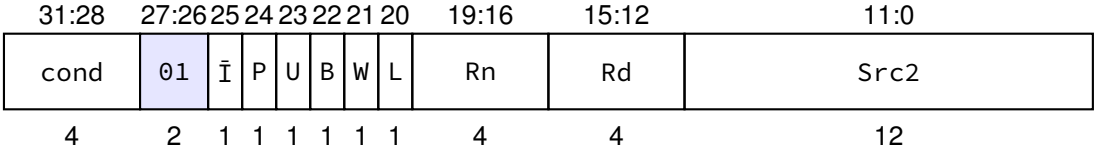
- `ldr r1, [r0]`
- `ldrb r1, [r0, #-15]`
- `str r1, [r0, #4]!`
- `strb r1, [r0], r3`

The reader should be able to describe the characteristics of each of the above instructions, paying particular attention to what happens to the *base register* `r0`.

### 9.4.1 Memory Instruction Layout

We first define the layout of the memory instruction. As you can see, the first 4-bits are the same as the data processing insructions, and bits `[27:26]` are hard-coded by the assembler to `01` which signifies a memory instruction.

Bits [25:20] contain the key characteristics of the memory instruction.




## Defining Ī, P, U, B, W, L

We will review each of the bits from [25:20] in turn here:

- Ī The not *I*mmediate bit. Wen $\bar{I} = 1$, an immediate is not present. When $\bar{I} = 0$, an immediate is present. See Section 9.4.1 for a detailed discussion on how Source 2 is used here. Where neither an immediate or register is used, *the assembler treats this as immediate #0*
- P The *P*re-index bit. There are two classes of pre-index: (a) offset and, (b) pre-index. Note, offset addressing is considered a form of pre-index addressing in that the final address is computed *before* memory is read or written.
- U *U*pward or downword bit. Used to define if the offset register or immediate value (where present) is subtracted or added to the base register. For subtraction, $U = 0$ for addition, $U = 1$
- B The *B*yte bit. Where $B = 0$ a word of memory is read or written from/to memory. Where $B = 1$ and the instruction is a type of `ldr`, the 8 least significant bits, (LSBs) are read from memory, zero extended to 32-bits, and placed into the destination register. When $B = 1$ and the instruction is a type of `str`, the 8 LSBs from the source register are placed into the memory and right aligned to the LSB. The remaining left 3 bytes of the memory address are unchanged.
- W The *W*riteback bit is used to differentiate between the offset and pre-index addressing modes. In the offset mode, $P = 1 \wedge W = 0$. In the pre-index mode, $P = 1 \wedge W = 1$. In the post-index mode, $P = 0 \wedge W = 0$. Note - there is no combination $P = 1 \wedge W = 1$.
- L The *L*oad register bit. Where $L = 0$ this is a store register instruction. When $L = 1$, it is a load register instruction.

## Examples of Ī, P, U, B, W, L

We can now briefly the examples from Section 9.4 and show the bit pattern for [25:20]:

- `ldr r1, [r0]`: `011000`. An especially interesting example. Note the bit pattern. It is the *same* bit pattern as `[r0, #0]`. Is this ambguous? No, because the assembler *always* converts `[r0, #0]` into `[r0]` and treats this as an immediate present, value 0.
- `ldrb r1, [r0, #-15]`: `010101`.
- `str r1, [r0, #4]!`: `011010`.
- `strb r1, [r0], r3`: `101100`. Another intersting pattern. Notice $U = 1$ regardless of the sign of the value in `r3`. [2] If `r3` is positive then we have the addition of two positive numbers, and if its negative, we have the addition of a positive and a nega-

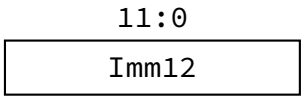
[2] not to be confused with `-r3` in which case $\bar{I} = 1$ and $U = 0$

tive, both cases are handled in the ALU, not here. The ALU and Microarchitecture topics are covered in more detail in Chapter 10

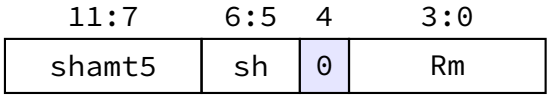
## Defining Src2

Finally, we should complete our look at memory instructions by looking at some examples of how Src2, `[11:0]` is encoded. In the case where $\bar{I} = 0$, this field is an unsigned 12-bit number. Therefore, the maximum addressable offset from the base register is $\pm\ 2^{12}$ bytes, which is $\pm\ 4096$ bytes or 4 megabytes of addressable offset forward or backward from the base register. Is this an upper bound on the size of an array in ARM32? No. We can easily assign the a new vaule to base register. For example, suppose we have a 12mb array. We can access positions $0, 4096, 8192$ as follows:

```
mov r5, #4096  // base register offset
ldr r1, [r0]   // access position 0
// ... use the r1 value
add r0, r0, r5 // jump to 4096
ldr r1, [r0]   // access position 4096
// etc ...
```

1. 

Example: `ldr r5, [r0], #4`, $\bar{I} = 0 \therefore$ **"Immediate"** variant:

2. 

Example: `ldr r5, [r0], r2`, $\bar{I} = 1$ and Immediate Shift $\therefore$ **"Immediate Shifted Register"** variant

## Examples of Source 2 Encodings

We will work through some examples of Source 2 to show `[11:0]` is encoded for various types of memory instructions. We will use the examples from Section 9.4 as the basis, but only discuss the `[11:0]` encoding for each instruction. We will then conclude with the complete instruction encoding for each of the examples.

- `ldr r1, [r0]`: $\bar{I} = 0$ - this is the **"Immediate Shifted Register"** variant. `Rm` = `r0` with `[11:0]` = `000000000000` as there is no shift. Therefore, the full instruction encoding is: `1110 01 011000 0000 0001 000000000000` ($e5901000$)
- `ldrb r1, [r0, #-15]`: $\bar{I} = 0$ - this is the **"Immediate"** variant. The immediate value is $15$ which is `000000001111` as an unsigned 12-bit number. Therefore, `[11:0]` = `000000001111`. The full instruction encoding is: `1110 01 010101 0000 0001 000000001111` ($e550100f$)
- `str r1, [r0, #4]!`: $\bar{I} = 0$ - this is the **"Immediate"** variant. The immediate value is $4$ which is `000000000100` as an unsigned 12-bit number. Therefore, `[11:0]` = `000000000100`. The full instruction encoding is: `1110 01 011010 0000 0001 000000000100` ($e5801004$)

- `strb r1, [r0], r3`: $\bar{I} = 1$ - this is the **"Immediate Shifted Register"** variant. `Rm` = `r3` with `[11:0]` = `000000000011` as there is no shift. Therefore, the full instruction encoding is: `1110 01 101100 0000 0001 000000000011` ($e6c01003$)

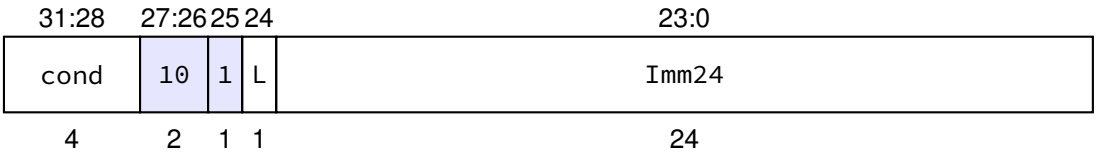
## A note on register naming

We should mention that the special register `sp` is just an alias for `r13`. Indeed, this is not the only special register. `r14` is also known as `lr` (link register) and `r15` is also known as `pc` (program counter) are also special registers. When `sp` is used in an instruction, the assembler will encode it as `r13`. For example, the instruction `ldr r5, [sp]` will be encoded as `e59d5000` where `r13` is used in place of `sp`. This is because the assembler recognizes `sp` as an alias for `r13` and encodes it accordingly. Therefore, when we see `sp` in an instruction, we can understand it as referring to `r13` in the binary encoding.

This is also true for `lr` and `pc`, which are aliases for `r14` and `r15` respectively. The assembler will encode these special registers as their corresponding general-purpose registers in the binary encoding of the instruction.

### 9.4.2 Branch Instruction Layout

Finally, we will look at the *branch* instruction class. This class consists of a single instruction, the branch instruction `b`, with some variants such as `bl` (branch with link) and conditional branches (for example, `beq`, `bne`, etc.). We first define the layout of the branch instruction. As you can see, the first 4-bits are the same as the data processing instructions, and bits `[27:25]` are hard-coded by the assembler to 101 which signifies a branch instruction:




## Defining `L` and `Imm24`

1. `L`: The *L*ink bit. Where $L = 0$ this is a plain branch instruction. When $L = 1$, it is a branch with link instruction, which means the return address (the address of the instruction following the branch) is stored in the link register `lr` (`r14`).
2. `Imm24`: The immediate field for the branch instruction. It is a signed 24-bit value that encodes the branch offset relative to the architectural program counter. For an ARM32 branch instruction, the PC value used in the calculation is the address of the current instruction plus 8. The target address is calculated as

   $$\text{Target Address} = (\text{address of current instruction} + 8) + (\text{SignExtend}(\texttt{Imm24}) \ll 2)\,.$$

   The left shift by 2, or multiplication by 4, occurs because ARM instructions are word-aligned and each instruction is 4 bytes. Since `Imm24` is signed, the branch range is approximately

   $$\pm 2^{23} \times 4 = \pm 2^{25} \text{ bytes},$$

   or about $\pm 32$ MB from the PC value used by the instruction.

### Examples of `Imm24`

`Imm24` is a signed 24-bit value that represents the branch offset from the architectural PC value to the target instruction. For an ARM32 branch instruction, the PC value used in the calculation is the address of the branch instruction plus 8. The offset can be either positive or negative, depending on whether the target instruction is located after or before the branch instruction in memory.

For example, if the branch instruction is located at address `0x00001000` and the target instruction is located at address `0x00002000`, then the byte offset is `0x00002000 - (0x00001000 + 0x00000008) = 0x00000FF8`. Because ARM branch offsets are encoded in words rather than bytes, this byte offset is divided by 4 before being stored in `Imm24`. Therefore, the encoded value is `0x00000FF8 / 4 = 0x000003FE`. This value is encoded in `Imm24` as a signed 24-bit integer.

Here is an example of a branch instruction that will result in imm24 being encoded with a positive 3:

```
b target

mov r1, #10
add r1, r1, #1
sub r1, r1, #2
mov r2, r1

target:
  mov r0, #0
```

Of course, in the above example, the 4 instructions following the branch are unreachable, and will never be executed. This is because the branch instruction is unconditional and will always jump to the label `target`, skipping over the instructions in between. [3]

In the next example, we show a branch back to a previous instruction, which will result in `Imm24` being encoded with a negative value:

```
loop:
  mov r1, #10
  add r1, r1, #1
  sub r1, r1, #2
  mov r2, r1
  b loop
```

Again, note that this is an infinite loop, and the instructions continuosly execute unless interrupted. We would never really write code like this, but it is useful to illustrate the method by which `Imm24` is calculated for a backward branch. [4]

## 9.5 Chapter Summary

In this chapter we examined how ARM assembly language instructions are represented internally as fixed-width 32-bit binary values. Although programmers normally write

[3] Realistically, a compiler or assembler would likely optimize this code by removing the unreachable instructions, or the programmer would restructure the code to avoid having unreachable code.

[4] In practice, a programmer would typically use a loop counter or conditional branch to exit the loop after a certain number of iterations, rather than creating an infinite loop.

textual assembly instructions, the processor executes only their encoded binary representation. The assembler is therefore responsible for translating each assembly instruction into its corresponding machine code representation.

A motivating example demonstrated that every instruction occupies exactly one 32-bit field, and that a sequence of assembly instructions ultimately becomes a sequence of 32-bit hexadecimal or binary values in the executable program. This fixed-width representation greatly simplifies instruction fetch and decoding, but also places strict limits on the amount of information that can be encoded within a single instruction.

To make instruction encoding manageable, ARM groups instructions into three broad classes:

- Data processing instructions, which perform arithmetic, logical, comparison, and register movement operations.
- Memory instructions, which transfer data between registers and memory using a variety of addressing modes.
- Branch instructions, which alter the normal sequential flow of execution.

For data processing instructions, we saw that the available 32 bits must encode information such as the operation to be performed, the source and destination registers, condition information, and any immediate values or operand modifiers. Because the available space is limited, careful design choices are required to balance expressiveness against encoding size.

Memory instructions follow a different structural pattern, allocating bits to represent load or store operations together with addressing information such as base registers, offsets, and indexing modes. Despite serving a different purpose, they obey the same fundamental constraint that every instruction must fit within a single 32-bit encoding.

Branch instructions use yet another encoding format, dedicating much of the available space to representing the displacement between the current instruction and the branch target. This allows efficient implementation of program control flow while maintaining the fixed instruction size characteristic of the ARM architecture.

Finally, the chapter demonstrated that machine code is not mysterious or arbitrary. Once the structural patterns associated with each instruction class are understood, the relationship between textual assembly language and its binary representation becomes systematic and predictable. This understanding provides valuable insight into assembler operation, executable programs, and the overall design of RISC instruction sets.

## 9.6 Exercises

**Exercise 9.6.1 (Instruction Classes)** *For each of the following instructions, identify whether it is a* data processing, memory, *or* branch *instruction.*

```
mov r0, #5
add r1, r2, r3
ldr r4, [r5]
str r6, [r7, #8]
cmp r1, r2
b loop
bl printf
```

**Exercise 9.6.2 (Hexadecimal to Binary)** *Convert the following hexadecimal values into their 32-bit binary representations.*

```
e3a01000
e3a0200a
e0813002
e1510003
```

**Exercise 9.6.3 (Machine Code Investigation)** *Write the following program and assemble it.*

```
mov r1, #0
mov r2, #10
add r3, r1, r2
cmp r3, #20
blt loop
```

*Use your assembler or disassembler to determine the hexadecimal machine code generated for each instruction.*

**Exercise 9.6.4 (Reverse Engineering)** *Using an assembler, disassembler, or simulator, determine the assembly instructions corresponding to the following machine code.*

```
e3a01000
e3a0200a
e1510002
bafffffb
```

*Verify your answers by reassembling the decoded instructions.*

**Exercise 9.6.5 (Comparing Instruction Formats)** *Select one data processing instruction, one memory instruction, and one branch instruction from your own programs.*

*For each instruction:*

- *Record its hexadecimal representation.*
- *Convert it to binary.*
- *Compare the layouts.*
- *Identify any obvious fields that differ between the three instruction classes.*

*Comment on the similarities and differences that you observe.*

**Exercise 9.6.6 (Encoding Constraints)** *Explain why every ARM instruction must fit into a single 32-bit word. Discuss how this restriction affects:*

- *the size of immediate values,*
- *the number of register fields,*
- *the complexity of instruction formats, and*
- *the overall design of the instruction set.*

**Exercise 9.6.7 (Counting Instructions)** *Suppose an ARM program contains 250 instructions.*

1. *How many bytes of machine code are required to store the instructions?*
2. *Approximately how many kilobytes is this?*
3. *If the same program contained 500 instructions, how much storage would be required?*

**Exercise 9.6.8 (Assembler Exploration)** *Write a short ARM program containing:*

- *two* `mov` *instructions,*
- *one arithmetic instruction,*
- *one comparison instruction,*
- *one conditional branch, and*
- *one unconditional branch.*

*Assemble the program and record both the assembly listing and the generated hexadecimal machine code. Compare your results with those of another student.*

# 10

# Single- and Multi-Cycle Microarchitecture

## 10.1 Overview

Every processor executes the same fundamental sequence of events: fetching an instruction, decoding it, executing it, and saving the result. A **Microarchitecture** is the internal hardware organization that implements the instruction set architecture. From the pedagogical point of view, and for the sake of simplicity, the study of microarchitecture begins with **Single-Cycle Microarchitecture**. Later in the chapter we will refine this model into the **Multi-Cycle Microarchitecture**.

These two approaches are briefly contrasted here:

1. The **Single-Cycle Microarchitecture**, in which each instruction takes the same amount of time to complete, and the clock cycle is the length of the longest instruction path.
2. The **Multi-Cycle Microarchitecture**, in which each instruction is broken into discrete stages, each stage taking one or more clock cycles.

Both designs implement the same ISA, but they trade off simplicity, speed, and hardware utilization in different ways. This chapter explores both approaches in detail, focusing first on a clear and intuitive undergraduate understanding.

## 10.2 From ISA to Microarchitecture

An **Instruction Set Architecture (ISA)** is the formal contract between software and hardware. It specifies *what* the processor must do when a well-formed instruction executes, without prescribing *how* the processor is built internally. In other words, the ISA is the interface that assembly programmers, compilers, and operating systems target; microarchitectures are free to implement the ISA in many different ways (single-cycle, multi-cycle, pipelined, out-of-order), as long as they preserve the externally visible behavior.

At the heart of an ISA is the *architectural state*: the set of elements a programmer can reason about or observe:

- A defined set of **registers** (general-purpose and special-purpose such as `pc`, `sp`, and status/condition flags).
- The **memory address space** and how instructions and data reside in it.
- **System-visible** registers or state (e.g., control/status registers, exception vectors) that affect traps, interrupts, or privilege.

When an instruction completes, the ISA says exactly how this architectural state may

change.

However, the ISA does *not* require a particular microarchitecture. It says nothing about pipeline depth, forwarding paths, cache sizes, or whether instructions complete in one or many cycles. Nor does it require a specific physical layout, timing, or power-management scheme. All of those are implementation choices hidden beneath the architectural contract.

## 10.3 What the Microarchitecture Implements

If the Instruction Set Architecture (ISA) defines *what* a processor must do, the **microarchitecture** defines *how* it is actually built to do it. It represents the internal organization and flow of data within the CPU, transforming instructions into sequences of hardware operations. While the ISA is fixed and visible to software, the microarchitecture is an engineering design choice that can vary widely among implementations of the same ISA.

### 10.3.1 Hardware building blocks.

A microarchitecture is constructed from three broad categories of hardware components:

- **Storage elements** such as registers, latches, flip-flops, and memory arrays. These components hold the state of the machine: the current instruction, operands, intermediate results, and the program counter.
- **Combinational logic** such as adders, multiplexers, decoders, shifters, and Arithmetic Logic Units (ALUs). These perform the computations, comparisons, and selections that process the stored data.
- **Control logic**, typically implemented as finite state machines or microprograms, that coordinates the movement of data and the sequencing of operations in time.

### 10.3.2 Freedom of implementation.

Two processors may implement the same ISA yet differ dramatically in their microarchitecture. For example:

- A simple educational RISC processor may use a **single-cycle** datapath: each instruction runs from start to finish in one long clock cycle.
- A higher-performance implementation may be **pipelined**, overlapping several instructions so that one completes on every clock tick.
- A superscalar design may contain multiple ALUs, multiple pipelines, and out-of-order execution logic that rearranges instructions dynamically to maximize throughput.

Despite these differences, all of these implementations produce the same observable results, because they obey the same architectural rules defined by the ISA.

### 10.3.3 Goals and trade-offs.

Microarchitecture design involves balancing conflicting objectives:

- **Performance:** how many instructions the processor can execute per second,
- **Area:** the amount of silicon (and thus cost) consumed,
- **Power consumption:** the energy efficiency of the implementation.

Designers can choose simpler or more complex datapaths, deeper or shallower pipelines, and various caching or control techniques, depending on these constraints. The study of single-cycle and multi-cycle designs provides a foundation for understanding these trade-offs before progressing to advanced architectures.

## 10.4 The Instruction Execution Cycle

Regardless of its internal complexity, every processor follows the same general pattern when executing instructions. This sequence, often called the **instruction execution cycle**, divides the process into smaller, logical stages that reveal the flow of information through the datapath.

### 10.4.1 Stage 1: Fetch (F)

The processor begins by reading the next instruction from memory. The address of this instruction is stored in the **Program Counter (PC)**. The fetched instruction bits are loaded into an *instruction register*, and the PC is incremented (usually by 4 bytes in a 32-bit RISC architecture) so that it points to the next sequential instruction. In case of a branch or jump, the PC may later be updated with a new target address.

### 10.4.2 Stage 2: Decode (D)

The binary pattern of the instruction is examined by the control logic to determine what type of operation it specifies. At the same time, the **register file** is accessed to read the values of the source operands identified by the instruction's register fields. This stage translates "what to do" and "what data to use" into actual signals and data paths within the CPU.

### 10.4.3 Stage 3: Execute (E)

The operation is performed in the Arithmetic Logic Unit (ALU). Depending on the instruction, this stage may:

- Perform arithmetic or logical computation (e.g., addition, subtraction, AND, OR),
- Compute an effective address for a memory operation (e.g., base + offset),
- Evaluate a branch condition by comparing two register values.

The output of the ALU is an immediate numerical result or an address that will be used in the next stage.

### 10.4.4 Stage 4: Memory (M)

If the instruction requires interaction with memory, the calculated address from the EX stage is used here.

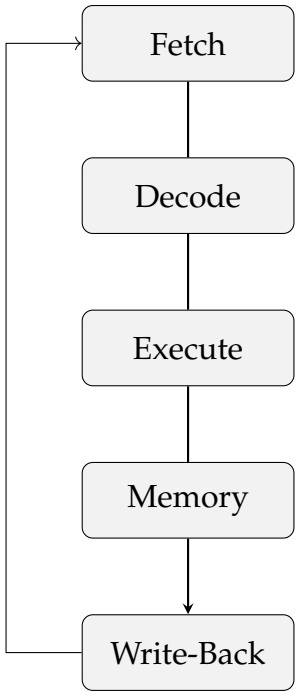
- For a `ldr` instruction, data is read from the data memory and temporarily stored in an internal register.
- For a `str` instruction, data from a register is written into memory at the computed address.

Instructions that do not access memory (such as arithmetic ones) simply bypass this stage.

### 10.4.5 Stage 5: Write-Back (W)

Finally, the result of the operation (either the ALU output or the data retrieved from memory) is written back into the destination register. This completes the instruction's lifecycle, returning the processor to a new stable state ready to begin the next fetch.

Together, these five stages define the flow of information through the processor. They separate the work of executing an instruction into manageable and reusable parts. The clear boundaries between stages allow engineers to reason about timing, optimize hardware reuse, and later extend the design into multi-cycle or pipelined implementations.




Note in the above diagram that the Write-Back stage feeds back into the Fetch stage, because the next instruction may depend on the result of the previous one. However, not every instruction will require all five stages. Some instructions (like branches or stores) may skip the Write-Back stage.

**▶ The 5-Stage Pipeline Live**

PlayARM animates exactly the five stages above on an interactive canvas. Open PlayARM, enter the program below, and press **Step** (→) once per clock cycle. Watch each instruction bubble move from **F** → **D** → **E** → **M** → **W** across the pipeline canvas.

```
mov r0, #3
mov r1, #7
add r2, r0, r1
sub r3, r1, r0
```

Check the **Pipeline Activity** panel on the right to see which instruction occupies each stage at every cycle. Notice that once the pipeline is full, a new result is produced every cycle.

**Historical note.** This five-stage model originates from early RISC architectures developed in the 1980s (such as MIPS, SPARC, and ARM), where clarity and simplicity of data movement were primary goals. Even today, these same conceptual stages remain the organizing principle for most CPU designs, from simple teaching processors to complex out-of-order superscalar cores.

**Conceptual importance.** For undergraduate students, understanding this model is essential. It not only clarifies how instructions progress through hardware, but also builds intuition for timing, performance bottlenecks, and later topics such as pipelining and hazard management. Whether a CPU is built as a single-cycle or a deeply pipelined machine, these five stages are the unchanging rhythm of computation.

## 10.5 Part I: Single-Cycle Microarchitecture

The single-cycle microarchitecture executes every instruction in one clock cycle. This section develops the datapath and control logic required to achieve this before examining individual functional units.

The **single-cycle microarchitecture** represents the simplest complete hardware organization capable of running a full instruction set. Its defining idea is that *each instruction* (from fetching in memory to writing the final result back into a register) is completed entirely within one clock period.

- Every instruction consumes exactly one clock cycle, so the **cycles per instruction (CPI)** equals 1.
- The clock period, however, must be long enough to accommodate the *slowest* instruction in the ISA.

This design is elegant for learning because all the moving parts of a processor can be seen operating once per instruction. The concept helps students visualize how data flows through the datapath, how control signals coordinate hardware components, and how an instruction's five stages connect. Yet this same simplicity introduces performance limitations: the entire processor must wait for the slowest possible instruction to finish before the next instruction begins.

**The "race of the slowest" effect.** Imagine three instructions:

- An `add` that only needs the ALU,
- A `ldr` that uses both the ALU and data memory,
- A `b` that also computes and tests a target address.

Because the `ldr` instruction touches more hardware and requires more propagation delay, the clock period must be long enough for it to complete safely. As a result, simpler instructions waste potential performance, idly waiting through that same long cycle.

Despite this inefficiency, the single-cycle model is a crucial foundation. It shows in one unified picture how the instruction fetch, decode, execute, memory, and write-back stages interconnect (concepts that remain valid for all modern CPU designs, regardless of later optimizations).

**Decoding is wiring, not computation.** Because every field occupies a fixed bit range, the hardware simply *connects* those wires to the appropriate inputs, no arithmetic is needed. The register-address fields (`Rn`, `Rd`, `Rm`) are connected directly to the address ports of the register file; the `Op/Funct` bits are connected to the control unit's lookup logic. Decoding in a fixed-length RISC design happens in one gate delay, which is one of the key reasons RISC architectures can sustain high clock frequencies.

## 10.6 Microarchitecture Components

In order to understand the single-cycle microarchitecture, we will first examine the components that make up the *datapath*. The datapath is the collection of hardware blocks that are used to implemet the instruction execution cycle. The datapath does not include the *control logic*, which is a separate set of circuits that generates the control signals for each instruction. The control logic is grouped into a specialized component called the *control unit*, which we will discuss in Section 10.6.5.

For each instruction, the datapath specifies how data flows through the processor. It describes only the movement and transformation of data, not the control signals that coordinate the processor's operation.

### 10.6.1 Harvard and Von Neumann Memory Models

Before examining the datapath, it is useful to distinguish between the two principal memory organizations used in computer systems: the von Neumann and Harvard architectures. They differ primarily in how instructions and data are accessed by the processor:

**Von Neumann architecture** Instructions and data share a common memory interface and address space. This means that the processor cannot fetch an instruction and read/write data simultaneously.

**Harvard architecture** Instructions and data use independent memory interfaces, enabling simultaneous access. This can improve performance, especially in pipelined designs, because the processor can fetch the next instruction while reading or writing data.

The core feature of the Harvard architecture is having physically separate memory spaces (or separate paths/caches) for instructions and data. Because the pathways are separate, the processor can fetch a new instruction and read or write data in memory at the exact same clock cycle without creating a bottleneck.

RISC architectures rely heavily on high instruction throughput and pipelining (aiming to execute one instruction per clock cycle). Harvard architecture — or modified Harvard architecture (using separate Instruction and Data L1 caches) — is standard in modern RISC processors (like ARM and RISC-V) to keep the pipeline moving without data access halting the instruction fetch step.

### 10.6.2 Datapath Components

In this section we define and then illustrate the major components of the datapath:

**Instruction Memory** This is read-only memory that contains the program instructions. For simplicity, we assume that the instruction memory is separate from the data memory, (following the Harvard architecture). But in practice, many processors

use a unified memory for both instructions and data, with separate access paths. On the left side of this block we see a 32-bit address input, (A) and on the right side we see a 32-bit instruction output (RD). The input is driven by the program counter, and the output is the instruction word that is fetched from memory.

**Data Memory** This is read/write memory that stores program data. For example, the stack, heap, and global variables reside in data memory. Therefore, there are two input lines to this block: a 32-bit address input (A) and a 32-bit write-data input (WD). There is one output line: a 32-bit read-data output (RD). The address input is the output line from the ALU. It's important to note that data memory is not accessed for every instruction. Even if it is accessed, it may be for a read or a write, but not both. The control unit generates the appropriate signal to govern this behaviour.

**Register File** This is a collection of registers that can be read and written by the processor. For the sake of simplicity, we assume that the register file has two read ports and one write port. [1]

The read ports are A1 and A2, are 4-bit address inputs that select which registers to read. The outputs of the read ports are 32-bit read-data outputs (RD1 and RD2). The register file has a 4-bit write address input (A3), a 32-bit write-data input (WD3), and a write-enable input (WE3). Both A3 and WD3 are considered together as the write port of the register file. WE3 is controlled by the control unit, which asserts it when an instruction writes to the register file. We will discuss this in more detail in the next section.

**ALU** This is the arithmetic logic unit, which performs arithmetic and logical operations on depending on the input signal asserted by the control unit. The ALU has two 32-bit input sources (`srcA` and `srcB`). It also has an input from the control unit that specifies which operation to perform. The ALU has a 32-bit output (`ALUResult`) line.

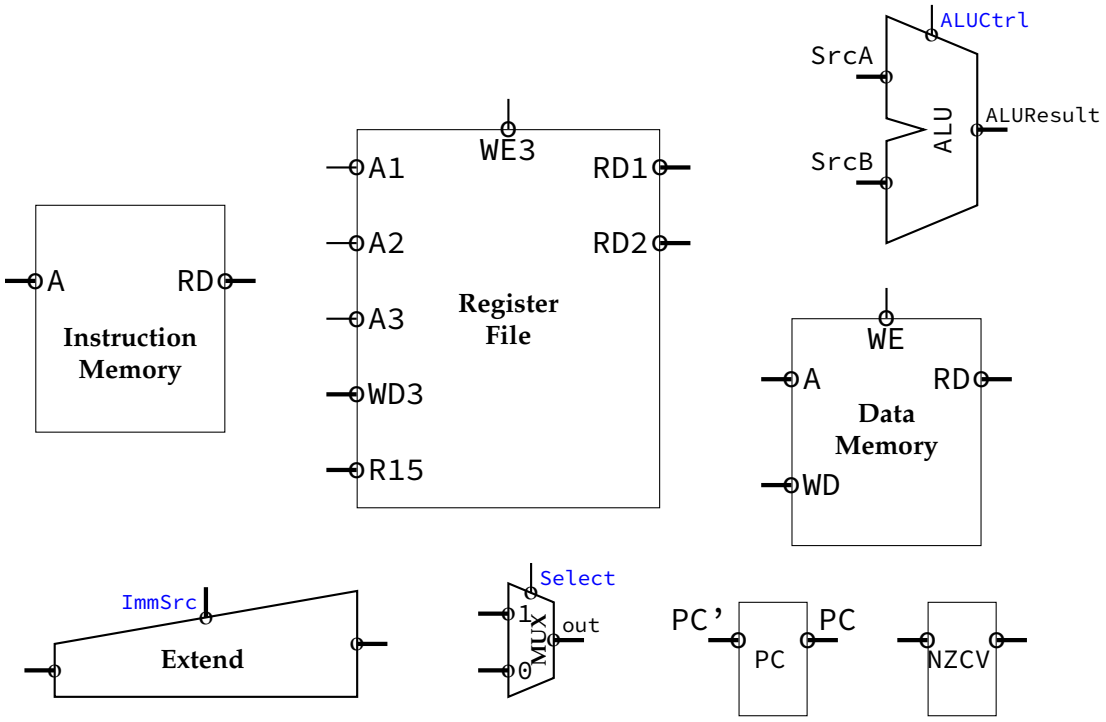
**Extend** This block takes a 12-bit immediate value from the instruction (src2) and extends it to a 32-bit value. The extension can be either sign-extension or zero-extension, depending on the instruction type. The output of this block is used as an operand for certain instructions, such as immediate arithmetic or load/store operations.

**Multiplexer (MUX)** The multiplexer (MUX) is a combinational circuit that selects one of several input signals and forwards the selected input to a single output line. In the diagram shown here, we see that the multiplexer has two 32-bit input lines, a *select* input line, and a 32-bit output line. In the example shown, the multiplexer selects between `0` and `1` based on the value of the select line. A multiplexer can have more than two inputs, but in this case, only two are shown.

**Program Counter (PC'/PC)** The program counter is a special-purpose register that holds the address of the next instruction to be fetched from instruction memory. It is updated from the modified address (`PC'`) to the actual address (`PC`) at the end of each clock cycle.

[1] This simplification has the consequence that the register file can only read two registers per instruction, meaning that an instruction such as `str r8, [r0, r1]` is not possible to implement in this single cycle model.

**NZCV** The NZCV block represents the special-purpose register (CPSR) that holds the 4-bits of interest to the programmer: the condition flags. These are Negative (N), Zero (Z), Positive (V), and Carry (C). These flags are set by the ALU after each arithmetic or logical operation, and they can be used to control conditional execution of instructions (as long as the instruction supports condition flags such as `adds`). Refer to Chapter 4 for a detailed discussion of the condition flags.




### 10.6.3 Register File Ports in Detail

The register file diagram above uses a compact notation; each labeled wire carries a specific signal. Table 10.1 defines every port and maps it back to the instruction fields described above.

Table 10.1: Register File Port Definitions

| Port | Width | Direction | Meaning |
|---|---|---|---|
| `CLK` | 1 bit | Input | Clock; write occurs on the rising edge when `WE3` is asserted |
| `A1` | 4 bits | Input | Address of first read port; driven by `Rn` [19:16] from the instruction |
| `A2` | 4 bits | Input | Address of second read port; driven by `Rm` [3:0] or `Rt` depending on instruction type |
| `A3` | 4 bits | Input | Address of the write port; driven by `Rd` [15:12] (or `Rt` for a load) |
| `WD3` | 32 bits | Input | Write data; the value to be stored into register `A3` |
| `WE3` | 1 bit | Input | Write enable; asserted by the control unit when the instruction writes a result |
| `RD1` | 32 bits | Output | Data read from the register addressed by `A1`; feeds ALU input 1 |
| `RD2` | 32 bits | Output | Data read from the register addressed by `A2`; feeds ALU input 2 or the store-data path |
| `R15` | 32 bits | Output | Always outputs the current value of `R15` (the PC); used for PC-relative loads and branch-link return addresses |

**How the ports connect to the datapath.**

- `RD1` is always routed to ALU input A.
- `RD2` reaches a multiplexer: when `ALUSrc` is asserted, the MUX selects the sign-extended immediate instead; otherwise `RD2` passes through.
- `WD3` comes from a second multiplexer: when `MemToReg` is asserted (a `ldr` instruction), the data read from memory is selected; otherwise the ALU result is selected.
- `WE3` is asserted for every instruction that modifies a register (data-processing and `ldr`), and de-asserted for `str` and branch instructions.
- `R15` is permanently tied to the PC register output so that instructions such as `ldr PC, [R0]` or `mov lr, pc` can read the current program counter without a separate path.

The read ports (`RD1`, `RD2`) are *asynchronous*: they respond to address changes combinationally within the same cycle. The write port (`WD3` via `A3`) is *synchronous*: it latches the new value only on the rising clock edge and only when `WE3 = 1`. This means a register can be read and written in the same cycle, but the new value is not visible until the *next* cycle, a deliberate choice that preserves the single-cycle semantics.

A minimal single-cycle CPU is composed of a small but complete set of hardware blocks. Each block performs a specific role in the instruction's journey from memory to execution and back.

**Program Counter (PC):** A dedicated register that stores the address of the current instruction. After each instruction is fetched, the PC normally increments by 4 bytes

(for a 32-bit ISA such as ARM), thereby pointing to the next sequential instruction. In case of a branch or jump, control logic overrides this default update and loads a new address.

**Instruction Memory:** A read-only memory (or instruction cache) that outputs the 32-bit instruction stored at the address given by the PC. It represents the program itself in machine code form. In hardware, this may be implemented as a simple ROM, flash, or instruction cache line connected to the PC's output.

**Register File:** A fast storage array containing all the general-purpose registers defined by the ISA (typically 32 registers for ARM or MIPS-style designs). It allows two source operands to be read simultaneously (`rs1`, `rs2`) and one destination register to be written (`rd`) within the same clock cycle. The register file acts as the primary working area for arithmetic and logic operations.

**Arithmetic Logic Unit (ALU):** The computational heart of the datapath. It performs integer arithmetic (addition, subtraction, multiplication by shifting) and logical operations (AND, OR, XOR, NOT). The ALU receives its inputs directly from the register file or from an immediate value embedded in the instruction, depending on control signals. Its output is either a final result to be written back or an address used to access data memory.

**Data Memory (DMEM):** A separate memory module that stores program data (variables, arrays, the stack). Its interface exposes four signals:

- `A`, the 32-bit byte address, produced by the ALU.
- `WD` (write data), the 32-bit value to store, sourced from register-file output `RD2`.
- `WE` (write enable), asserted by the control unit for a `str` instruction; de-asserted for a `ldr`.
- `RD` (read data), the 32-bit value fetched from address `A`, routed to the `WD3` multiplexer of the register file.

Like the register file's write port, DMEM writes are synchronous (clocked), while reads are asynchronous. Separating instruction and data memory follows the **Harvard architecture** principle common in RISC designs, allowing simultaneous instruction fetch and data access in the same cycle.

**Control Unit:** The combinational logic that converts instruction bits into the set of boolean control signals that orchestrate every other component. Given the `Class` [27:26], `Op/Funct` [25:20], and the condition flags from the ALU, the control unit asserts or de-asserts each of the following signals:

| Signal | Effect when asserted |
|---|---|
| `RegWrite` | Enables write to register file (`WE3`) |
| `MemWrite` | Enables write to data memory (`WE` on DMEM) |
| `MemToReg` | Selects memory read data as write-back source (vs. ALU result) |
| `ALUSrc` | Selects sign-extended immediate as ALU second operand (vs. `RD2`) |
| `RegSrc` | Selects which instruction field drives `A2` (e.g., `Rm` vs. `Rt`) |
| `ImmSrc` | Selects the immediate extension scheme (8-bit rotated vs. 12-bit offset) |
| `ALUControl` | 2–3 bit code telling the ALU which operation to perform |
| `PCSrc` | Overrides the default PC+4 update with a branch target |

Because these signals are generated combinationally in a single-cycle design, the control unit is simply a *lookup table*: for each instruction encoding, a fixed pattern of outputs is hard-wired. There is no feedback, no clock, and no state, the signals stabilize within one gate delay of the instruction word arriving.

**How these blocks interact.** Each component plays its part once per cycle:

1. The `pc` supplies an address to the instruction memory.
2. The instruction memory outputs the corresponding instruction word.
3. The control unit interprets this instruction and sets the control lines.
4. The register file outputs operands for the ALU.
5. The ALU computes either a data result or an effective address.
6. The data memory is accessed if required.
7. The result is written back into the register file, and the PC updates for the next fetch.

### 10.6.4 Dataflow of an Instruction (Example)

To understand how all of the processor's blocks work together, it helps to follow one instruction as it travels through the datapath. Consider a simple arithmetic instruction in a RISC architecture such as ARM or MIPS:

```
add r3, r1, r2
```

This instruction adds the contents of register `R1` and `R2`, and stores the result in `R3`. Even though it looks simple, it touches nearly every major part of the datapath. Let's examine each action, step by step, as it occurs during the single clock cycle.

1. **Instruction Fetch (IF).** The **Program Counter (PC)** holds the address of the next instruction. At the beginning of the cycle, the PC's output is sent to the **Instruction**

**Memory**, which retrieves the 32-bit binary encoding of `add R3, R1, R2`. The fetched instruction is passed simultaneously to the **Control Unit** for decoding and to various parts of the datapath that will soon need its fields.

2. **Instruction Decode and Register Read (ID).** The **Control Unit** interprets the opcode bits and recognizes that this is an `add` instruction (an R-type operation requiring two register operands and one destination register). From the instruction fields, it extracts:

   - `rs1` = R1 (first source),
   - `rs2` = R2 (second source),
   - `rd` = R3 (destination).

   The **Register File** simultaneously outputs the values contained in R1 and R2 onto two internal buses. These become the ALU's inputs for this cycle.
3. **Execute (EX).** Guided by the control signals, the **ALU** is configured to perform an addition operation. It receives the two operand values from the register file and computes the sum:

$$\texttt{ALU\_Result} \leftarrow \texttt{R1\_Value} + \texttt{R2\_Value}.$$

   The resulting value now exists on the ALU's output bus, ready to be written back. Because this instruction is purely arithmetic, no memory access is required.
4. **Memory Access (MEM).** Since the `add` instruction does not involve a load or store, the data memory remains idle during this part of the cycle. Still, the datapath physically includes this stage because other instructions, such as `ldr` or `str`, will use it to interact with data memory.
5. **Write-Back (WB).** The output from the ALU is sent to the **Register File**'s write port. Control signals assert the `RegWrite` line, and the destination register field (`rd` = R3) determines which register receives the value. On the next rising clock edge, R3 is updated with the sum computed by the ALU.
6. **PC Update.** In parallel with the above operations, the PC's incrementer calculates the address of the next instruction:

$$\mathsf{PC}_{\text{next}} = \mathsf{PC}_{\text{current}} + 4.$$

   Unless a branch or jump overrides it, this new value is loaded into the PC at the end of the cycle, ensuring continuous sequential execution.

All six of these actions occur within one clock period. There are no intermediate registers to pause or pipeline the work; data flows smoothly through the combinational hardware of the datapath. This one-cycle "heartbeat" makes the design conceptually easy to trace but electrically demanding: the clock must be slow enough for the longest possible path (from PC through instruction memory, decoding, register file, ALU, and data memory) before the next rising edge arrives.

**Key Observation**

In a single-cycle processor, **the clock period is determined by the worst-case instruction path**. For example, a `LOAD` instruction must:

1. Fetch the instruction from memory,
2. Decode and read registers,
3. Compute an effective address in the ALU,
4. Access the data memory,
5. Write the loaded value back to the register file.

Because all of this happens in one cycle, the clock must be long enough to accommodate this slowest path. Consequently, even simple arithmetic instructions like `add` must wait through that same long period, limiting the processor's overall performance.

### 10.6.5 Control Unit Overview

## 10.7 Part II: Multi-Cycle Microarchitecture

### 10.7.1 Motivation

The single-cycle microarchitecture, while conceptually elegant, suffers from a fundamental performance limitation: the clock period must be long enough for the *slowest* instruction in the instruction set. This means that even the simplest operations are forced to wait through an unnecessarily long cycle. The processor performs many operations sequentially within that one period (fetching, decoding, computing, accessing memory, and writing results) when most of those steps do not need to occur simultaneously.

The key idea behind the **multi-cycle microarchitecture** is to divide the work of executing an instruction into several smaller steps, each performed in its own clock cycle. By doing so, each cycle now contains fewer operations, the combinational delay per cycle becomes much shorter, and the clock can run at a higher frequency. Instead of completing one instruction in one long cycle, we complete it over several short cycles.

Single-cycle: One long step per instruction $\Rightarrow$ Multi-cycle: Several short steps per instruction.

The total time per instruction might still be comparable, but the hardware is used more efficiently and scales better with complex operations.

In the single-cycle design, every major unit (such as the ALU, the instruction memory, and the data memory) was active once per instruction. This required separate hardware blocks for each role: one memory for instructions and another for data, one adder for PC incrementing and another inside the ALU, and so on. In a multi-cycle design, the same hardware resources can be **reused** across different cycles of the same instruction.

For example:

- The **ALU** can first compute `PC + 4` during the fetch cycle and later compute an effective address for a `LOAD`.
- A single **memory unit** can be used both for fetching instructions (during one cycle) and for accessing data (during a later cycle).

This reuse greatly reduces hardware cost and power consumption, especially for teaching processors or small embedded designs.

By restricting each cycle to a smaller amount of logic, we can make the clock much faster. However, because different instructions may require different numbers of steps, the number of cycles per instruction (**CPI**) is no longer fixed at 1. A simple arithmetic instruction might take three cycles (fetch, decode, execute), whereas a load instruction might take five (fetch, decode, address calculation, memory access, write-back).

The average instruction time therefore becomes:

$$T_{\text{avg}} = \text{CPI}_{\text{avg}} \times T_{\text{cycle}},$$

where $T_{\text{cycle}}$ is now much shorter than in the single-cycle case, but $\text{CPI}_{\text{avg}} > 1$. The overall performance gain depends on whether the product of these two terms is smaller than the single-cycle clock period.

Suppose the single-cycle CPU has a long clock period of $850\,\text{ps}$ (set by the slowest load instruction). In a multi-cycle version, the work might be divided into five shorter steps, each with a cycle time of $250\,\text{ps}$. A load instruction might require all five cycles ($5 \times 250 = 1250\,\text{ps}$), but a simple add might finish in three ($3 \times 250 = 750\,\text{ps}$). While the slow instruction becomes slightly slower, the frequent short ones become significantly faster, improving average throughput.

Moving to a multi-cycle design introduces several new engineering considerations:

- The datapath must now include temporary **intermediate registers** to hold values between cycles (such as the instruction, operands, and ALU results).
- The **control unit** must evolve from a simple combinational decoder to a finite state machine (FSM) that sequences through the proper control signals cycle by cycle.
- Timing analysis must now consider per-cycle critical paths rather than a single end-to-end delay.

### 10.7.2 Additional Internal Registers

When execution of a single instruction is stretched across multiple clock cycles, the processor must retain intermediate results between cycles. In the single-cycle design, temporary values naturally flow through combinational hardware within one long cycle and are not stored anywhere. However, once execution is split into distinct short cycles, the datapath needs dedicated **intermediate registers** to "remember" partial results that will be used in later cycles.

**Purpose of internal registers.** These internal registers serve as short-term holding areas between the main stages of the instruction cycle. They decouple the combinational logic of one stage from the next, allowing the processor to reuse hardware components (such as the ALU or memory) over time without losing intermediate values. Effectively, they act as "checkpoints" between cycles, enabling the CPU to pause and continue the same instruction safely on the next clock edge.

The most common intermediate registers are described below.

- **IR (Instruction Register):** During the *fetch* cycle, the memory outputs the instruction word addressed by the Program Counter (PC). To avoid losing this instruction when the memory is reused for data access in later cycles, the instruction must be stored

in a dedicated register called the **Instruction Register (IR)**. The control unit reads the opcode and other fields from the IR in subsequent cycles to generate appropriate control signals. This separation allows the memory to be reused for data operations while the instruction remains safely latched.

- **MDR (Memory Data Register):** The **Memory Data Register** (sometimes called *Memory Buffer Register*) holds data that has been read from, or is about to be written to, the memory. When a `LOAD` instruction fetches a word from memory, the value first enters the MDR, from which it will later be written into the destination register. Similarly, during a `STORE`, the value to be written is placed into the MDR before the actual memory write occurs. This mechanism stabilizes the data and provides a clean interface between the CPU core and the memory subsystem.
- **A and B (Operand Registers):** In the multi-cycle design, the **Register File** is accessed only during the *decode* stage. Once the operands are read, they are stored in two temporary registers, traditionally named **A** and **B**. These hold the source operands for use in later cycles, particularly during execution or memory address calculation. By capturing operand values early, the ALU can perform operations in later cycles even when the register file is busy with other tasks.
- **ALUOut (ALU Output Register):** After the ALU performs a computation (such as evaluating an address for a load/store or producing the result of an arithmetic operation), the result is latched into the **ALUOut** register. This stored value then becomes available for use in subsequent cycles. For example, during a `LOAD`, the address calculated in one cycle (stored in ALUOut) will be used in the next cycle to access memory. Similarly, for arithmetic instructions, ALUOut holds the final result until the write-back stage.

**How these registers enable hardware reuse.** Without these intermediate registers, each instruction would need its own dedicated ALU and memory path active simultaneously. By introducing them, the same hardware can perform multiple roles in different cycles:

- In the first cycle, the ALU computes `PC + 4` (to advance to the next instruction).
- In the next cycle, the same ALU computes an effective address (`base + offset`) for a data access.
- Later, the same ALU might perform an arithmetic operation (`R1 + R2`).

Each intermediate result is stored temporarily (in ALUOut, IR, or MDR) until it is needed again. Thus, the addition of these small registers transforms the datapath from a "one-pass conveyor belt" into a flexible, time-multiplexed machine.

**Analogy.** An intuitive way to visualize this is to imagine a small workshop where only one worker (the ALU) and one toolbox (the memory) are available. To build something complex, the worker performs one step at a time, saving intermediate parts on the workbench (the internal registers) before moving to the next step. This strategy allows the same tools to be reused efficiently instead of buying new ones for every task.

**Design note.** These registers are not visible to programmers; they belong to the microarchitecture, not the ISA. They do not correspond to architectural registers like

`R0-R31`, but exist purely to coordinate timing and resource reuse. Their correct operation depends entirely on the sequencing logic of the control unit, which determines when each one should be loaded or read during instruction execution.

In summary, adding the **IR**, **MDR**, **A**, **B**, and **ALUOut** registers is what makes a multi-cycle design possible. They store the right pieces of information at the right times, enabling a single ALU and a single mem

### 10.7.3 State-by-State Operation

In the multi-cycle microarchitecture, the processor executes each instruction as a sequence of **states**, where each state corresponds to one short clock cycle. During a given state, specific control signals are asserted, allowing one well-defined operation (such as fetching an instruction, reading registers, or performing an ALU computation) to take place. At the end of each cycle, intermediate results are stored in internal registers (IR, A, B, ALUOut, MDR) to be used in the next state.

A typical controller for a simple RISC-style processor (such as MIPS or ARM-like subset) can be implemented with five key states. This model captures all the essential behavior of arithmetic, memory, and branch instructions, while demonstrating how hardware is reused efficiently from one cycle to the next.

**State 0 – Instruction Fetch (IF).** The processor begins every instruction by fetching its binary encoding from memory. The address used is the current value of the Program Counter (PC). During this cycle:

- The memory address input is connected to the PC.
- The memory performs a read operation, and the 32-bit instruction word is retrieved.
- The fetched instruction is latched into the **Instruction Register (IR)** so that it will be available even when the memory is later reused for data access.
- The ALU, operating in parallel, adds 4 to the current PC value, producing `PC + 4`. This result is stored back into the PC in preparation for sequential execution.

**Control summary:**

- Inputs: PC $\rightarrow$ Memory address bus
- Outputs: Memory data $\rightarrow$ IR
- Operations: ALU performs `PC + 4`; PC updated

This state uses both the memory and the ALU, but these components will later be reused for other functions. The IR now holds the instruction for subsequent decoding.

**State 1 – Instruction Decode and Register Read (ID).** With the instruction safely stored in the IR, the processor next determines what kind of operation is required. The control unit inspects the opcode and function fields, and simultaneously, the register file is read to obtain source operands.

- The two source registers specified in the IR (`rs1` and `rs2`) are read from the register file.
- Their values are stored into temporary registers **A** and **B**, respectively.

- If the instruction is a branch, the target address can be partially computed by adding the sign-extended immediate field to `PC + 4`. This helps reduce latency in the next state.

**Control summary:**

- IR opcode bits drive control decoding.
- Register file read enabled; A ← R[rs1], B ← R[rs2].

At the end of this state, the processor has decoded the instruction type and prepared all necessary operands for the execute stage.

**State 2 – Execution / Address Calculation (EX).** This state performs the core arithmetic or address computation required by the instruction. Different instruction types use the ALU for different purposes:

- **Arithmetic/Logic instructions (R-type):** The ALU executes the operation specified by the function field in the IR. Example:

$$\texttt{ALUOut} \leftarrow A \ \texttt{op} \ B,$$

  where `op` might be addition, subtraction, AND, OR, etc.
- **Load/Store instructions:** The ALU computes an effective address by adding a base register (A) to the sign-extended offset from the instruction. Example:

$$\texttt{ALUOut} \leftarrow A + \texttt{SignExt}(Immediate).$$

- **Branch instructions:** The ALU performs a subtraction (`A - B`) to check for equality. The Zero output of the ALU becomes a condition flag that determines whether the PC should be updated with the branch target address.

All of these operations reuse the same ALU hardware, demonstrating efficient resource sharing. The result of this computation is stored in **ALUOut**, ensuring it remains available for the next state.

**Control summary:**

- ALU operation selected based on opcode/function.
- Destination: ALUOut ← ALU result.

**State 3 – Memory Access (MEM).** In this state, the processor interacts with the data memory: either reading from it (for loads), writing to it (for stores), or updating the PC (for branches).

- **Load instruction:** The address stored in ALUOut is sent to the memory address bus. The memory performs a read, and the returned data word is stored in the **Memory Data Register (MDR)**.
- **Store instruction:** The data value stored in temporary register B is sent to the memory's data input, while ALUOut provides the target address. A write signal is asserted, causing the memory to store the value.

- **Branch instruction:** If the ALU's zero flag from the previous state is true (meaning the branch condition is satisfied), the PC is updated with the branch target address computed earlier. Otherwise, the PC retains its sequential value (PC + 4).

**Control summary:**

- For load: MemRead=1, MDR ← Mem[ALUOut].
- For store: MemWrite=1, Mem[ALUOut] ← B.
- For branch: Conditional PC update based on Zero flag.

At this point, the memory access or control flow change has been completed. Arithmetic instructions skip this stage because they do not involve data memory.

**State 4 – Write-Back (WB).** The final state of the instruction returns results to the architectural register file, making them visible to software. Depending on the instruction type:

- **R-type arithmetic/logic instructions:** The result previously stored in ALUOut is written to the destination register specified by the IR's `rd` field.

$$R[\texttt{rd}] \leftarrow \texttt{ALUOut}.$$

- **Load instructions:** The data fetched from memory, stored in the MDR during the previous cycle, is written back to the destination register:

$$R[\texttt{rd}] \leftarrow \texttt{MDR}.$$

No write-back occurs for store or branch instructions, since their results are either memory-side effects or PC updates.

**Control summary:**

- RegWrite signal asserted.
- Source of data selected: ALUOut (for ALU ops) or MDR (for loads).
- Destination register determined by `rd` field in IR.

After this state, the instruction is complete, and the control unit returns to State 0 to begin fetching the next instruction.

**Putting the states together.** These five states (IF, ID, EX, MEM, and WB) form the **control sequence** for all instructions. Each instruction type follows a different path through this state machine:

- Arithmetic instructions: `IF → ID → EX → WB`
- Load instructions: `IF → ID → EX → MEM → WB`
- Store instructions: `IF → ID → EX → MEM`
- Branch instructions: `IF → ID → EX → MEM (conditional PC update)`

Although each instruction requires multiple cycles, each cycle is short and efficient, and the same hardware is reused across states. This sequencing demonstrates the key virtue of the multi-cycle architecture: **doing more with less**, by carefully orchestrating time and control.

### 10.7.4 Advantages of the Multi-Cycle Approach

**1. Shorter Clock Period.** Each cycle includes only a portion of the datapath, so the clock period can be reduced dramatically.

**2. Hardware Reuse.** The same ALU and memory can be used across several cycles, saving area and cost.

**3. Flexibility.** Different instruction types can take different numbers of cycles.

**4. Foundation for Pipelining.** Splitting the instruction execution into well-defined steps sets the stage for overlapping those steps later.

### 10.7.5 Performance Comparison

Let $T_{\text{single}}$ be the long single-cycle clock period and $T_{\text{multi}}$ be the short multi-cycle period.

If the average number of cycles per instruction (CPI) is 4, and

$$T_{\text{multi}} = \frac{1}{3} T_{\text{single}},$$

then the average instruction time is:

$$4 \times \frac{1}{3} T_{\text{single}} = \frac{4}{3} T_{\text{single}}.$$

So even though each instruction takes more cycles, the shorter period nearly compensates. More importantly, this design scales better as instruction complexity increases.

### 10.7.6 Comparison Summary

Table 10.2: Single-cycle vs. Multi-cycle Comparison

| Feature | Single-Cycle | Multi-Cycle |
|---|---|---|
| Clock period | Long (slowest instruction) | Short (per stage) |
| Cycles per instruction | 1 | Variable (3–5 typical) |
| Hardware reuse | Limited | High (same ALU reused) |
| Control | Simple, combinational | FSM-based, more complex |
| Performance scaling | Poor | Better for complex ISAs |
| Hardware cost | Higher | Lower (shared resources) |

## 10.8 Putting It All Together

### 10.8.1 Evolution Toward Pipelining

Both architectures teach the same principle: instructions naturally divide into stages. In single-cycle, all stages happen sequentially within one period; in multi-cycle, stages occur over multiple periods; and in pipelining, stages overlap across multiple instructions.

Thus, multi-cycle design serves as the conceptual bridge between simple single-cycle and advanced pipelined processors.

**▶ See the Pipeline Fill Up**

Run a longer program and use PlayARM's **Play** button with **Normal** speed. Watch the **Pipeline Activity** panel after the first few cycles the pipeline becomes *full*: five different instructions occupy all five stages simultaneously, producing one result per cycle.

```
mov R0, #1
mov R1, #2
add R2, R0, R1
add R3, R1, R2
add R4, R2, R3
sub R5, R4, R0
cmp R5, R3
bne #-4
```

Count how many cycles pass before the first `add` reaches **WB**. Then observe how subsequent instructions complete every single cycle after that this is the throughput advantage of pipelining over single-cycle and multi-cycle designs.

## 10.9 Summary

- The **single-cycle** design executes one instruction per long cycle. It is conceptually simple but inefficient for complex ISAs.
- The **multi-cycle** design divides execution into shorter cycles, allowing hardware reuse and a faster clock.
- Both achieve functional correctness and implement the same ISA, but they differ significantly in performance and complexity.

Understanding these two models is essential before studying pipelining, which builds directly upon the cycle-based decomposition introduced here.

## 10.10 Exercises

**Exercise 10.10.1 (Signal Control Mapping)** *Draw a table listing, for each instruction type (R-type, Load, Store, Branch), which control signals are asserted in each stage for both single-cycle and multi-cycle architectures.*

**Exercise 10.10.2 (Critical Path Calculation)** *Given component delays: IMEM = 200 ps, RegFile = 120 ps, ALU = 150 ps, DMEM = 250 ps, WB = 40 ps, estimate the clock period and frequency for single-cycle vs multi-cycle designs.*

**Exercise 10.10.3 (Datapath Sketch)** *Sketch a simplified datapath showing all key signals (PC, IR, A, B, ALUOut, MDR) and explain the data flow of a Load instruction in the multi-cycle design.*

**Exercise 10.10.4 (Hardware Efficiency)** *Why does hardware reuse matter in real-world CPU design? Give two examples where hardware reuse helps reduce chip area or energy consumption.*

**Exercise 10.10.5 (Conceptual Bridge)** *Explain in your own words why the multi-cycle architecture is considered the "bridge" between single-cycle and pipelined designs.*

# 11

# Pipeline Microarchitecture

**Learning Objectives**

By the end of this chapter you should be able to:

- Explain how pipelining increases throughput without reducing individual instruction latency.
- Identify RAW, WAW, and WAR data hazards in an instruction sequence and determine which require hardware intervention.
- Trace the forwarding multiplexer select signals (`ForwardAE`, `ForwardBE`) for any pair of consecutive instructions.
- Explain why a `ldr` followed immediately by a dependent instruction cannot be resolved by forwarding, and describe the stall mechanism that fixes it.
- Derive the stall and flush signals (`StallF`, `StallD`, `FlushE`, `FlushD`) from the complete Hazard Unit logic.
- Compute effective CPI given load-use hazard and branch misprediction rates.

## 11.1 Overview and Motivation

The single-cycle and multi-cycle designs studied in Chapter 10 execute one instruction at a time. Even the multi-cycle design, which shortens the clock period by breaking execution into stages, still completes one instruction before beginning the next. **Pipelining** breaks this constraint: by overlapping the execution of multiple instructions, the processor keeps all five stages busy simultaneously, dramatically increasing throughput.

The five execution stages (Fetch, Decode, Execute, Memory, and Write-Back) each operate on different hardware resources. Once an instruction leaves the Fetch stage, that stage is idle while the instruction moves to Decode. A pipelined processor immediately fills that idle slot with the next instruction's fetch. Up to five instructions can therefore be *in-flight* at any moment, one in each stage.

**An assembly-line analogy.** A car factory is a useful comparison. Rather than one team of workers building each car from start to finish before touching the next, a modern factory splits the work into stations: one bolts on the engine, the next paints the body, the next installs the electronics, and so on. As soon as a car moves from station 1 to station 2, a fresh car rolls in behind it. At peak capacity, every station is working on a different car at the same time. The factory's *throughput* (cars finished per hour) is far higher than it would be if each car were built sequentially from scratch, even though the time to build any individual car has not changed.

A pipelined processor works the same way. Each pipeline stage is a station and each instruction is a car. Once the pipeline is full (after the first five cycles), one instruction

finishes every single clock cycle.

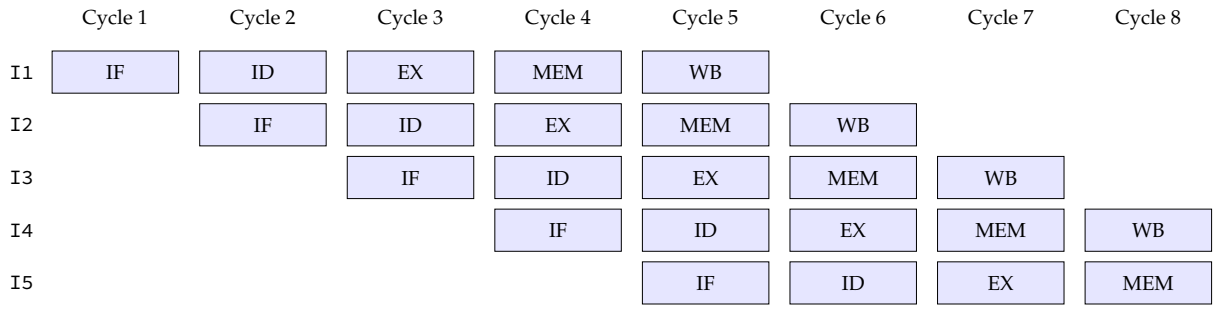



Figure 11.1: **Pipeline execution diagram.** Five instructions fill the five stages; once the pipeline is full, every stage is active every clock cycle.

**Throughput vs. latency.** Pipelining does *not* make any single instruction execute faster. Its latency (the number of cycles from fetch to write-back) stays at five. What changes is **throughput**: once the pipeline is full, one instruction completes every cycle, giving a steady-state CPI of 1. Compare this with the single-cycle design (CPI = 1 but a very long clock period) and the multi-cycle design (CPI ≈ 4–5, shorter period).

**Pipelining Improves Throughput, Not Latency**

A common misconception is that pipelining makes instructions run faster. It does not. An `add r1, r2, r3` still passes through all five stages and takes five clock cycles from fetch to write-back. What pipelining improves is **throughput**: once the pipeline is full, one instruction *completes* every cycle, even though each instruction individually still takes five cycles. Back to the car factory: adding more workers to each station does not build any car faster, but more cars come off the line per hour.

**The cost: hazards.** Overlapping instructions creates **hazards**, situations where a later instruction needs a result or a decision that an earlier instruction has not yet produced. The three classes of hazards (data, control, and structural) and their solutions make up most of this chapter.

## 11.2 The Five Pipeline Stages

We design a pipelined processor by subdividing the single-cycle processor into five pipeline stages and separating them with **pipeline registers** that store the signals between stages. Each stage is named by a single letter that becomes a suffix on all signals within it: **F** (Fetch), **D** (Decode), **E** (Execute), **M** (Memory), **W** (Write-back).

**Fetch (F):** The processor reads the 32-bit instruction from instruction memory at the address held in the Program Counter (`PCF`). `PCPlus4F` is computed and written back to the PC register so that the next instruction's address is ready for the following cycle.

**Decode (D):** The instruction word (`InstrD`) is examined by the control unit, which produces all control signals for the instruction. Simultaneously, the register file

is read: bits [19:16] of `InstrD` select register A1, and bits [3:0] select register A2, producing `RD1D` and `RD2D`. The 12-bit immediate field is sign- or zero-extended to 32 bits (`ExtImmD`).

**Execute (E):** The ALU receives its two inputs (which may be forwarded from later stages; see Section 11.6) and computes a result (`ALUResultE`). For load/store instructions the result is the effective memory address; for arithmetic instructions it is the final value. Branch conditions are also evaluated here.

**Memory (M):** If the instruction is a `LDR`, data memory is read at `ALUResultM` to produce `ReadDataM`. If the instruction is a `STR`, `WriteDataM` is written to that address. All other instructions pass through this stage without touching memory.

**Write-back (W):** The result, either `ALUResultW` for arithmetic or `ReadDataW` for a load, is selected by the `MemToReg` multiplexer and written into the destination register (`WA3W`) of the register file. `RegWriteW` must be asserted by the control unit for the write to occur.

### 11.2.1 Tracing One Instruction Through All Five Stages

The following walkthrough follows `add r1, r2, r3` from fetch to write-back, one stage at a time. It shows exactly what each piece of hardware does and why signals must travel with the instruction through every pipeline register. Assume the instruction enters the Fetch stage in cycle 1.

**Cycle 1, Fetch (IF):** The processor reads the 32-bit instruction word from instruction memory at the address held in `PCF`. The encoding `0xE0811002` (condition AL, opcode ADD, Rn=r2, Rd=r1, Rm=r3) is placed into the IF/ID pipeline register. In the same cycle, `PCPlus4F = PCF + 4` is written back to the PC so the next instruction's address is ready. Note that the processor has not yet interpreted the instruction at this point; it is just a 32-bit pattern.

**Cycle 2, Decode (ID):** The control unit reads `Op[27:26] = 00` (data-processing group) and `Funct[24:20] = 00100` (ADD) from `InstrD`. It asserts `RegWrite=1`, `ALUSrc=0` (register second operand, not immediate), and `ALUControl = ADD`. The register file is read at the same time: bits [19:16] = `0010` select `r2`, giving `RD1D`; bits [3:0] = `0011` select `r3`, giving `RD2D`. The destination address `WA3D = 0001` (register `r1`) comes from bits [15:12]. The data values `RD1D`, `RD2D`, control signals, and `WA3D` are all latched into the ID/EX pipeline register.

**Cycle 3, Execute (EX):** The ALU receives `SrcAE` (the value of `r2`, possibly forwarded) and `SrcBE` (the value of `r3`). It computes `ALUResultE = r2 + r3` and sets the condition flags (N, Z, C, V). This is the cycle where the result is ready. Any subsequent instruction that needs `r1` and has a forwarding path can receive it starting next cycle. `ALUResultE`, `WA3E = 1`, and the remaining control signals move into the EX/MEM pipeline register.

**Cycle 4, Memory (MEM):** Since this is an ADD instruction (not `ldr` or `str`), no data memory access occurs. `ALUResultM` passes through unchanged. `WA3M = 1` and the control signals move into the MEM/WB pipeline register. Meanwhile, the Fetch stage is loading a completely different instruction. The ADD knows nothing about this.

**Cycle 5, Write-back (WB):** `MemToReg = 0`, so the result multiplexer picks `ALUResultW` as `ResultW`. With `RegWriteW = 1`, the register file writes `r2 + r3` into register `r1` (selected by `WA3W = 1`). The instruction is now complete. The write happens on the falling clock edge, so any instruction simultaneously in Decode that reads `r1` will see the new value in the second half of this cycle.

### Why WA3 Must Travel Through Every Stage

The destination register address `WA3` starts as `WA3D` in Decode and must be carried through the ID/EX, EX/MEM, and MEM/WB pipeline registers to reach Write-back as `WA3W`. If the hardware read `InstrD[15:12]` directly at Write-back time, it would get whatever instruction happens to be in Decode at that moment, which is a completely different instruction. The write would land in the wrong register. Every signal produced in one stage but consumed in a later one must be pipelined with its instruction for exactly this reason.

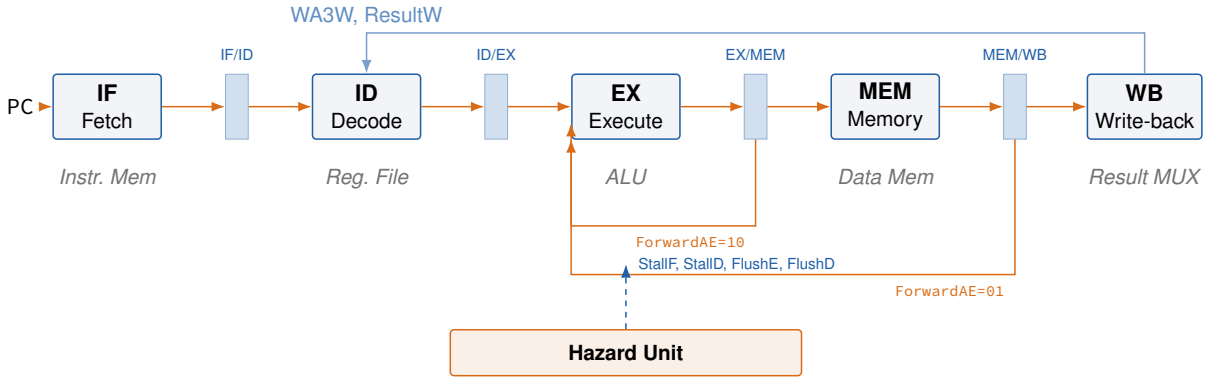



Figure 11.2: **Pipelined ARM datapath.** Pipeline registers (shaded blue) latch all signals between stages; orange forwarding paths route ALU results back to bypass the register file; the Hazard Unit (bottom) drives stall and flush signals.

### ▶ The Pipeline Live

Enter the following program and press **Step** repeatedly. Watch each instruction bubble move left-to-right through the **Pipeline Activity** panel.

```
ldr r2, [r0, #40]
add r3, r9, r10
sub r4, r1, r5
and r5, r12, r13
str r6, [r1, #20]
orr r7, r11, #42
```

Once the pipeline is full (after cycle 5), notice that a new instruction completes every single cycle. This is the throughput advantage of pipelining.

## 11.3 Pipelined Datapath

### 11.3.1 Pipeline Registers and Signal Naming

Between every adjacent pair of stages sits a **pipeline register** that captures all signals needed by the downstream stage. The four pipeline registers are named **IF/ID**, **ID/EX**, **EX/MEM**, and **MEM/WB**. On every rising clock edge, each register latches the outputs of the stage to its left and presents them to the stage on its right.

Every signal carries a stage suffix to make its position in the pipeline explicit. For example, the write-address for the register file starts as `WA3D` (in Decode), is passed through the pipeline registers to become `WA3E`, then `WA3M`, and finally `WA3W` in Write-back. This naming discipline makes it immediately clear, for any signal, exactly which stage's value it represents.

A fundamental rule follows from this: *all signals associated with a particular instruction must advance through the pipeline in unison*. Using, say, a Decode-stage address together with a Write-back-stage value produces incorrect results.

**The Pipeline Register Rule**

Every signal that belongs to an instruction must travel through the pipeline registers alongside that instruction, advancing by exactly one stage per clock cycle. Never use a signal that has already been overwritten by a newer instruction in an earlier stage. The stage suffix on every signal name (F, D, E, M, W) makes this rule visible in the code: only mix signals with the same suffix.

### 11.3.2 Pipelining the Write Address (WA3)

A subtle bug in a naive pipelined datapath illustrates why signal synchronization matters.

Consider the instruction `ldr r2, [r0, #40]` followed by `and r5, r12, r13`. The write address for the load (`r2`, i.e., `WA3 = 2`) comes from bits [15:12] of the instruction. In the Write-back stage the correct destination is `r2`, but if the write address is taken directly from the Decode stage (`InstrD[15:12]`), by the time the load reaches Write-back, `InstrD` holds the *and* instruction, whose destination is `r5`. The result of the load would be incorrectly written to `r5`.

**Fix: pipeline WA3 alongside the instruction.** The destination register address must travel through all pipeline registers together with the data:

$$\texttt{WA3D} \xrightarrow{\text{ID/EX}} \texttt{WA3E} \xrightarrow{\text{EX/MEM}} \texttt{WA3M} \xrightarrow{\text{MEM/WB}} \texttt{WA3W}$$

In the Write-back stage, `WA3W` and `ResultW` are fed back together to the register file. This ensures the correct register is always written, regardless of what other instructions are simultaneously in earlier stages.

**Signal Synchronization: The Most Common Pipeline Design Mistake**

Forgetting to pipeline a signal alongside its instruction is one of the most frequent mistakes when first building a pipelined datapath. If a signal is *produced* in one stage but *consumed* in a later stage, it must travel through every pipeline register between them. The stage suffix on signal names (F, D, E, M, W) is the guide: signals with mismatched suffixes on opposite sides of an operation almost always indicate a bug.

### 11.3.3 Register File Write and Read Timing

In the pipelined processor the register file is *written* in the Write-back stage and *read* in the Decode stage, and both can happen in the *same cycle* once the pipeline is full.

To make this work safely, the register file is designed so that writes occur on the **falling edge** of CLK and reads occur in the **second half** of the cycle (after the write has settled). This means a value written in Write-back in cycle $n$ is visible to a Decode read in the second half of cycle $n$, eliminating one potential source of stalls.

The shading convention in pipeline diagrams reflects this:

- Shading on the *right* half of an RF cell means the register file is **read** in that cycle.
- Shading on the *left* half means the register file is **written** in that cycle.
- No shading means the register file is idle.

### 11.3.4 PC Increment and Optimization

Each time an instruction is fetched, `PCPlus4F` is written simultaneously to the PC register and to the IF/ID pipeline register. On the next cycle, the ID stage sees this value as `PCPlus8D` (it is two instructions ahead of the current Fetch, because the Fetch stage has already incremented once more).

Rather than carrying a separate adder and pipeline register to compute `PCPlus8D` from scratch, the optimized datapath reuses `PCPlus4F` directly: `PCPlus8D = PCPlus4F` at that point in time. This eliminates one adder and one pipeline register, reducing area and power.

## 11.4 Pipelined Control

### 11.4.1 Deriving Control Signals

The pipelined processor uses the *same* control unit as the single-cycle design. In the Decode stage, the control unit examines the `Op` field (bits [27:26]) and `Funct` field (bits [25:20]) of `InstrD` to produce the full set of control signals: `RegWrite`, `MemWrite`, `MemToReg`, `ALUSrc`, `RegSrc`, `ImmSrc`, `ALUControl[2:0]`, and `PCSrc`.

Table 11.1 summarises what each signal does and which stage ultimately uses it.

Table 11.1: **Control signal summary.** Each signal's source stage and the stage(s) that consume it.

| Signal | Consumed in | Purpose |
|---|---|---|
| `ALUControl[2:0]` | Execute (EX) | Selects ALU operation (ADD, SUB, AND, ...) |
| `ALUSrc` | Execute (EX) | Chooses register vs. immediate as ALU input B |
| `MemWrite` | Memory (MEM) | Enables data memory write (`str`) |
| `MemToReg` | Write-back (WB) | Selects load data vs. ALU result to write back |
| `RegWrite` | Write-back (WB) | Enables register file write |
| `PCSrc` | Write-back (WB) | Triggers a PC update (branch or `bx`) |

The control unit also checks whether the destination register is `R15` (the PC), because a write to `R15` is treated as a branch and requires special hazard handling.

### 11.4.2 Pipelining the Control Signals

Just like data signals, control signals must be pipelined along with the instruction they belong to. A control signal generated in the Decode stage that governs Write-back behavior (`RegWrite`, `MemToReg`) must be carried through the ID/EX, EX/MEM, and MEM/WB pipeline registers so that it arrives at Write-back at the same time as the data it controls.

For example, `RegWriteW` feeds back from the Write-back stage to the register file's write-enable port. If `RegWrite` were not pipelined, the write-enable for the *ldr* instruction would arrive before its data, potentially writing a garbage value.

**Concrete example.** Consider `ldr r3, [r0, #8]`. The control unit produces `RegWrite=1` and `MemToReg=1` in the Decode stage (cycle 2). The instruction does not reach Write-back until cycle 6. Without pipelining, `RegWrite=1` would enable a register write in cycle 2, four cycles before the data is ready. By carrying `RegWrite` through ID/EX, EX/MEM, and MEM/WB, the write-enable arrives at the register file in cycle 6 together with the loaded data, so the write happens at the right time.

## 11.5 Data Hazards

### 11.5.1 The Read-After-Write Problem

A **data hazard** occurs when an instruction tries to read a register that has not yet been written back by a previous instruction. The most common form is **Read-After-Write (RAW)**: an instruction produces a result that a subsequent instruction needs before Write-back has completed.

Consider the following sequence:

```
add r1, r2, r3    ; writes R1 in WB (cycle 5)
and r4, r1, r5    ; reads R1 in ID (cycle 3) -- WRONG VALUE
orr r6, r1, r7    ; reads R1 in ID (cycle 4) -- WRONG VALUE
sub r8, r1, r9    ; reads R1 in ID (cycle 5, 2nd half) -- OK
```

The `add` writes `R1` at the end of cycle 5 (Write-back). `and` reads `R1` in cycle 3, which is two cycles too early. `orr` reads `R1` in cycle 4, one cycle too early. Only `sub`, which reads in the second half of cycle 5, gets the correct value thanks to the falling-edge write timing described earlier.

The timeline below makes this concrete. Say the original value of `R1` is 7 and `add` produces 15.

| Instruction | C1 | C2 | C3 | C4 | C5 | C6 | C7 |
|---|---|---|---|---|---|---|---|
| `add r1,r2,r3` | IF | ID | EX | MEM | WB (writes 15) | | |
| `and r4,r1,r5` | | IF | ID reads 7! | EX | MEM | WB | |
| `orr r6,r1,r7` | | | IF | ID reads 7! | EX | MEM | WB |
| `sub r8,r1,r9` | | | | IF | ID reads 15 (OK) | EX | MEM |

correct  stale value read (old `R1` = 7, not yet updated)

Figure 11.3: **RAW hazard timeline (no forwarding).** `and` and `orr` read the register file before `add` has written back, so they see the stale value 7 instead of the correct value 15.

**Why forwarding fixes this.** By cycle 4, when `and` is in the Execute stage and needs `R1`, the result 15 is already sitting in the **EX/MEM** pipeline register (labelled `ALUOutM`). The forwarding multiplexer can route `ALUOutM` directly to the ALU input, bypassing the register file entirely. By cycle 5, when `orr` is in the Execute stage, the same result has moved to the **MEM/WB** register (`ResultW`), and forwarding is still possible from there. Neither instruction needs to stall.

A second type, **Write-After-Write (WAW)**, occurs when two instructions both write to the same destination register. In an in-order 5-stage pipeline this resolves itself automatically: whichever instruction is farther along the pipeline writes first, and the later instruction's Write-back always overwrites it in the correct order. However, WAW becomes a serious problem in out-of-order processors, where instructions can complete out of sequence.

A third type, **Write-After-Read (WAR)**, also called an *anti-dependence*, occurs when a later instruction writes a register that an earlier instruction still needs to read. In a simple in-order pipeline, the reader always reaches the Decode (read) stage before the writer reaches Write-back, so WAR hazards do not arise. They become relevant in out-of-order designs where the writer might overtake the reader.

Table 11.2: **The three pipeline hazard types.** Structural, data, and control hazards with their causes and whether this pipeline must address each one.

| Type | Trigger | Example | This pipeline |
|---|---|---|---|
| RAW (true dependence) | Read before write completes | `add r1,r2,r3` then `and r4,r1,r5` | Must handle (forwarding or stall) |
| WAW (output dependence) | Two writes to same register | `add r1,r2,r3` then `sub r1,r4,r5` | Auto-resolved (in-order) |
| WAR (anti-dependence) | Write before earlier read | `add r4,r1,r2` then `sub r1,r5,r6` | Auto-resolved (in-order) |

### 11.5.2 Result Forwarding

Rather than stalling the pipeline until Write-back, the hardware can **forward** (bypass) the result directly from the stage where it is computed to the stage that needs it.

The `add` instruction above computes its ALU result at the end of cycle 3 (Execute stage). At cycle 4, when `and` reaches the Execute stage and needs `R1`, the result of `add` is sitting in the EX/MEM pipeline register (`ALUOutM`). A multiplexer can select `ALUOutM` instead of the register-file output, forwarding the value one stage early.

By cycle 5, when `orr` is in the Execute stage, `add`'s result has reached the MEM/WB register (`ResultW`), and forwarding is still possible.

Forwarding eliminates most RAW hazards without any stalls. There is one important exception, covered in Section 11.7.

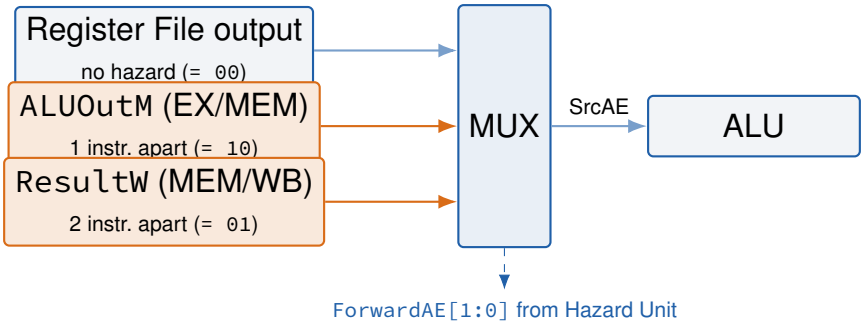



Figure 11.4: **Forwarding multiplexer for ALU input A (`SrcAE`).** `ForwardAE` selects the register-file value, the one-cycle-old EX/MEM result, or the two-cycle-old MEM/WB result; an identical mux handles input B via `ForwardBE`.

**Forwarding Rule**

Always forward from the *earliest* stage that holds the correct result. If both the Memory stage and the Write-back stage contain a matching destination register, the Memory stage has priority because its value is more recent. Fall back to the register file only when neither stage has a pending write to the source register.

## 11.6 The Hazard Unit

### 11.6.1 Architecture

The **Hazard Unit** is a small combinational block added to the pipelined datapath. It detects hazards and drives the forwarding-multiplexer select signals along with the stall and flush signals so that the pipeline always produces correct results.

Two multiplexers sit in front of the ALU, one per ALU input, each with a three-way select:

- `00`: take the value from the register file (no hazard).
- `01`: forward `ResultW` from the Write-back stage.
- `10`: forward `ALUOutM` from the Memory stage.

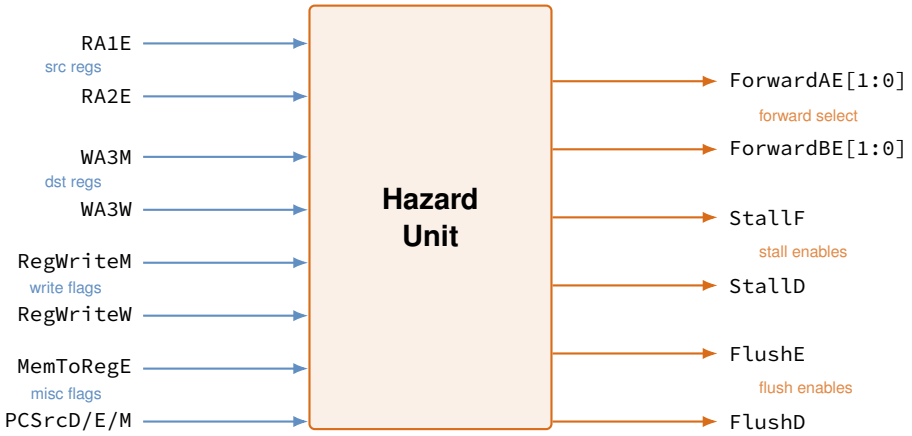



Figure 11.5: **Hazard Unit interface.** Inputs are in-flight register addresses and control flags; outputs drive the forwarding-select, stall, and flush signals.

### 11.6.2 Match Signals

The Hazard Unit computes four **match signals** by comparing the source register addresses in the Execute stage (`RA1E`, `RA2E`) against the destination register addresses in the Memory and Write-back stages (`WA3M`, `WA3W`):

$$\begin{aligned}\texttt{Match\_1E\_M} &= (\texttt{RA1E} == \texttt{WA3M})\\ \texttt{Match\_1E\_W} &= (\texttt{RA1E} == \texttt{WA3W})\\ \texttt{Match\_2E\_M} &= (\texttt{RA2E} == \texttt{WA3M})\\ \texttt{Match\_2E\_W} &= (\texttt{RA2E} == \texttt{WA3W})\end{aligned}$$

To understand what these signals mean, recall where instructions are in the pipeline when the Hazard Unit checks them:

- The **current instruction** is in the Execute stage: it reads `RA1E` and `RA2E` as its source registers.
- The instruction **one step ahead** is in the Memory stage: it will write to `WA3M`.
- The instruction **two steps ahead** is in the Write-back stage: it will write to `WA3W`.

When a match signal is asserted, the instruction currently in Execute needs a value that an older instruction has computed but not yet written to the register file. The Hazard Unit then activates the appropriate forwarding path so the result is taken from a pipeline register rather than the register file.

`Match_2E_M` is the tightest RAW dependency the pipeline can still resolve without stalling, the one-instruction-apart case:

```
add r5, r4, r2     ; WA3M = r5
add r6, r4, r5     ; RA2E = r5  -- forwarded from Memory stage
```

### 11.6.3 Forwarding Logic

If both Memory and Write-back stages contain a matching destination register, Memory has priority because it holds the *more recently executed* instruction. Forwarding from the wrong stage (Write-back when Memory has the fresher value) would produce incorrect results. The forwarding logic for ALU input A is:

```
if (Match_1E_M && RegWriteM)
    ForwardAE = 10;   // SrcAE = ALUOutM  (1 instr apart)
else if (Match_1E_W && RegWriteW)
    ForwardAE = 01;   // SrcAE = ResultW  (2 instr apart)
else
    ForwardAE = 00;   // SrcAE from register file (no hazard)
```

The `&& RegWriteM` and `&& RegWriteW` guards are important: just because two instructions share a register address does not mean a hazard exists. If the instruction in the Memory stage has `RegWrite=0` (e.g., a `str` or a `cmp`), it is not writing any result, so there is nothing to forward. Without these guards, the hardware would forward garbage values to the ALU.

The logic for `ForwardBE` is identical except it checks `Match_2E_M` and `Match_2E_W`.

**Forwarding the store data.** A subtlety often missed: `ForwardBE` serves double duty for a `str` instruction. In an ARM `str rT, [rBase, #off]`, register `rT` is the *data to be stored*, not an ALU operand. It enters the pipeline as the register-file read on port B (`RD2E`) and travels through the datapath alongside the computed address. If a preceding instruction wrote `rT` and has not yet reached Write-back, `ForwardBE` selects the forwarded value as `WriteDataE`, ensuring the correct data reaches data memory in the MEM stage. Without this path, a `str` immediately following an arithmetic instruction that produces its store data would write a stale value to memory.

**Example trace.** Consider the sequence:

```
add r5, r4, r2     ; Instruction A
sub r6, r5, r1     ; Instruction B (needs r5)
and r7, r5, r3     ; Instruction C (also needs r5)
```

When instruction B is in Execute (cycle 4), instruction A is in Memory. `Match_1E_M` is asserted (`RA1E = r5 = WA3M`) and `RegWriteM = 1`, so `ForwardAE = 10` and instruction B's ALU reads `ALUOutM` from the EX/MEM register. When instruction C is in Execute (cycle 5), instruction A is in Write-back. `Match_1E_W` is asserted and `RegWriteW = 1`, so `ForwardAE = 01` and instruction C's ALU reads `ResultW`. Neither B nor C stalls; both hazards are resolved by forwarding.

## 11.7 Load-Use Hazard and Pipeline Stalls

### 11.7.1 Why `ldr` Cannot Be Forwarded

For arithmetic instructions the ALU result is available at the end of the Execute stage and can be forwarded to the *next* instruction's Execute stage one cycle later. A `ldr` instruction, however, does not produce its result until the end of the Memory stage, one cycle later than the ALU result. A `ldr` followed immediately by a dependent instruction therefore creates a hazard that forwarding alone cannot fix.

```
ldr r2, [r0, #40]   ; result available end of cycle 4 (MEM)
orr r3, r2, r1      ; needs r2 at start of cycle 4 (EX) --
   impossible
```

The cycle count below shows why:

| Instruction | C1 | C2 | C3 | C4 | C5 | C6 |
|---|---|---|---|---|---|---|
| `ldr r2,[r0,#40]` | IF | ID | EX | MEM *result ready here* | WB | |
| `orr r3,r2,r1` | | IF | ID | EX *needs r2 here!* | MEM | WB |

normal stage  conflict: value needed before it is ready

Figure 11.6: **Load-use timing conflict.** The `ldr` result is ready at the end of MEM (cycle 4), but `orr` needs it at the start of EX in the same cycle, so forwarding alone cannot help.

The root problem is that the `ldr` result comes from data memory, which is accessed in the MEM stage. An arithmetic instruction's result comes from the ALU in the EX stage, one cycle earlier. This one-cycle difference is exactly what makes the load-use hazard impossible to resolve with forwarding alone.

The `ldr` has a **two-cycle latency**: a dependent instruction cannot use its result until two cycles after the load. One pipeline stall cycle must be inserted.

**Load-Use Hazard Rule**

A `ldr` instruction followed immediately by an instruction that uses the loaded register cannot be resolved by forwarding alone. The hardware inserts exactly one stall cycle (a bubble). The easiest software fix is to reorder code so that an independent instruction fills the slot between the load and its first consumer. This is called *load-delay slot scheduling*.

### 11.7.2 Stalls and Bubbles

A **stall** freezes the Fetch and Decode pipeline registers (their contents hold unchanged on the next clock edge) while simultaneously **flushing** the Execute pipeline register by clearing all its control signals to zero. The flushed Execute stage propagates through the pipeline as a **bubble**, a NOP-like slot that performs no action and modifies no architectural state.

| Instruction | C1 | C2 | C3 | C4 | C5 | C6 | C7 |
|---|---|---|---|---|---|---|---|
| `ldr r2,[r0,#40]` | IF | ID | EX | MEM | WB | | |
| `orr r3,r2,r1` | | IF | ID | *stall* | EX | MEM | WB |
| `sub r4,r5,r6` | | | IF | *stall* | ID | EX | MEM |

Figure 11.7: **Load-use hazard with one stall cycle.** `orr` and `sub` are each delayed one cycle; a bubble propagates through Execute while Fetch and Decode are frozen.
active stall/bubble

Stalling one stage always requires stalling *all earlier stages* too, so that no instruction already in the pipeline is lost. The pipeline register immediately after the stalled stage is cleared (flushed) to prevent the current instruction from being executed a second time.

**Why All Earlier Stages Must Stall Together**

When the pipeline stalls to wait for a result, it is not enough to freeze just the one waiting stage. If only the Decode stage were frozen while Fetch kept running, the instruction in the Fetch stage would overwrite the stalled instruction in the IF/ID pipeline register on the next clock edge, and that instruction would be lost. A stall works like a traffic jam: when one car stops, every car behind it must stop too. Every stage *upstream* of the stall point (including Fetch) holds its current instruction until the hazard is resolved.

### 11.7.3 Hazard Unit: LDR Stall Logic

The Hazard Unit detects the load-use hazard by inspecting the instruction in the Execute stage. If it is a `ldr` (`MemToRegE` is asserted) and its destination register (`WA3E`) matches either source of the instruction currently in the Decode stage, a stall is required:

```
Match_12D_E = (RA1D == WA3E) || (RA2D == WA3E);
LDRstall    = Match_12D_E && MemToRegE;

StallF  = LDRstall;
StallD  = LDRstall;
FlushE  = LDRstall;
```

The `EN` (enable) inputs of the Fetch and Decode pipeline registers are de-asserted when `StallF` and `StallD` are set, freezing those stages. The Execute pipeline register has a synchronous `CLR` input that zeroes all control signals when `FlushE` is set.

### 11.7.4 The Compiler's Role: Avoiding Hazards in Software

The hardware must handle every hazard correctly, but the *compiler* (or a careful assembly programmer) can often eliminate hazards entirely by reordering instructions. This is called **instruction scheduling** or **load-delay slot scheduling**.

**Before scheduling (1 stall cycle):**

```
ldr  r2, [r0, #40]   @ load r2
orr  r3, r2, r1      @ uses r2 immediately -- STALL inserted
   here
add  r4, r3, r5
```

**After scheduling (0 stall cycles):**

```
ldr  r2, [r0, #40]   @ load r2
add  r4, r3, r5      @ independent -- moved up to fill the
   slot
orr  r3, r2, r1      @ uses r2 (now ready, no stall needed)
```

By moving the independent `add` into the slot between the load and its first consumer, the compiler eliminates the stall without changing the program's result. This technique is called **filling the load-delay slot**.

**When scheduling is not possible.** Sometimes there is no independent instruction to fill the slot. For example, at the start of a function the first thing is often loading several arguments from memory; there may be nothing to interleave. In these cases the hardware stall is unavoidable. Modern compilers try very hard to schedule loads early in a basic block so that the result is ready by the time it is needed.

**Hardware vs. Software Hazard Handling**

The Hazard Unit always guarantees correctness. The programmer never needs to worry about getting wrong results from a pipeline hazard. What the hardware cannot avoid entirely is the time cost of stalls. A well-written compiler (or careful assembly programmer) reduces stalls by scheduling independent instructions between a load and its first use, keeping the pipeline as busy as possible.

## 11.8 Control Hazards

### 11.8.1 The Branch Problem

A **control hazard** arises because the pipelined processor does not know which instruction to fetch next when a branch enters the pipeline. The branch decision (whether to take the branch and, if so, what the target address is) is not resolved until the Execute stage, which is cycle 3 for a branch fetched in cycle 1. By then, two instructions have already been fetched behind the branch.

The same problem applies to any instruction that modifies the PC:

```
bx lr           ; PC update known in Writeback (cycle 5)
ldr pc, [sp]    ; PC update known in Writeback (cycle 5)
```

These cases require the pipeline to be stalled until the new PC is known, which can cost up to four cycles per occurrence.

**An everyday analogy.** Picture driving down a highway toward a fork. A sign tells you which lane to take, but it only becomes legible 50 metres before the split, yet the lane you are in was chosen 100 metres back. Branch prediction works the same way. The processor commits to fetching two instructions before knowing which path is correct. If the guess is wrong, those two fetched instructions must be discarded (flushed) and the correct instructions fetched instead. The goal is simply to be right as often as possible.

### 11.8.2 Predict-Not-Taken

Rather than stalling every time a branch is encountered, the processor employs a simple **branch prediction** strategy: *predict that branches are not taken* and continue fetching the sequential instructions.

If the prediction is correct (branch not taken), no penalty is incurred. If the branch *is* taken, the two instructions fetched behind the branch are wrong and must be **flushed** from the pipeline by clearing the Decode and Execute pipeline registers (`FlushD` and `FlushE`). These discarded instruction cycles are called the **branch misprediction penalty**.

| Instruction | C1 | C2 | C3 | C4 | C5 | C6 | C7 |
|---|---|---|---|---|---|---|---|
| `b target` *(taken)* | IF | ID | EX | MEM | WB | | |
| `add r6,...` *(wrong path)* | | *flush* | *flush* | | | | |
| `sub r7,...` *(wrong path)* | | | *flush* | | | | |
| `target: mov ...` *(correct)* | | | | IF | ID | EX | MEM |

active stage  flushed (wrong-path instruction discarded)

Figure 11.8: **Branch misprediction penalty (predict-not-taken).** Two wrongly-fetched instructions are flushed when the taken branch is resolved in EX (end of cycle 3), giving a 2-cycle penalty.

**Predict-Not-Taken Strategy**

Assume every branch is *not taken* and continue fetching instructions sequentially. If the branch turns out to be not taken, no cycles are lost. If the branch *is* taken, flush the two wrong-path instructions from the Fetch and Decode stages and redirect the PC to the branch target. With Execute-stage resolution this costs exactly **2 cycles** per misprediction.

### 11.8.3 Resolving Branches in the Execute Stage

Adding a multiplexer before the PC register allows the branch destination address to be selected from `ALUResultE` in the Execute stage (cycle 3), rather than waiting until

Write-back (cycle 5). The control signal `BranchTakenE` is asserted when the branch condition is satisfied.

When `BranchTakenE` is asserted:

- The PC is loaded with the branch target from `ALUResultE`.
- The two instructions in the Fetch and Decode stages are flushed (`FlushD = FlushE = 1`).

This cuts the misprediction penalty from four cycles to **two cycles**.

### 11.8.4 Handling PC Writes from Write-Back

Instructions that write to `R15` (`bx lr`, `pop {pc}`) produce the new PC value only in Write-back. While such an instruction works its way through the pipeline, the Fetch stage must be **stalled** to avoid fetching from the wrong address. `PCWrPendingF` signals that a PC-write is in flight somewhere in the pipeline:

```
PCWrPendingF = PCSrcD || PCSrcE || PCSrcM;
```

When `PCSrcW` is finally asserted in Write-back, `StallF` is released and the correct instruction is fetched, but `FlushD` is held for one extra cycle to discard whatever was in the Fetch stage during the stall.

### 11.8.5 Complete Hazard Unit Control Logic

Combining data and control hazard handling, the full set of stall and flush signals is:

```
PCWrPendingF = PCSrcD || PCSrcE || PCSrcM;

StallD  = LDRstall;
StallF  = LDRstall || PCWrPendingF;
FlushE  = LDRstall || BranchTakenE;
FlushD  = PCWrPendingF || PCSrcW || BranchTakenE;
```

**`StallD`:** Hold the Decode stage (stop `InstrD` from advancing) during a load-use hazard.

**`StallF`:** Hold the Fetch stage during a load-use hazard or while a PC-write is pending in the pipeline.

**`FlushE`:** Clear the Execute stage to insert a bubble after a stall or a correctly-predicted-not-taken branch that was in fact taken.

**`FlushD`:** Clear the Decode stage when a PC write is completing or a taken branch is detected.

### 11.8.6 Reducing the Branch Penalty Further

With additional logic, the branch decision can be moved from the Execute stage to the **Decode stage**, reducing the misprediction penalty to just **one cycle**. The branch target is computed as:

$$\texttt{PCBranchD} = \texttt{PCPlus8D} + \texttt{ExtImmD}$$

and `BranchTakenD` is evaluated using the ALU flags produced by the *previous* instruction (`ALUFlagsE`).

The trade-off is a potential increase in the critical path if the flag signals arrive late from the preceding Execute stage. Whether this optimization improves overall performance depends on the branch frequency and the clock timing budget.

### 11.8.7 Beyond Predict-Not-Taken: Smarter Branch Predictors

Predict-not-taken is simple but limited. It fails every time a branch is actually taken, which is about 55–65% of the time in typical loop-heavy code. Real processors use more sophisticated predictors that reach 95%+ accuracy.

**2-bit saturating counter.** Rather than always predicting not-taken, the processor keeps a small table of **2-bit counters**, one per branch (indexed by the branch's PC address). Each counter has four states:

Table 11.3: **2-bit saturating counter states.** Two consecutive surprises are required before the prediction flips, making the predictor robust to isolated outliers.

| State | Prediction | Transition |
|---|---|---|
| 11 (Strongly Taken) | Taken | Taken $\rightarrow$ 11, Not-taken $\rightarrow$ 10 |
| 10 (Weakly Taken) | Taken | Taken $\rightarrow$ 11, Not-taken $\rightarrow$ 01 |
| 01 (Weakly Not-Taken) | Not taken | Taken $\rightarrow$ 10, Not-taken $\rightarrow$ 00 |
| 00 (Strongly Not-Taken) | Not taken | Taken $\rightarrow$ 01, Not-taken $\rightarrow$ 00 |

One misprediction does not flip the prediction. A loop branch that is taken 99 times and falls through on the last iteration will be mispredicted just once. The counter needs two consecutive surprises before it changes its prediction, which is why loop branches are handled well.

**Branch history table (BHT).** A **branch history table** stores one 2-bit counter per branch, indexed by the low bits of the branch's PC address. A 1 KB BHT with 2-bit entries holds 4,096 predictions, which covers most branches in a typical program.

**Global history.** The most accurate predictors correlate the outcome of the current branch with the outcomes of recent previous branches using a global history register. This lets the predictor notice patterns such as: whenever branch A was taken and branch B was not, branch C is almost always taken. Modern ARM Cortex-A processors use variants of this idea and reach misprediction rates below 5% on many workloads.

**Why Branch Prediction Matters More with Deeper Pipelines**

The deeper the pipeline, the more instructions are fetched on the wrong path before the misprediction is detected, so the penalty grows. This 5-stage pipeline has a 2-cycle penalty. A modern out-of-order processor with a 14-stage pipeline can pay 14–20 cycles per misprediction. At that scale, going from 90% to 95% prediction accuracy roughly halves the branch penalty overhead. Branch predictors are one of the most actively researched areas in processor design for exactly this reason.

## 11.9 Exceptions in a Pipelined Processor

### 11.9.1 The Problem: Multiple In-Flight Instructions

An **exception** (or **interrupt**) is an event that disrupts the normal flow of execution: a memory fault, an undefined instruction, a divide-by-zero, or an external hardware signal. In a single-cycle processor, handling one is straightforward: finish the current instruction, then jump to the exception handler.

In a pipelined processor, however, up to five instructions are executing simultaneously. When an exception occurs, the processor must answer several difficult questions:

- Which instruction caused the exception?
- What should happen to the instructions that were already fetched and partially executed *after* the faulting instruction?
- What state was the processor in when the exception occurred, and can execution be resumed correctly after the handler returns?

### 11.9.2 Precise vs. Imprecise Exceptions

The gold standard for exception handling is a **precise exception**: the processor saves its state as if every instruction *before* the faulting instruction had fully completed, and *no* instruction after it had executed at all. This gives the illusion that the pipeline does not exist: from the software's point of view, execution stopped cleanly at the faulting instruction.

**Imprecise exceptions** (used in some older or simpler designs) allow some instructions after the fault to complete, and some before it to remain incomplete. Imprecise exceptions are much harder to reason about and make it nearly impossible to resume execution after an exception.

**Precise Exceptions**

A precise exception guarantees that at the moment the handler is invoked, the processor state reflects exactly all instructions before the faulting instruction (committed) and none of the instructions after it (squashed). This is the model used by all modern ARM processors.

### 11.9.3 Implementing Precise Exceptions in a 5-Stage Pipeline

When an exception is detected in stage $S$, the processor must:

1. **Flush** all instructions in stages *after* $S$ (they must not modify architectural state).

2. **Complete** all instructions in stages *before* $S$ normally (they were fetched before the faulting instruction and must commit).
3. **Save the PC** of the faulting instruction (so the handler knows where to resume or report the fault).
4. **Jump to the exception vector**, a fixed address in memory where the operating system's exception handler begins.

In the ARM architecture, the processor saves the current PC into the **Link Register** (`r14`) and the processor status flags into the **Saved Program Status Register (SPSR)**, then switches to a privileged mode and begins executing the handler.

**Example: undefined instruction.** Suppose the Decode stage encounters a 32-bit pattern it does not recognise as a valid ARM instruction.

- The instruction in Fetch (behind the bad instruction) is flushed immediately.
- The bad instruction itself is flushed from Decode after the exception address is saved.
- Any instruction in Execute, Memory, or Write-back that was fetched *before* the bad instruction completes normally.
- The PC of the bad instruction is written to the link register, and control transfers to the undefined-instruction exception vector.

**Exceptions Are Not the Same as Pipeline Flushes for Branches**

A branch misprediction flush (Section 11.8) is a *performance* correction: the processor speculatively fetched the wrong instructions and discards them before they do any harm. An exception flush is a *correctness* event: something truly wrong happened, and the operating system must be notified. The mechanism (clearing pipeline registers) looks similar in hardware, but the purpose and software handling are entirely different.

## 11.10 Structural Hazards

A **structural hazard** occurs when two instructions simultaneously require the *same hardware resource* and the hardware cannot satisfy both requests in the same cycle. Unlike data hazards (wrong value) or control hazards (wrong instruction), structural hazards arise purely from resource contention.

### 11.10.1 The Memory Conflict

The most common structural hazard in a pipeline is a **memory access conflict**: the Fetch stage reads an instruction from memory in the same cycle that the Memory stage reads or writes data.

Table 11.4: **Structural hazards in the 5-stage ARM pipeline.** Each hazard and the hardware solution that eliminates it.

| Potential conflict | Stage A | Stage B | Resolved by |
|---|---|---|---|
| Memory access | Fetch reads *instruction* memory (IF) | Memory stage reads/writes *data* memory (MEM) | Harvard architecture: separate instruction and data memories |
| Register file R/W | Decode *reads* register file (ID) | Write-back *writes* register file (WB) | Falling-edge write, second-half read (Section 11.3) |

### 11.10.2 Harvard Architecture Eliminates the Memory Hazard

The introduction says the pipeline has *separate instruction and data memories*. This is the **Harvard architecture**: instruction fetch and data memory access use independent memory banks (in real processors, separate L1 instruction and data caches). Because the two accesses go to entirely different hardware, there is no contention, and no structural hazard arises.

**Why Harvard Architecture Matters**

A single unified memory would create a structural hazard every time a load or store instruction is in the Memory stage at the same cycle that any other instruction is being fetched. Separate instruction and data memories (or caches) allow both accesses to proceed in parallel, eliminating this hazard without any stalls.

### 11.10.3 Register File Read/Write in the Same Cycle

Write-back and Decode both access the register file in the same cycle once the pipeline is full. This would be a structural hazard if both operations required the port simultaneously in the same clock phase. Writing on the **falling clock edge** and reading in the **second half** of the cycle means the write settles before the read samples the value, so both can proceed without conflict. This is described in detail in Section 11.3.

### 11.10.4 Summary

In the 5-stage ARM pipeline as designed in this chapter, *no structural hazard requires a stall*. Both potential conflicts are eliminated by the choice of Harvard architecture and the dual-phase register file. This is why the chapter's hazard discussion focuses entirely on data hazards and control hazards.

## 11.11 Real-World ARM Pipelines

The 5-stage pipeline studied in this chapter is a clean, pedagogical design that captures the essential ideas. Real ARM processors extend this foundation in several important ways.

**From 5 Stages to Modern ARM Pipelines**

| Processor | Stages | Key additions over 5-stage |
|---|---|---|
| ARM7TDMI (1994) | 3 | No forwarding; classic RISC simplicity |
| ARM9TDMI (1997) | 5 | Closest to this chapter's design |
| ARM Cortex-A5 (2009) | 8 | Superscalar 2-wide, branch predictor |
| ARM Cortex-A7 (2011) | 8 | Dual-issue in-order, improved predictor |
| ARM Cortex-A53 (2012) | 8 | Advanced predictor, 2-wide decode |
| ARM Cortex-A76 (2018) | 13 | 4-wide out-of-order, deep speculation |
| ARM Cortex-X4 (2023) | 15+ | 6-wide decode, large instruction window |

Each additional pipeline stage allows a *shorter clock period* (higher frequency) by splitting the slowest stage into two. The Cortex-A7's 8-stage pipeline runs at up to 1.6 GHz, while the ARM9's 5-stage design peaked around 300 MHz. More stages allow a shorter clock period, but a deeper pipeline means a larger branch misprediction penalty, typically 8–15 cycles instead of 2. This is exactly why sophisticated branch predictors (Section 11.8) are so important in modern processors.

**What "superscalar" means.** A **superscalar** processor issues more than one instruction per cycle by duplicating pipeline resources (two ALUs, two decode units, etc.). A 2-wide superscalar has an ideal CPI of 0.5, meaning two instructions complete every cycle. Most modern ARM Cortex-A processors are 2-wide to 6-wide superscalar, which is why their throughput far exceeds the $5\times$ speedup that a simple 5-stage pipeline can offer.

**Out-of-order execution.** In the pipeline covered in this chapter, instructions always execute in program order. An **out-of-order** processor can execute instruction 5 before instruction 4 if instruction 4 is waiting for a result but instruction 5 already has all its inputs ready. This hides latency and keeps execution units busy, at the cost of significantly more complex hardware for tracking dependencies and maintaining precise exceptions.

## 11.12 Complete Pipeline Trace: Putting It All Together

The best way to consolidate everything from this chapter is to trace a short program completely, cycle by cycle, and identify every forwarding path, every stall, and every flush. This is the style of problem most commonly seen on exams.

### 11.12.1 The Program

```
; Assume r0 = base address, r9 = 5
ldr  r1, [r0, #0]    @ I1: load
add  r2, r1, r9      @ I2: uses r1  (load-use hazard with I1)
sub  r3, r2, r9      @ I3: uses r2  (RAW, forwardable from I2
   )
```

```
cmp  r3, #0          @ I4: uses r3  (RAW, forwardable from I3
   )
bne  skip            @ I5: branch (taken)
add  r4, r3, r2      @ I6: wrong path -- will be flushed
sub  r5, r1, r9      @ I7: wrong path -- will be flushed
skip:
mov  r6, #1          @ I8: correct branch target
```

### 11.12.2 Cycle-by-Cycle Table

| Instr. | C1 | C2 | C3 | C4 | C5 | C6 | C7 | C8 | C9 | C10 | C11 | C12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| I1 `ldr r1` | IF | ID | EX | MEM | WB | | | | | | | |
| I2 `add r2,r1` | | IF | ID | *stall* | EX *fwd* | MEM | WB | | | | | |
| I3 `sub r3,r2` | | | IF | *stall* | ID | EX *fwd* | MEM | WB | | | | |
| I4 `cmp r3` | | | | *stall* | IF | ID | EX *fwd* | MEM | WB | | | |
| I5 `bne skip` | | | | | | IF | ID | EX | MEM | WB | | |
| I6 `add r4` *(wrong)* | | | | | | | *flush* | *flush* | | | | |
| I7 `sub r5` *(wrong)* | | | | | | | | *flush* | | | | |
| I8 `mov r6,#1` | | | | | | | | | IF | ID | EX | MEM |

active stage  stall / bubble  flushed (wrong-path)  *fwd* = forwarded result

Figure 11.9: **Complete pipeline trace (eight instructions).** One load-use stall, three forwarded results, and a 2-cycle branch flush across twelve clock cycles.

### 11.12.3 Pipeline Snapshot at Cycle 6

A **pipeline snapshot** shows the state of every stage at one instant in time. Reading the table column by column gives the snapshot; reading row by row gives the instruction's journey. At cycle 6 (highlighted column in Table 11.9), the pipeline contains:

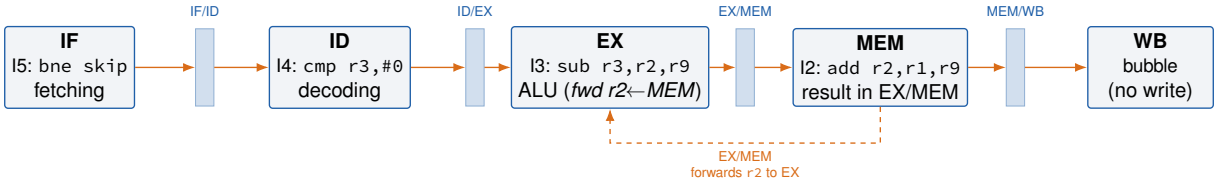



Figure 11.10: **Pipeline snapshot at cycle 6.** Four instructions occupy four different stages simultaneously: I5 fetching in IF, I4 decoding in ID, I3 executing in EX (receiving I2's result via EX/MEM forwarding), I2 in MEM, and the load-use bubble in WB.

### 11.12.4 Annotated Walkthrough

**Cycle 3 (I1 in EX, I2 in ID):** The Hazard Unit sees `MemToRegE = 1` (I1 is a load) and `RA1D = r1 = WA3E`. `LDRstall` is asserted: I2 and I3 are frozen; a bubble is inserted into EX. This stall also pushes I3 and I4 back by one cycle each.

**Cycle 5 (I1 in WB, I2 in EX):** I1's load result is now in the MEM/WB register (`ResultW`). `Match_1E_W` is asserted; `ForwardAE = 01`. I2's ALU reads `ResultW` directly; the load value arrives just in time.

**Cycle 6 (I2 in MEM, I3 in EX):** I3 needs `r2` (written by I2). I2's result is in EX/MEM; `Match_1E_M` is asserted; `ForwardAE = 10`.

**Cycle 7 (I3 in MEM, I4 in EX):** I4 (`cmp r3, #0`) needs `r3` from I3. I3's result is in the EX/MEM register; `ForwardAE = 10`. Meanwhile I5 is in ID being decoded, and I6 is being speculatively fetched.

**Cycle 8 (I4 in MEM, I5 in EX):** I5 (`bne skip`) evaluates its branch condition in the Execute stage. `BranchTakenE` is asserted at the end of this cycle. `FlushD = FlushE = 1`: I6 (in Decode) and I7 (in Fetch) are erased. The PC is redirected to the address of `skip`.

**Cycle 9 onwards:** I8 begins fetching from the correct branch target. No further hazards.

**Counting cycles and CPI.**

- Useful instructions committed: I1, I2, I3, I4, I5, I8 = **6 instructions**. I6 and I7 are flushed and never modify architectural state.
- I8 enters IF in cycle 10 and completes Write-back in cycle 14, giving a total execution time of **12 cycles** from first fetch to when I8 reaches MEM (the last column in the trace above). Using the full execution window from C1 to I8's WB:

$$\text{CPI}_{\text{trace}} = \frac{14}{6} \approx 2.3$$

- For a long program the pipeline-fill overhead is negligible and the standard formula applies:

$$\text{CPI}_{\text{actual}} = 1 + \frac{\text{wasted cycles}}{\text{instructions}} = 1 + \frac{1+2}{6} = 1.5$$

  where the 3 wasted cycles are 1 load-use stall and 2 branch-flush cycles.
- This dense cluster of hazards is a worst case; typical programs spread hazards out and achieve CPI $\approx$ 1.2–1.5.

## 11.13 Performance Analysis

### 11.13.1 Ideal CPI and Throughput

In the absence of hazards, a pipelined processor achieves CPI = 1: one instruction completes every clock cycle. The clock period is set by the longest single pipeline stage (not the longest instruction), so the period is much shorter than in the single-cycle design.

If the single-cycle clock period is $T_{\text{single}}$ and the pipeline has $N = 5$ stages, the ideal pipeline period is approximately $T_{\text{pipe}} \approx T_{\text{single}}/N$. The ideal speedup over the single-cycle design is therefore $N = 5\times$.

### 11.13.2 Pipeline Efficiency

**Pipeline efficiency** measures how close the actual pipeline performance is to the ideal:

$$\text{Efficiency} = \frac{\text{Ideal CPI}}{\text{Actual CPI}} = \frac{1}{\text{CPI}_{\text{actual}}}$$

Efficiency of 1.0 (100%) means no stall cycles were wasted. Efficiency of 0.78 means 22% of all cycles were spent stalling.

Using the worked example from Section 11.13.4 (CPI = 1.28):

$$\text{Efficiency} = \frac{1}{1.28} \approx 0.78 = 78\%$$

This is a reasonable figure for a simple in-order pipeline. Modern out-of-order processors achieve effective efficiencies above 90% on many workloads by hiding latency through dynamic scheduling and deep branch prediction.

Alternatively, pipeline efficiency can be expressed as:

$$\text{Efficiency} = \frac{\text{Busy stages}}{\text{Total stage-cycles}} = \frac{N \times I}{N \times (I + \text{Stall cycles})}$$

where $I$ is the number of instructions and $N$ is the pipeline depth. As $I$ grows large (long programs), the startup transient (first $N$ cycles) becomes negligible and efficiency approaches $1/\text{CPI}_{\text{actual}}$.

### 11.13.3 Stall Penalties

Stalls degrade throughput. Each stall cycle adds 1 to the effective CPI:

$$\text{CPI}_{\text{actual}} = 1 + \text{stalls per instruction}$$

Typical stall sources and their approximate frequencies in real programs:

- **Load-use hazards (LDR stall):** approximately 5–10% of instructions are loads followed immediately by a dependent instruction, contributing $\approx 0.05$–$0.10$ stall cycles per instruction.
- **Branch mispredictions (2-cycle penalty):** branches occur in roughly 15–20% of dynamic instructions; with a predict-not-taken strategy and a 50–60% taken rate, the misprediction penalty contributes $\approx 0.15$–$0.25$ stall cycles per instruction.

Even with these penalties, a pipelined processor typically achieves an effective CPI of 1.2–1.5, which is far superior to the 4–5 CPI of the multi-cycle design.

### 11.13.4 Worked Example: Computing Effective CPI

Let us walk through a concrete calculation using realistic program statistics.

**Given a program with the following profile:**

- Total dynamic instructions: 100,000
- LDR instructions: 10% of all instructions (10,000 loads)
- Load-use hazards: 60% of loads are followed immediately by a dependent instruction ($0.10 \times 0.60 = 0.06$ stall cycles per instruction on average)

- Branch instructions: 20% of all instructions (20,000 branches)
- Taken branches: 55% of branches (misprediction rate = 55%, since we predict not-taken)
- Branch misprediction penalty: 2 cycles per misprediction

**Step 1: Calculate stall cycles per instruction.**

$$\begin{aligned}\text{LDR stall contribution} &= 0.10 \times 0.60 \times 1 \text{ cycle} = 0.060\\ \text{Branch stall contribution} &= 0.20 \times 0.55 \times 2 \text{ cycles} = 0.220\\ \text{Total stalls per instruction} &= 0.060 + 0.220 = 0.280\end{aligned}$$

**Step 2: Effective CPI.**

$$\text{CPI}_{\text{actual}} = 1 + 0.280 = \mathbf{1.28}$$

**Step 3: Speedup over single-cycle.** If the single-cycle design has clock period $T_s$ and CPI = 1, while the pipelined design has clock period $T_s/5$ and CPI = 1.28:

$$\text{Speedup} = \frac{1 \times T_s}{1.28 \times T_s/5} = \frac{5}{1.28} \approx \mathbf{3.9}\times$$

Despite hazard penalties consuming 0.28 out of an ideal CPI of 1, the shorter clock period still yields nearly a 4× speedup over the single-cycle design. This illustrates why pipelining is so effective even in the presence of hazards.

**Reducing Stalls Matters More at Scale**

In this example, branch mispredictions (0.22 stalls/instr) cost more than load-use hazards (0.06 stalls/instr). This is a realistic ratio in programs with many conditional branches (like loops). Real processors invest heavily in branch predictors precisely because the branch penalty dominates. Even a 2% improvement in prediction accuracy translates directly into a measurable speedup at scale.

### 11.13.5 Comparison Summary

Table 11.5: **Performance comparison: single-cycle, multi-cycle, and pipelined.**

| Feature | Single-Cycle | Multi-Cycle | Pipelined |
|---|---|---|---|
| Clock period | Long ($T_s$) | Short ($T_s/5$) | Short ($T_s/5$) |
| Ideal CPI | 1 | 4–5 | 1 |
| Hazard overhead | None | None | 0.2–0.5 CPI |
| Hardware reuse | Low | High | Medium |
| Control complexity | Simple | FSM | Hazard Unit + FSM |
| Throughput (instructions/s) | $1/T_s$ | $1/(4 \cdot T_s/5)$ | $\sim 5/T_s$ |

**▶ Observing Stalls and Forwarding**

Enter the following program in PlayARM and step through it cycle by cycle.

```
ldr  r2, [r0, #40]
orr  r3, r2, r1       @ load-use hazard: stall inserted
add  r4, r2, r3
sub  r5, r4, r2
b    skip
add  r6, r3, r4       @ branch misprediction: flushed
add  r7, r4, r5       @ branch misprediction: flushed
skip:
mov  r8, #0
```

In the **Pipeline Activity** panel:

- After the `ldr`, watch for a *bubble* (grey slot) inserted in the Execute stage.
- After the `b skip`, watch for two instructions flushed from Fetch and Decode.
- For the `add` and `sub`, verify in the **Forwarding** panel that results are being forwarded rather than read from the register file.

## 11.14 Summary

- **Pipelining** overlaps the execution of multiple instructions by dividing the datapath into five stages (IF, ID, EX, MEM, WB) separated by pipeline registers. It improves *throughput*, not the latency of any individual instruction.
- All signals are suffixed by their pipeline stage (F, D, E, M, W) and must advance through the pipeline in unison with their instruction. Mismatched suffixes in an expression almost always indicate a bug.
- The **WA3** write-address must be pipelined through all stages to prevent incorrect register writes. Without this, Write-back would use the destination register of whatever instruction happens to be in Decode at that moment.
- The register file writes on the **falling clock edge** so that a Write-back write and a Decode read can coexist within the same cycle without a structural hazard.
- **Data hazards** (RAW: Read-After-Write) arise when an instruction reads a register that a preceding instruction has not yet written. The **Hazard Unit** resolves most RAW hazards by *forwarding* results from the EX/MEM or MEM/WB pipeline register directly to the ALU inputs.
- **Forwarding priority**: if both the Memory stage and Write-back stage have a matching destination register, always forward from Memory, since it holds the more recent value.
- A **load-use hazard** (`ldr` followed immediately by a dependent instruction) cannot be resolved by forwarding because the loaded value is not available until the end of MEM, which is the same cycle the dependent instruction needs it in EX. One stall cycle (bubble) is inserted.
- **Control hazards** arise from branch instructions; a **predict-not-taken** strategy with Execute-stage branch resolution limits the misprediction penalty to **two cycles**. If prediction succeeds, zero cycles are wasted.

- PC-modifying instructions (`bx`, `pop {pc}`) stall the Fetch stage until the new PC is available in Write-back, using the `PCWrPendingF` signal.
- **Structural hazards** (resource conflicts) are eliminated by design: separate instruction and data memories remove the Fetch/MEM memory conflict; dual-phase register-file timing removes the Decode/Write-back conflict. No structural stalls are needed.
- The complete stall/flush logic is encoded in five signals: `StallF`, `StallD`, `FlushE`, `FlushD`, and `PCWrPendingF`.
- Realistic programs achieve an effective CPI of roughly 1.2–1.5, giving a speedup of approximately 3–4× over the single-cycle design after accounting for hazard penalties.

## 11.15 Exercises

**Exercise 11.15.1 (Pipeline Diagram)** *Draw a cycle-by-cycle pipeline diagram for the following six-instruction sequence, marking any stall cycles and flushed instructions. Assume a predict-not-taken branch policy with Execute-stage resolution.*

```
ldr r1, [r0, #0]
add r2, r1, r3
cmp r2, #10
bne loop
sub r4, r2, r1
mov r5, #0
```

**Exercise 11.15.2 (Forwarding Paths)** *For each instruction pair below, state whether forwarding is sufficient (no stall needed), a stall is required, or there is no hazard at all. Justify your answer by identifying which match signal is asserted.*

(a) `add r5, r2, r3` *followed immediately by* `sub r7, r5, r1`
(b) `ldr r5, [r0, #4]` *followed immediately by* `and r6, r5, r2`
(c) `mov r3, #42` *followed two instructions later by* `add r4, r3, r1`

**Exercise 11.15.3 (Hazard Unit Logic)** *Write out the complete forwarding logic for* `ForwardBE` *(ALU input B) in pseudo-code, following the same structure as the* `ForwardAE` *logic shown in Section 11.6.*

**Exercise 11.15.4 (Branch Penalty Calculation)** *A program has 10,000 instructions. 18% are branches; of those, 55% are taken. The processor uses predict-not-taken with a 2-cycle misprediction penalty. Compute:*

(a) *The total number of wasted cycles due to branch mispredictions.*
(b) *The effective CPI (assuming no other hazards).*
(c) *The speedup compared with the same program on a stall-on-every-branch implementation (4-cycle penalty, always stalling).*

**Exercise 11.15.5 (Design Trade-off)** *Moving branch resolution from the Execute stage to the Decode stage reduces the misprediction penalty from two cycles to one cycle. Explain why this optimization might* not *improve performance in all cases, and describe the condition under which it could actually* hurt *performance.*

**Exercise 11.15.6 (Signal Tracing)** *Trace the instruction* `ldr r4, [r1, #12]` *through all five pipeline stages. For each stage, list:*

*(a) which hardware unit is active (instruction memory, register file, ALU, data memory, or result MUX),*
*(b) what value is computed or transferred, and*
*(c) which signals are latched into the outgoing pipeline register.*

*How does your trace change if the instruction immediately after is* `add r5, r4, r2` *(a load-use hazard)? Draw the modified pipeline diagram showing the inserted stall.*

**Exercise 11.15.7 (Forwarding Decision Table)** *For each of the following three-instruction sequences, determine the value of* `ForwardAE` *and* `ForwardBE` *when the third instruction is in the Execute stage. Show which match signals are asserted and whether* `RegWrite` *is set.*

*(a)* `add r1, r2, r3` → `sub r5, r6, r7` → `and r8, r1, r9`
*(b)* `str r1, [r0]` → `mov r3, #5` → `orr r4, r1, r2`
*(c)* `add r1, r2, r3` → `add r1, r4, r5` → `sub r6, r1, r7`

*In case (c), which instruction's value does* `ForwardAE` *select, and why?*

**Exercise 11.15.8 (Hazard Identification Drill)** *For each of the ten instruction pairs below, classify the relationship as one of:* ***(N)*** *No hazard,* ***(F)*** *RAW hazard resolved by forwarding (no stall),* ***(S)*** *RAW hazard requiring a stall, or* ***(C)*** *Control hazard. Briefly justify each answer.*

| # | *Instruction A* | *Instruction B (immediately after)* | *Type?* |
|---|---|---|---|
| 1 | `add r1, r2, r3` | `sub r4, r1, r5` | |
| 2 | `ldr r1, [r0, #4]` | `add r2, r1, r3` | |
| 3 | `mov r1, #42` | `orr r5, r6, r7` | |
| 4 | `ldr r1, [r0, #4]` | `str r1, [r0, #8]` | |
| 5 | `add r3, r1, r2` | `cmp r3, #0` | |
| 6 | `cmp r3, #0` | `beq done` | |
| 7 | `str r4, [r0]` | `add r5, r4, r6` | |
| 8 | `ldr r2, [r1, #0]` | `ldr r3, [r2, #0]` | |
| 9 | `add r1, r2, r3` | `add r1, r4, r5` | |
| 10 | `mul r5, r3, r4` | `sub r6, r5, r1` | |

**Exercise 11.15.9 (CPI Optimisation by Instruction Scheduling)** *The following code fragment has two load-use hazards.*

```
ldr  r1, [r0, #0]    @ load array[0]
add  r2, r1, r9      @ uses r1  (hazard 1)
ldr  r3, [r0, #4]    @ load array[1]
sub  r4, r3, r9      @ uses r3  (hazard 2)
add  r5, r2, r4      @ uses r2 and r4
```

*(a) Draw the pipeline diagram for this code fragment and mark the two stall cycles. How many total cycles does it take?*

*(b) Reorder the instructions to eliminate both stalls (without changing the program's result). Verify that your reordered sequence has no hazards.*

*(c) Draw the pipeline diagram for your reordered sequence. How many cycles does it now take?*

*(d) Calculate the CPI for the original code and the reordered code. What is the speedup?*

# 12

# CPU Caches

**Learning Objectives**

By the end of this chapter you should be able to:

- Describe the ideal memory and explain why it cannot be built with today's technology.
- Explain the memory hierarchy and why it is organised as a pyramid of speed versus capacity.
- Define temporal and spatial locality and give concrete examples of each in real programs.
- Decompose a memory address into tag, index, and block-offset fields for a direct-mapped or set-associative cache.
- Trace the cache READ algorithm step-by-step for both a hit and a miss.
- Compare direct-mapped, set-associative, and fully associative organisations and articulate the trade-offs of each.
- Distinguish write-through from write-back and write-allocate from no-write-allocate.
- Compute Average Memory Access Time (AMAT) and execution-time speedup from cache hit rate, hit time, and miss penalty.
- Classify a cache miss as compulsory, capacity, or conflict and state which design lever fixes each type.

## 12.1 The Ideal Memory

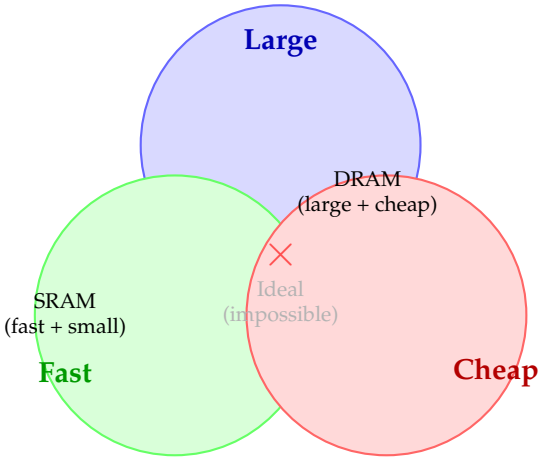
Before studying what caches are, it helps to ask: what would we *want* from a memory system if we could have anything?

**The Three Wishes for Memory**

An ideal memory would be simultaneously:

- **Infinitely large**, enough capacity to hold every program and every data set, no matter how big.
- **Infinitely fast**, zero latency; every read and write completes in the same cycle as any ALU instruction.
- **Free**, negligible cost per bit so that cost is never a constraint on size.

Unfortunately, these three properties are mutually exclusive with today's technology. Memory that is large is slow. Memory that is fast is expensive and small. The goal of a memory system is to *approximate* the ideal as closely as possible within real physical constraints.

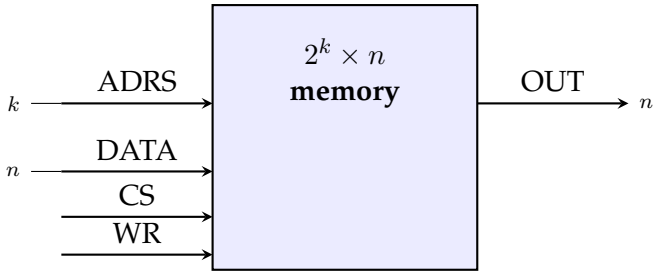



*No technology simultaneously achieves all three.*

Figure 12.1: The three properties of an ideal memory are mutually exclusive. SRAM is fast and small; DRAM is large and cheap; no current technology sits in the centre.

## 12.2 Basic Memory Structure

Before studying caches, it helps to understand how RAM itself is structured. Think of computer memory as one big array of data. The **address** acts like an array index: it selects which word to read or write.




**Memory Operation**

| CS | WR | Operation |
|---|---|---|
| 0 | x | None (disabled) |
| 1 | 0 | Read selected word |
| 1 | 1 | Write selected word |

Figure 12.2: The RAM interface. **CS** (Chip Select) enables the chip. **ADRS** selects the word. **WR** chooses read (0) or write (1). This is the same Data Memory block seen in the single-cycle microarchitecture (Chapter 10).

The key insight is that *every memory access goes through this interface*. When pipelining was introduced we treated memory as instantaneous. In reality, accessing DRAM through this interface takes 50–200 clock cycles. The cache sits between the processor and DRAM to hide this latency.

## 12.3 The Memory Hierarchy

Since no single memory technology delivers large, fast, and cheap storage simultaneously, computer systems build a **memory hierarchy**: a pyramid of storage technologies where each level is faster, smaller, and more expensive per byte than the level below it.

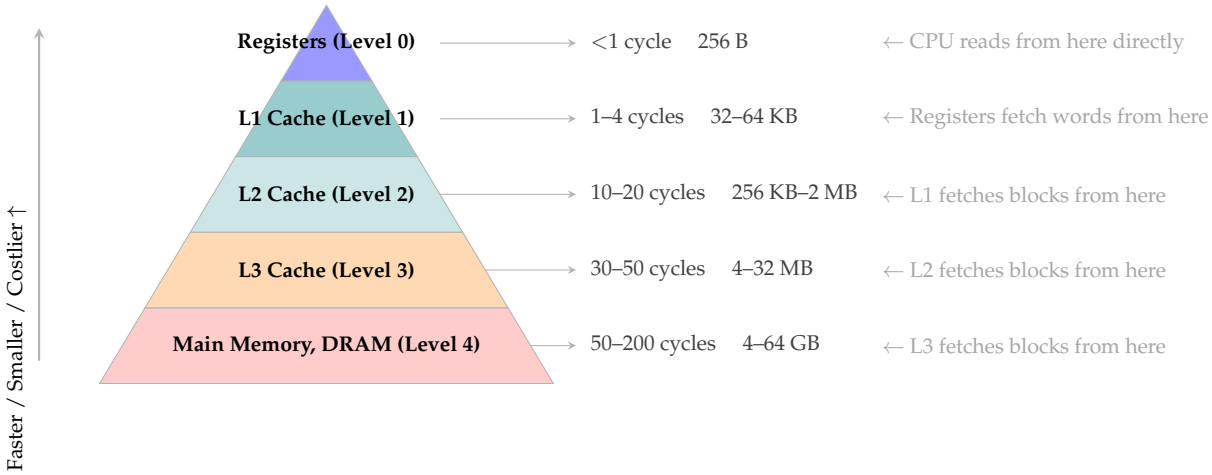



Figure 12.3: The memory hierarchy pyramid. Each level is faster, smaller, and more expensive than the one below. On a miss, a **block** of data is fetched from the next level and installed at the current level.

**Why registers are at the top.** CPU registers hold individual words and are accessed in one clock cycle. They are part of the processor itself and do not use memory addresses — they are identified by register name (`r0`, `x1`, etc.). There are only 16–32 of them; anything not currently in a register must be fetched from the cache hierarchy.

**Why there are three levels of cache.**

- **L1** must be fast enough to supply operands every cycle. Small size (32–64 KB) ensures 1–4 cycle access, one per core.

- **L2** acts as a backstop for L1 misses, trading some speed for larger capacity (256 KB – 2 MB), still one per core.
- **L3** is shared across all cores and absorbs most of what L2 misses, providing a large (4–32 MB) last resort before the slow DRAM bus.

**Larger Memories Are Always Slower**

Due to the physics of chip design, larger memories require longer wire runs and more complex decoding circuitry. This is not a software problem or a design choice, it is a fundamental physical constraint. A 64 KB SRAM array can be accessed in 1 cycle; a 64 GB DRAM module requires 50–200 cycles. The hierarchy exists precisely to work around this constraint.

### 12.3.1 SRAM versus DRAM: Why the Technology Gap Exists

Understanding *why* cache memory is fast requires a brief look at the underlying semiconductor technology.

**SRAM (Static RAM).** Each bit in an SRAM cell is stored by a **cross-coupled pair of inverters**, six transistors that latch into a stable 0 or 1. The cell holds its value as long as power is applied, with no need for periodic refresh. Because the data is always ready at the output of the latch, reading takes only a few gate delays, typically 0.5–2 ns on a modern process node.

The cost: six transistors per bit. At roughly 6–8 $F^2$ per transistor (where F is the minimum feature size), a single SRAM bit occupies $\approx 50\,F^2$ of silicon. A 32 KB L1 cache requires about 2 million transistors, expensive, but feasible on-chip.

**DRAM (Dynamic RAM).** Each DRAM bit uses only **one transistor and one capacitor**. The capacitor holds the charge that represents a 1 or 0, but capacitors leak, the charge dissipates in a few milliseconds. The memory controller must **refresh** every row thousands of times per second just to prevent data loss. Reading is also *destructive*: reading a row discharges the capacitors, which must then be recharged (a *precharge* step), adding latency.

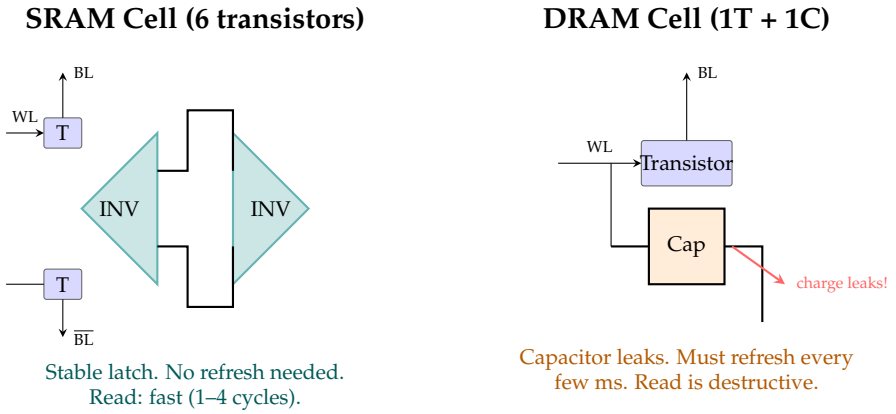



Figure 12.4: SRAM (left) uses 6 transistors per bit in a stable latch, fast but area-expensive. DRAM (right) uses 1 transistor and 1 capacitor per bit, compact and cheap but the capacitor leaks, requiring periodic refresh and making reads destructive and slow.

Table 12.1: SRAM versus DRAM: a head-to-head comparison.

| Property | SRAM | DRAM |
|---|---|---|
| Transistors per bit | 6 | 1 (+ 1 capacitor) |
| Access time | 0.5–2 ns | 50–100 ns |
| Refresh required | No | Yes (∼64 ms cycle) |
| Read destructive | No | Yes (must recharge) |
| Cost per bit | High (≈10×) | Low |
| Density | Low | High |
| Power consumption | Low (static) | Higher (refresh) |
| Used for | Caches (L1, L2, L3) | Main memory |

The 50–200 cycle gap between SRAM and DRAM is not a software bug or a design flaw, it is a direct consequence of the physics of capacitor charging and the cost of building six-transistor cells at scale. Caches exist to bridge this gap by keeping the most-used data in SRAM while storing everything else in cheap DRAM.

## 12.4 The Principle of Locality

Caches only work because real programs do not access memory randomly. They exhibit two key patterns collectively called the **Principle of Locality**:

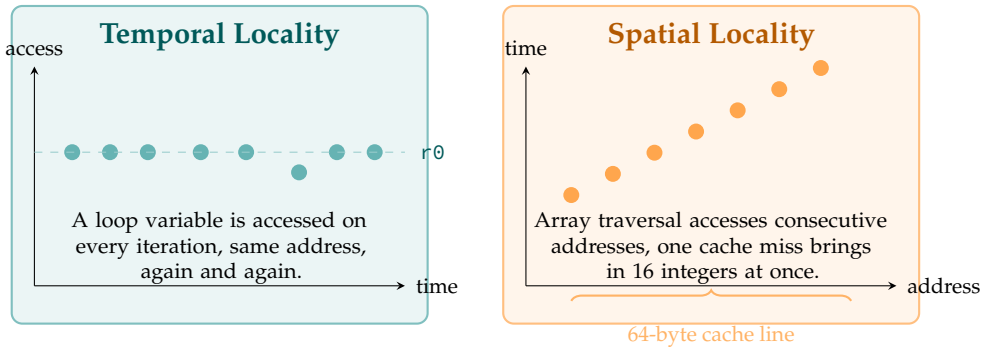



Figure 12.5: Temporal locality (left): the same address is reused many times. Spatial locality (right): addresses close together in memory are accessed in sequence. Both patterns make caches effective.

**Programs naturally exhibit locality.**

- **Temporal locality** arises from *loops*: the same instructions and variables are used on every iteration.
- **Spatial locality** arises from *sequential execution* (the next instruction to execute is at the next address) and from *data structures* (arrays, structs, and stacks are laid out contiguously in memory).

**Locality is a Prediction, Not a Guarantee**

Moving data to the cache is a bet: "this data will be reused before it is evicted." The bet is correct 95–99% of the time for instruction fetches and 75–90% of the time for data accesses in typical programs. Code with poor locality, such as pointer-chasing through a randomly-ordered linked list, loses the bet frequently and benefits little from a cache.

## 12.5 Cache Fundamentals

### 12.5.1 Cache Capacity and Block Size

Two numbers define the basic size of a cache:

- **Capacity** $C$: the total number of data words the cache can hold.
- **Block size** $b$: the number of words in one *cache line* (the unit of transfer between cache and memory).

These two numbers together determine the number of blocks:

$$B = \frac{C}{b}$$

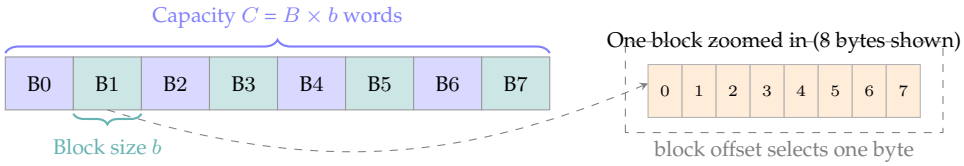



Figure 12.6: A cache with $B$ blocks, each of size $b$ bytes. When a miss occurs, one whole block is fetched from the next memory level. The block offset field of the address selects the desired byte within the fetched block.

**Why transfer whole blocks?** Fetching a whole line exploits *spatial locality*: neighbouring bytes arrive for free, ready for the accesses that are likely to follow. On a modern system, transferring 64 bytes costs only marginally more than transferring 4 bytes because the memory bus is wide and the setup latency dominates.

### 12.5.2 What the Cache Stores: Valid, Tag, Data, and Dirty Bits

Each cache line stores more than just the data bytes. The hardware needs extra bookkeeping bits to manage correctness:

Table 12.2: Fields stored per cache line in a write-back cache.

| Field | Width | Purpose |
|---|---|---|
| Valid bit | 1 bit | Set to 1 when this line contains meaningful data. On power-up all valid bits are 0 (cache is empty). |
| Tag | $t$ bits | The upper bits of the address; confirms that the correct memory line is cached here. |
| Data | $b \times 8$ bits | The actual cached bytes ($b$ = block size, typically 64 bytes = 512 bits). |
| Dirty bit | 1 bit | Set to 1 when the cached copy has been written but not yet written back to memory. Only needed for write-back caches. |

**Storage overhead calculation.** For a 32 KB, 4-way set-associative cache with 64-byte lines and 32-bit addresses:

- Number of sets: $S = 32768/(4 \times 64) = 128$ sets. $\Rightarrow$ index field: $s = \log_2 128 = 7$ bits.
- Offset field: $\log_2 64 = 6$ bits.
- Tag field: $32 - 7 - 6 = 19$ bits per way.
- Overhead per way: $1$ (valid) $+ 1$ (dirty) $+ 19$ (tag) $= 21$ bits.
- Total overhead for all ways: $4 \times 128 \times 21 = 10{,}752$ bits $\approx 1.3$ KB.

The actual data storage is 32 KB; the overhead is about 4%, small but not zero, and it must be fast SRAM like the data itself.

### 12.5.3 Address Decomposition

Every memory address is split into three fields by the hardware to locate data in the cache:

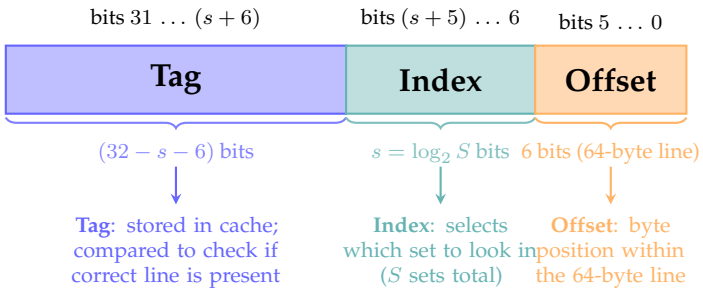



Figure 12.7: 32-bit address decomposition for a cache with $S$ sets and 64-byte lines. The low 6 bits are always the offset (since $\log_2 64 = 6$); the next $s = \log_2 S$ bits are the index; the rest form the tag.

**A worked decomposition.** Suppose a cache has $S = 256$ sets and 64-byte lines. Then $s = \log_2 256 = 8$ bits for the index and 6 bits for the offset. For a 32-bit address:

| Field | Bits | Width | Purpose |
|---|---|---|---|
| Tag | $31:14$ | 18 bits | Distinguishes lines mapped to same set |
| Index | $13:6$ | 8 bits | Selects one of 256 sets |
| Offset | $5:0$ | 6 bits | Byte within the 64-byte line |

### 12.5.4 Direct-Mapped Cache

In a **direct-mapped** cache each memory address maps to exactly one cache set, and each set holds exactly one block. The mapping rule is:

$$\text{cache set} = \text{address} \bmod S \quad \text{(implemented by the index field)}$$

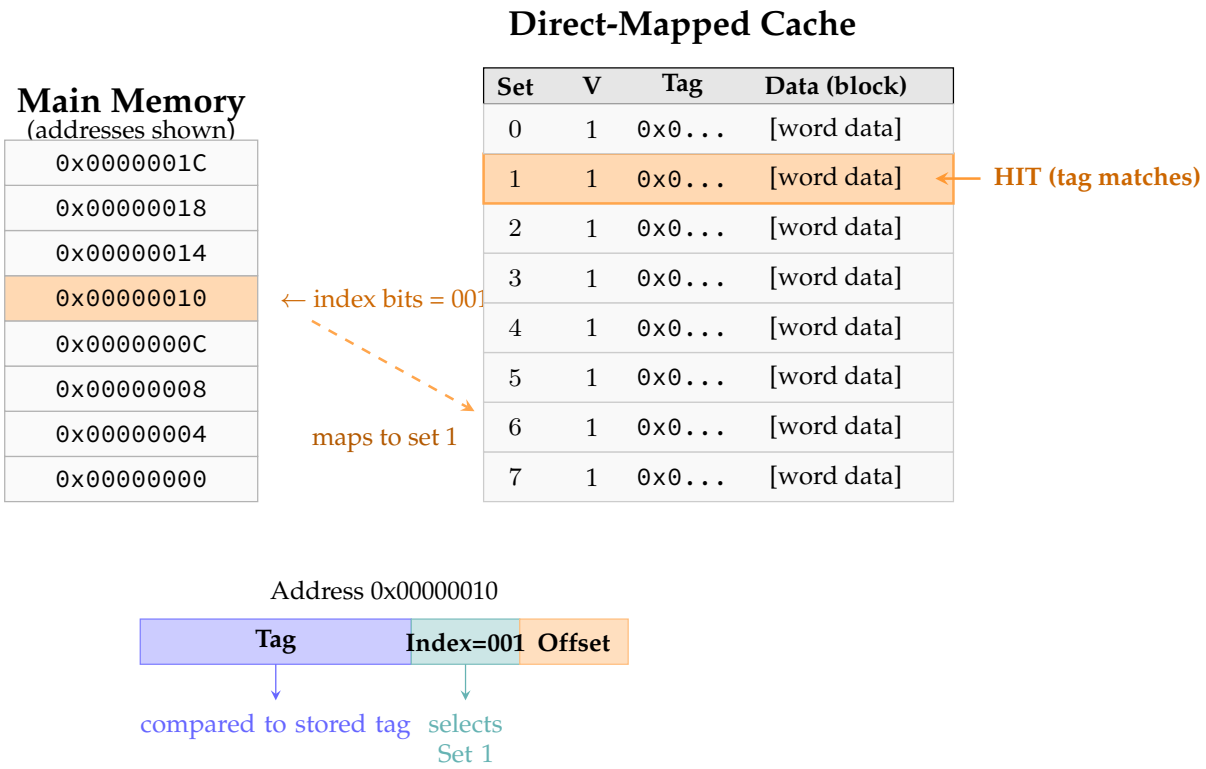



Figure 12.8: Direct-mapped cache lookup. Address `0x00000010` has index bits `001`, selecting Set 1. The tag is compared with the stored tag. A valid match is a **hit**; a mismatch or invalid entry is a **miss**.

**Conflict misses.** The weakness of direct-mapped caches is **conflict misses**: two addresses with the same index bits (e.g. `0x00000010` and `0x00001010` if both have index bits `001`) must share a single set. If a program alternates between them, every access evicts the other — a phenomenon called *thrashing*, even though most of the cache is empty.

**Thrashing worked example.** Consider a direct-mapped cache with $S = 8$ sets and 16-byte lines (4-bit offset, 3-bit index, remaining bits = tag). Addresses `A = 0x000` and `B = 0x080` both have index bits `000` (they differ only in their tag).

Table 12.3: Cache trace showing thrashing between two conflicting addresses. Every access misses, the cache is completely ineffective.

| Access | Addr | Index | Tag | Set 0 holds | Result |
|---|---|---|---|---|---|
| 1 | A (0x000) | 0 | 0x00 | , (cold) | **MISS**, install A |
| 2 | B (0x080) | 0 | 0x08 | A (tag=0x00) | **MISS**, evict A, install B |
| 3 | A (0x000) | 0 | 0x00 | B (tag=0x08) | **MISS**, evict B, install A |
| 4 | B (0x080) | 0 | 0x08 | A (tag=0x00) | **MISS**, evict A, install B |
| 5 | A (0x000) | 0 | 0x00 | B (tag=0x08) | **MISS**, evict B, install A |
| 6 | B (0x080) | 0 | 0x08 | A (tag=0x00) | **MISS**, evict A, install B |

Every single access is a miss, even though only 2 out of 8 sets are contested. The rest of the cache sits unused. A 2-way set-associative cache would resolve this immediately: both A and B fit in Set 0's two ways, and every access after the initial compulsory misses would hit.

### 12.5.5 Set-Associative Cache

An $N$**-way set-associative** cache adds $N$ **ways** to each set. An address still maps to exactly one set via the index bits, but it can occupy any of the $N$ ways within that set. A conflict miss occurs only if all $N$ ways are occupied by different addresses.

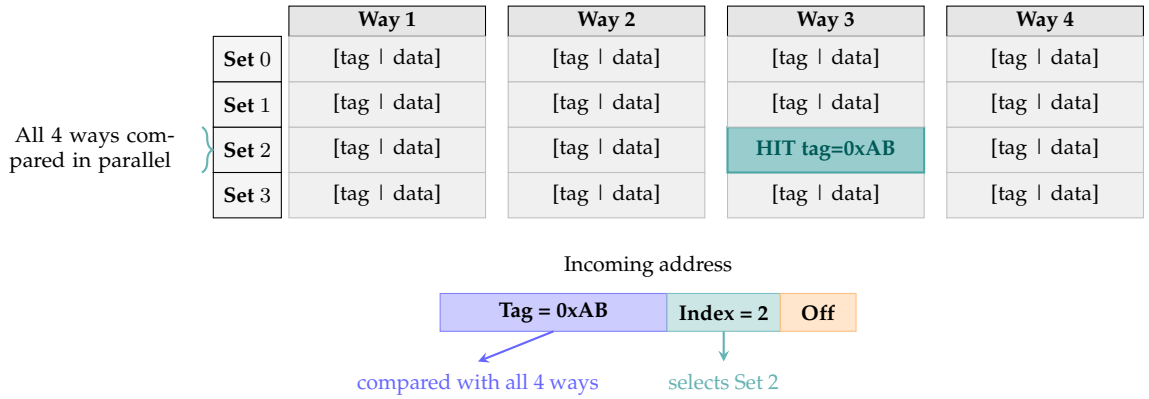



Figure 12.9: 4-way set-associative cache with 4 sets. The index field selects Set 2; all four ways are compared simultaneously. Way 3 matches (teal), this is a **hit**.

**Direct-Mapped vs. $N$-Way Set-Associative**

| Property | Direct-Mapped | $N$-Way Set-Assoc. |
|---|---|---|
| Sets per cache | $B$ (one block/set) | $B/N$ |
| Ways per set | 1 | $N$ |
| Comparators needed | 1 | $N$ (parallel) |
| Conflict misses | High | Reduced by $N$ |
| Hit time | Faster | Slightly slower |
| Typical use | L1 I-cache | L1 D-cache, L2, L3 |

### 12.5.6 Fully Associative Cache

A **fully associative** cache has a single set containing all blocks. Any address maps to any block, there are no conflict misses. The address has no index field: it is entirely tag and offset.

**Fully Associative Is Rare for Large Caches**

Comparing all tags in parallel requires one dedicated comparator per block. A 32 KB cache with 64-byte lines contains 512 blocks, 512 comparators. The hardware cost and power make this impractical for caches larger than a few dozen entries. Fully associative organisation is used for small, critical structures: Translation Lookaside Buffers (TLBs, 16–64 entries) and victim caches.

## 12.6 The Cache READ Algorithm

Every read from the processor triggers the following hardware algorithm:

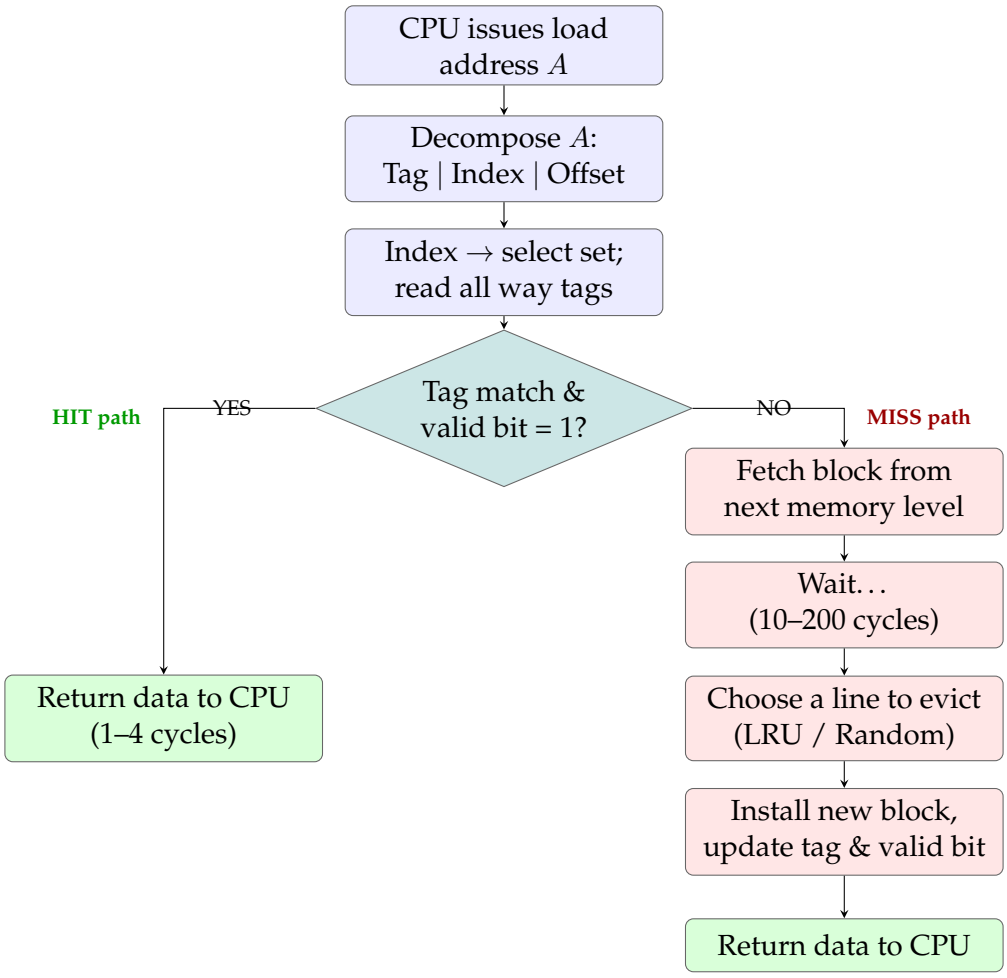



Figure 12.10: The cache READ algorithm. A hit returns data in 1–4 cycles. A miss triggers a block fetch from the next memory level (10–200 cycles depending on which level), followed by eviction and installation of the new block.

**Key question on a miss: which line to evict?** If all ways in the selected set are occupied, the hardware must choose one to discard. Common policies, LRU, Pseudo-

LRU, and Random, are covered in Section 12.7.3.

## 12.7 Hit, Miss, and the Three Types of Miss

### 12.7.1 Terminology

Table 12.4: Cache performance terminology.

| Term | Definition |
|---|---|
| Hit | The requested data is found in the cache. |
| Miss | The data is not in the cache; it must be fetched from lower memory. |
| Hit Rate | Fraction of memory accesses that are hits: HR = hits/total accesses. |
| Miss Rate | Fraction of accesses that miss: MR $= 1 -$ HR. |
| Hit Time | Time to service a cache hit (cycles). |
| Miss Penalty | *Extra* time to fetch a block on a miss (cycles). |

### 12.7.2 The Three Classes of Miss (The 3 Cs)

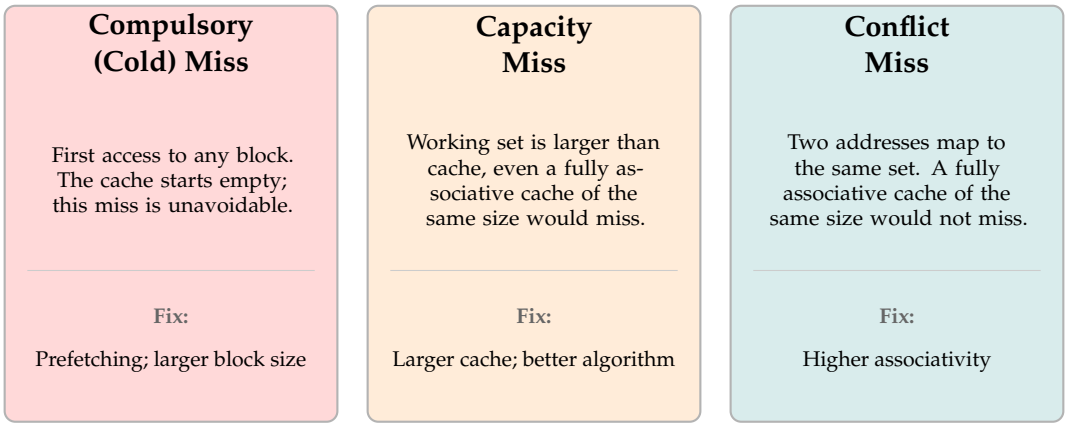



Figure 12.11: The Three Cs of cache misses. Each type has a different root cause and a different remedy.

### 12.7.3 Replacement Policies

When a miss occurs and all ways in a set are occupied, one must be evicted:

**Least Recently Used (LRU).** Evict the way accessed furthest in the past. Optimal for strong temporal locality; requires $N - 1$ bits per set to track order. Used in most L1 and L2 caches.

**Pseudo-LRU (PLRU).** Approximates LRU with a binary tree of 1-bit flags, costing only $\log_2 N$ bits per set. Used in ARM Cortex-A L1.

**Random.** Evict a randomly selected way. Requires no state and avoids the LRU anomaly (see below). Used in some ARM L3 caches.

**FIFO.** Evict the line that has been present the longest. Simple but can cycle pathologically when the working set has exactly $N + 1$ lines.

**The LRU Anomaly**

If the working set contains exactly $N + 1$ distinct lines and the cache is $N$-way, LRU evicts whichever line is needed *next* on every access — a 100% miss rate. Random replacement avoids this because it has no deterministic victim. This is a practical reason why real processors do not implement exact LRU.

**LRU replacement: step-by-step trace.** Consider a 2-way set-associative cache, one set, accessing lines A, B, C, A, B, C in order.

Table 12.5: LRU replacement trace: 2-way cache, one set, accesses A B C A B C.

| Step | Access | Way 1 | Way 2 | Result |
|---|---|---|---|---|
| 1 | A | **A** (MRU) | , | Miss (cold) |
| 2 | B | A | **B** (MRU) | Miss (cold) |
| 3 | C | **C** (MRU) | B | Miss, evict A (LRU) |
| 4 | A | C | **A** (MRU) | Miss, evict B (LRU) |
| 5 | B | **B** (MRU) | A | Miss, evict C (LRU) |
| 6 | C | B | **C** (MRU) | Miss, evict A (LRU) |

With 3 lines and a 2-way cache, every access is a miss, the LRU anomaly. A 3-way cache would hold all three lines simultaneously (compulsory misses only on steps 1–3, then all hits).

## 12.8 Write Policies

Reads are symmetric with the algorithm above. Writes introduce a choice: *when* do we propagate the change to lower memory?

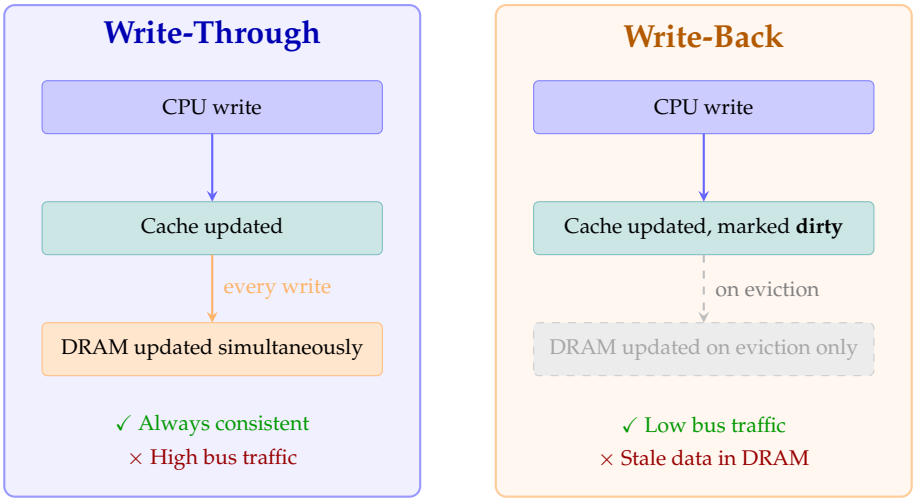



Figure 12.12: Write-through (left) updates DRAM on every write, keeping memory always consistent at the cost of bus traffic. Write-back (right) defers the DRAM update until the line is evicted, dramatically reducing bus traffic.

**Write-allocate vs. no-write-allocate.** On a write *miss*, should the block be fetched into the cache first?

- **Write-allocate (fetch on write):** fetch the block, then modify the cached copy. Natural partner of write-back. Good for data that will be read and written multiple times.
- **No-write-allocate (write around):** write directly to the next level without fetching. Natural partner of write-through. Avoids polluting the cache with write-once data.

**Standard Pairings in Practice**

Almost all caches use one of:

- **Write-back + write-allocate**, standard for L1 data caches and L2. Maximises hit rate by keeping written data in cache.
- **Write-through + no-write-allocate**, simpler coherence. Used for some L1 instruction caches and embedded systems.

### 12.8.1 The Write Buffer

Write-through caches generate a memory write on every store instruction. If the CPU can execute a store every cycle but memory takes 100 cycles to accept a write, the CPU would stall after each store.

A **write buffer** solves this by sitting between the cache and main memory as a small FIFO queue (typically 4–16 entries). The CPU deposits each write into the buffer and continues executing without waiting. The buffer drains into memory in the background.

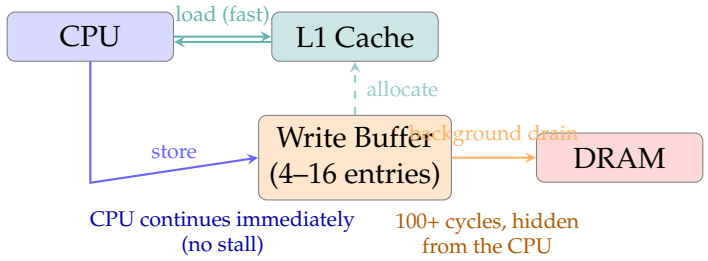



Figure 12.13: A write buffer decouples the CPU from memory write latency. Stores are placed in the buffer and the CPU continues immediately. The buffer drains to DRAM in the background.

**Write Buffer Hazard**

If a load follows a store to the same address and the store is still in the write buffer (not yet in the cache or memory), the load must read the value from the buffer rather than the stale cache. This is called a **write-buffer forwarding** hazard and is handled in hardware by checking the write buffer address before completing a load.

## 12.9 Cache Topology

**Cache topology** describes how many caches exist and how they are interconnected between the CPU core(s) and main memory.

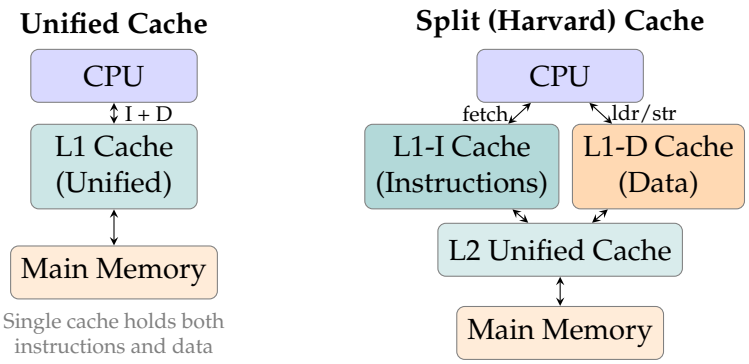



Figure 12.14: Unified cache (left): one cache holds instructions and data. Split (Harvard) cache (right): separate L1 caches for instructions and data allow the CPU to fetch an instruction and load data simultaneously, eliminating the structural hazard seen in the pipelined datapath.

**Unified.** A single L1 cache stores both instructions and data. Simpler hardware, but the CPU cannot simultaneously fetch an instruction and access a data operand, a structural hazard in the pipeline.

**Split (Harvard at L1).** Separate L1-I (instruction) and L1-D (data) caches. The CPU can read an instruction from L1-I and load a data word from L1-D in the same cycle. This is the standard design for all modern high-performance processors. L2 and L3 are typically unified.

## 12.10 Caches and the Pipeline

Chapter 11 built a five-stage pipeline that achieved CPI $\approx 1$ in the absence of hazards. That analysis assumed memory access took one clock cycle, an assumption that holds only when the L1 cache hits. When a cache miss occurs, the pipeline must stall until the data arrives from the next level.

### 12.10.1 Cache Miss as a Pipeline Stall

A load instruction (`ldr`) in the Memory stage of the pipeline requests data from the L1 cache. If the line is present (**hit**), the data returns in the same clock cycle and the pipeline continues without interruption. If the line is absent (**miss**), the Memory stage *stalls*:

- The pipeline freezes Fetch, Decode, and Execute (they hold their current instruction).
- The Memory stage signals the cache controller to fetch the missing block from L2 (or L3, or DRAM).
- After the miss penalty (10–200 cycles), the block arrives, the Memory stage completes, and the pipeline resumes.

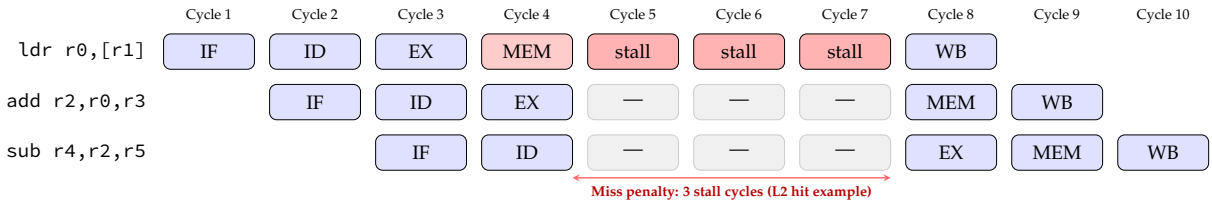



Figure 12.15: A cache miss on `ldr` stalls the pipeline. The instruction sits in the Memory stage until the block arrives (3 stall cycles shown for an L2 hit; up to 150+ for a DRAM miss). Downstream instructions freeze as bubbles.

### 12.10.2 Impact on CPI

Each cache miss adds stall cycles to the effective CPI:

$$\text{CPI}_{\text{actual}} = \text{CPI}_{\text{ideal}} + \text{MR}_{\text{data}} \times f_{\text{mem}} \times t_{\text{miss}}$$

where $f_{\text{mem}}$ is the fraction of instructions that access memory (typically 25–40%) and $\text{MR}_{\text{data}}$ is the data cache miss rate.

**Example.**

- Ideal CPI = 1.
- 30% of instructions are loads or stores ($f_{\text{mem}} = 0.30$).
- L1 data miss rate = 3%.
- L2 hit; miss penalty = 12 cycles.

$$\text{CPI}_{\text{actual}} = 1 + 0.03 \times 0.30 \times 12 = 1 + 0.108 = \mathbf{1.11}$$

An 11% CPI overhead from cache misses alone, and that is with an L2 hit. If the access falls through to DRAM (100-cycle penalty), the same 3% miss rate would give $\text{CPI} = 1 + 0.03 \times 0.30 \times 100 = 1.9$, nearly doubling execution time.

**L1 Cache is on the Critical Path**

The L1 cache access is part of the pipeline's MEM stage and thus on the critical path that determines the maximum clock frequency. A larger L1 cache has a longer access time and forces the clock period to increase, slowing down every instruction, not just the misses. This is why L1 caches are kept small (32–64 KB) even though a larger cache would reduce miss rate: the hit-time penalty of a larger cache would cost more than the miss-rate benefit.

## 12.11 Hardware Prefetching

A cache miss stalls the pipeline. **Hardware prefetching** tries to eliminate misses by fetching blocks *before* the processor requests them, based on detected access patterns.

### 12.11.1 How Prefetching Works

The processor contains a dedicated **prefetch engine** that monitors the stream of cache misses. When it detects a pattern, for example, sequential accesses to addresses `0x100`, `0x140`, `0x180` (stride 64), it issues a prefetch request for `0x1C0` in the background, before the CPU gets there.

**Sequential (stream) prefetcher.** Detects linear address sequences and fetches the next $N$ lines ahead. Ideal for array traversal and instruction fetch. All ARM Cortex-A cores include this.

**Stride prefetcher.** Detects constant-stride access patterns (e.g. accessing every 8th element of an array). Used in ARM Cortex-A76 and later.

**Next-line prefetcher.** Simply prefetches the line immediately after every demand fetch. Very cheap to implement; effective for instruction caches.

**Prefetching in ARM64 assembly.** The ISA also provides a software prefetch instruction for cases where the programmer or compiler can predict what will be needed:

```
// ARM64: hint that the cache line at address x0 will be needed
   soon
PRFM PLDL1KEEP, [x0]    // Prefetch for Load into L1, keep after
   use
PRFM PLDL2KEEP, [x0, #128] // Prefetch next stride into L2

// Example: prefetch ahead in a loop
loop:
  PRFM PLDL1KEEP, [x1, #64]   // bring next line into L1 now
  ldr  w2, [x1], #4            // load current element
  add  w3, w3, w2
  subs w4, w4, #1
  bne  loop
```

**Prefetching Can Hurt**

Prefetching is not always beneficial. If the prefetched line is not used before it is evicted (cache pollution), or if the memory bus is already saturated, prefetching wastes bandwidth and can increase miss rates. Always measure before adding software prefetch hints.

## 12.12 Cache Coherence in Multi-Core Systems

Modern processors have multiple cores, each with its own L1 and L2 cache. When two cores hold copies of the same cache line and one writes to it, the other core's copy becomes **stale**. **Cache coherence** is the guarantee that all cores always see a consistent view of memory.

### 12.12.1 The Coherence Problem

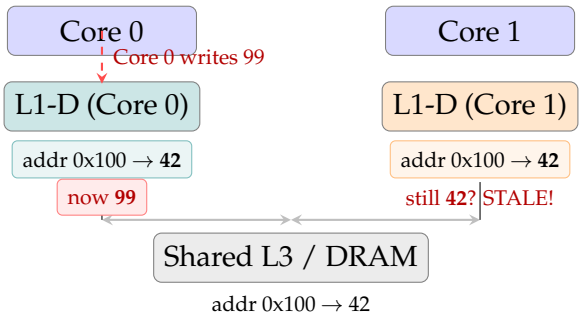



Figure 12.16: The coherence problem: Core 0 writes 99 to address `0x100` but Core 1's L1 cache still holds the old value 42. Without a coherence protocol, Core 1 reads stale data.

### 12.12.2 The MESI Protocol

ARM processors use the **MESI** protocol (or extensions of it) to maintain coherence automatically in hardware. Every cache line carries a 2-bit state tag:

Table 12.6: MESI cache line states.

| State | Name | Meaning |
|---|---|---|
| **M** | Modified | This cache has the only valid copy; it has been written and differs from memory. Must write back before another core can read. |
| **E** | Exclusive | This cache has the only valid copy; it is clean (identical to memory). Can be written without notifying others. |
| **S** | Shared | Multiple caches hold this line; all copies are clean. Cannot write without first invalidating others. |
| **I** | Invalid | This line is not present (or has been invalidated by another core's write). |

**How MESI prevents stale reads.** When Core 0 writes to a **Shared** line:

1. Core 0 broadcasts an **Invalidate** message on the interconnect.
2. All other caches with that line transition from S to **Invalid**.
3. Core 0's line transitions to **Modified**.
4. If Core 1 now reads the line, it is Invalid, it misses, fetches the updated value from Core 0 (or L3), and Core 0's line transitions from M to Shared.

This guarantees that Core 1 never reads stale data, at the cost of an extra cache miss (an **invalidation miss**).

**False Sharing: A Cache Coherence Performance Trap**

Two variables can share a cache line even if they are logically independent. If Core 0 frequently writes variable A and Core 1 frequently writes variable B, and A and B happen to live in the same 64-byte cache line, every write by either core invalidates the other's copy, even though neither core reads the other's variable. This **false sharing** can drop performance by 10–100× on multi-core workloads. The fix is to pad structures so that frequently written fields land in separate cache lines (`alignas(64)` in C11/C++11).

## 12.13 Cache Performance Analysis

### 12.13.1 The Cost of No Cache: A Motivating Example

Before computing AMAT, let us quantify how badly memory latency hurts without a cache, using a concrete example from this course.

**Setup.**

- Processor: 1 GHz clock ($T_{\text{cycle}} = 1\,\text{ns}$).
- Main memory: 10 ns access time ($= 10$ cycles).
- 35% of all instructions are loads or stores.
- Program: 1 billion instructions.

**Execution time without a cache.** Every load/store must wait 10 cycles for DRAM:

$$T_{\text{no cache}} = \big(1\,\text{ns} + 0.35 \times 10\,\text{ns}\big) \times 10^9 = 4.5 \times 10^9\,\text{ns} = \mathbf{4.5}\,\text{s}$$

**Execution time with a perfect cache.** If every load/store hits in 1 cycle:

$$T_{\text{perfect cache}} = 1\,\text{ns} \times 10^9 = \mathbf{1.0}\,\text{s}$$

**Execution time with a real cache (90% hit rate).**

$$\begin{aligned} T_{\text{cache}} &= \big(1\,\text{ns} + 0.35 \times (1 - 0.90) \times 10\,\text{ns}\big) \times 10^9 \\ &= \big(1 + 0.35 \times 0.1 \times 10\big)\,\text{ns} \times 10^9 = 1.35\,\text{s} \end{aligned}$$

**The Speedup a Cache Provides**

$$\text{Speedup} = \frac{T_{\text{no cache}}}{T_{\text{cache}}} = \frac{4.5\,\text{s}}{1.35\,\text{s}} \approx \mathbf{3.33}\times$$

A cache with a 90% hit rate reduces a 4.5-second program to 1.35 seconds. Real caches achieve 95–99% hit rates for instruction fetches and 75–90% for data, yielding speedups of 3–10× depending on the workload.

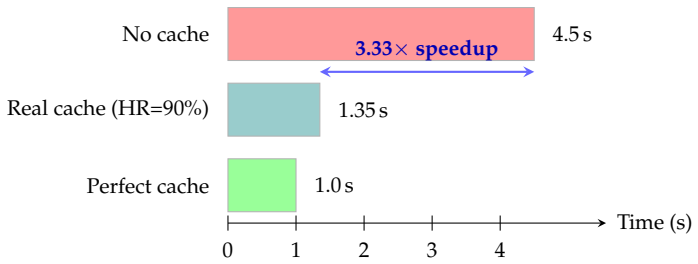



Figure 12.17: Execution time for a 1-billion-instruction program on a 1 GHz processor with 10 ns DRAM and 35% memory instructions. A 90% hit-rate cache reduces execution time from 4.5 s to 1.35 s.

### 12.13.2 Average Memory Access Time (AMAT)

The general formula for effective memory latency is:

$$\text{AMAT} = t_{\text{hit}} + \text{MR} \times t_{\text{miss}}$$

Table 12.7: AMAT sensitivity to miss rate ($t_{\text{hit}} = 2$ cycles, $t_{\text{miss}} = 100$ cycles).

| Hit Rate | Miss Rate | Miss cycles/access | AMAT |
|---|---|---|---|
| 99% | 1% | 1.0 | 3.0 cycles |
| 98% | 2% | 2.0 | 4.0 cycles |
| 95% | 5% | 5.0 | 7.0 cycles |
| 90% | 10% | 10.0 | 12.0 cycles |
| 75% | 25% | 25.0 | 27.0 cycles |

Even dropping from 99% to 95% hit rate more than doubles the AMAT — confirming that hit time is the most important parameter for L1 caches.

### 12.13.3 Two-Level AMAT

With L1 and L2 caches:

$$\text{AMAT} = t_{L1} + \text{MR}_{L1} \times \left(t_{L2} + \text{MR}_{L2} \times t_{\text{DRAM}}\right)$$

**Worked example.**

- $t_{L1} = 2$ cycles, $\text{MR}_{L1} = 5\%$
- $t_{L2} = 12$ cycles, $\text{MR}_{L2} = 20\%$ (local)
- $t_{\text{DRAM}} = 100$ cycles

$$\text{AMAT} = 2 + 0.05 \times (12 + 0.20 \times 100) = 2 + 0.05 \times 32 = \mathbf{3.6} \text{ cycles}$$

Compared with a single-level cache giving $\text{AMAT} = 2 + 0.05 \times 100 = 7$ cycles, the L2 cache nearly halves the effective latency.

## 12.14 Multi-Level Caches in Real ARM Processors

Table 12.8: Cache parameters for representative ARM Cortex cores. All use 64-byte cache lines.

| Core | L1-I | L1-D | L2 | L3 | Write Policy |
|---|---|---|---|---|---|
| Cortex-A53 | 8–32 KB | 8–32 KB | 128–512 KB | Optional | WB + WA |
| Cortex-A76 | 64 KB | 64 KB | 256–512 KB | 1–4 MB | WB + WA |
| Cortex-X4 | 192 KB | 96 KB | 512 KB | 32 MB | WB + WA |
| Neoverse V2 | 64 KB | 64 KB | 1 MB | 64 MB | WB + WA |

**Cache maintenance in ARM assembly.** When the software must guarantee coherence (DMA I/O, self-modifying code), ARM provides explicit cache control instructions:

```
// ARM64: flush a cache line containing address in x0
DC CIVAC, x0   // Clean and Invalidate D-cache line to Point of
   Coherency
IC IVAU,  x0   // Invalidate I-cache line to Point of Unification
DSB SY         // Data Sync Barrier: ensure all cache ops complete
ISB            // Instruction Sync Barrier: flush pipeline fetch
```

**▶ Observing Cache Behaviour in PlayARM**

Run the following in PlayARM with the **Memory Access** panel open:

```
ldr r0, =array       // load base address of array
ldr r1, [r0, #0]     // access 1: MISS  - fetches entire 64-byte
    line
ldr r2, [r0, #4]     // access 2: HIT   - within same line
ldr r3, [r0, #8]     // access 3: HIT
ldr r4, [r0, #12]    // access 4: HIT
ldr r5, [r0, #64]    // access 5: MISS  - next 64-byte line
.data
array: .word 1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16
```

Watch accesses 1 and 5 show **cache miss** (memory fetch required) while accesses 2–4 show **cache hit** (served from the same line brought in on access 1). This is spatial locality in action.

## 12.15 Cache-Friendly Programming

Understanding caches allows you to write code that minimises misses.

**Access arrays sequentially.** Iterating `a[0], a[1], ...` generates one miss for every 16 integers (in a 64-byte line of 4-byte ints). Iterating with stride 16 generates a miss on *every* access.

```
// Cache friendly: sequential (1 miss per 16 elements)
for (int i = 0; i < N; i++)   sum += a[i];

// Cache unfriendly: stride-16 (1 miss per element)
for (int i = 0; i < N; i += 16)  sum += a[i];
```

**Prefer structure-of-arrays over array-of-structures.** If only one field is accessed in a loop, an array-of-structures wastes cache capacity loading unused fields.

```
// Array of structures: poor cache use if only .x is needed
struct Point { float x, y, z, w; } pts[1000];
// 4 floats per point = 16 bytes; 4 points per 64-byte line

// Structure of arrays: only x[] loaded, 16 x-values per line
float x[1000], y[1000], z[1000], w[1000];
```

**Keep the working set small.**

- Working set $\leq$ L1 (32 KB): peak performance.
- Working set $\leq$ L2 (2 MB): $\approx 5\times$ slower than L1.
- Working set $\leq$ L3 (32 MB): $\approx 20\times$ slower than L1.
- Working set in DRAM: $\approx 100\times$ slower than L1.

**Matrix multiplication example.** The inner loop of naive matrix multiplication accesses column `j` of matrix `B` with a row stride, a cache killer for large $N$:

```
// Naive: B[k][j] has stride N between consecutive k -- cache
   unfriendly
for (i=0; i<N; i++)
  for (j=0; j<N; j++)
    for (k=0; k<N; k++)
      C[i][j] += A[i][k] * B[k][j];   // B accessed column-wise

// Loop interchange (i,k,j): B[k][j] now accessed row-wise -- cache
    friendly
for (i=0; i<N; i++)
  for (k=0; k<N; k++)
    for (j=0; j<N; j++)
      C[i][j] += A[i][k] * B[k][j];
```

## 12.16 Summary

- The **ideal memory** is infinitely large, infinitely fast, and free, all three properties are mutually exclusive with current technology. The memory hierarchy approximates the ideal by combining multiple technologies.
- The hierarchy is a pyramid: **registers** $\rightarrow$ **L1** $\rightarrow$ **L2** $\rightarrow$ **L3** $\rightarrow$ **DRAM**. Each level is faster, smaller, and more expensive than the one below. On a miss, a full **block** is fetched from the next level.

- Caches exploit **temporal locality** (reuse of the same address) and **spatial locality** (use of nearby addresses). These are not guarantees, they are statistical tendencies of real programs.
- A memory address is divided into **tag**, **index**, and **offset**. Index selects a set; tag confirms the correct line; offset picks the byte within the line.
- **Direct-mapped** (1-way): simple, fast, high conflict misses. $N$**-way set-associative**: $N$ parallel comparators, fewer conflict misses. **Fully associative**: no conflict misses, only practical for very small structures.
- On a **read hit**: return data in 1–4 cycles. On a **read miss**: fetch block ($\geq 10$ cycles), evict if needed, install, return data.
- The **Three Cs**: compulsory (first access, fix with prefetching), capacity (working set too large, fix with bigger cache), conflict (same set contention, fix with more ways).
- **Write-back + write-allocate** is the standard policy for L1-D and L2 caches; write-through for simpler coherence.
- AMAT $= t_{\text{hit}} + \text{MR} \times t_{\text{miss}}$. A 90% hit-rate cache on a 1 GHz system with 10 ns DRAM reduces execution time from 4.5 s to 1.35 s — a **3.33× speedup**.
- Modern ARM cores use **split L1** (separate I-cache and D-cache), **unified L2**, and a **shared L3**. All levels use 64-byte cache lines.
- Cache-friendly code: access memory *sequentially*, keep *working sets small*, use *structure-of-arrays* when only one field is needed, and *interchange loops* in matrix algorithms to restore row-major access order.

## 12.17 Exercises

**Exercise 12.17.1 (Address Decomposition)** *A direct-mapped cache has 256 sets and a line size of 64 bytes. The processor uses 32-bit addresses.*

1. *How many bits form the block offset? The index? The tag?*
2. *Which cache set does address* `0x00001240` *map to?*
3. *Two addresses* `0x00001240` *and* `0x00002240` *are accessed alternately. Do they conflict? Why or why not?*

**Exercise 12.17.2 (Cache Simulation)** *A direct-mapped cache has 4 sets and a line size of 16 bytes (word-addressed). Trace these byte-addresses and mark H (hit) or M (miss). The cache is cold.*

$$0,\ 16,\ 4,\ 0,\ 64,\ 16,\ 0,\ 4$$

*State which set each address maps to and what is evicted on each miss.*

**Exercise 12.17.3 (Speedup Calculation)** *A 2 GHz processor runs a program of 500 million instructions. 20% of instructions are loads or stores. Main memory has a 15 ns latency.*

1. *Calculate the execution time without a cache.*
2. *Calculate the execution time with a cache having a 95% hit rate.*
3. *What is the speedup?*
4. *What hit rate would be needed to achieve a 4× speedup over the no-cache case?*

**Exercise 12.17.4 (AMAT Calculation)** *A system has the following parameters:* $t_{L1} = 2$ *cycles,* $MR_{L1} = 4\%$, $t_{L2} = 14$ *cycles,* $MR_{L2} = 25\%$, $t_{DRAM} = 120$ *cycles.*

1. *Compute AMAT for the two-level hierarchy.*
2. *If $MR_{L1}$ is halved (to 2%) by doubling L1 size, what is the new AMAT?*
3. *Which is more beneficial: halving $MR_{L1}$ or halving $MR_{L2}$? Show your calculations.*

**Exercise 12.17.5 (Write Policies)** *Explain write-through versus write-back. For each scenario below, choose the more appropriate policy and justify:*

1. *A microcontroller with a single core writing sensor data to a buffer.*
2. *A 16-core server processor where multiple cores share data structures.*
3. *A tight inner loop that writes to an accumulator 10 million times before the result is read.*

**Exercise 12.17.6 (Miss Classification)** *Classify each miss as compulsory, capacity, or conflict:*

1. *The first time a program accesses a newly allocated array.*
2. *A sorting algorithm that accesses a 256 MB array on a machine with a 32 MB L3 cache.*
3. *Two $512 \times 512$ float matrices whose base addresses differ by exactly the direct-mapped cache size, causing every access to one to evict a line from the other.*

**Exercise 12.17.7 (Cache-Friendly Matrix Multiply)** *The naive $ijk$ matrix multiplication loop:*

```
for (i=0; i<N; i++)
  for (j=0; j<N; j++)
    for (k=0; k<N; k++)
      C[i][j] += A[i][k] * B[k][j];
```

1. *Identify which array access pattern is cache unfriendly and why.*
2. *Rewrite using loop interchange to fix the problem.*
3. *For $N = 512$ and 32-bit floats, how many 64-byte cache lines does one row of matrix `B` occupy?*
4. *Estimate how many cache misses the naive loop generates for matrix `B` versus the fixed loop, for $N = 512$ and a 32 KB L1 cache.*

# 13

# Virtual Memory

# 14

# Parallel Architectures

# 15

# Conclusion

This concludes our introduction to computer architecture.